Prognostic Modelling

pea700 · MSc Physics of the Earth and Atmosphere · Universities of Bonn and Cologne

Stefan Kollet

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Preface

This script accompanies the lecture *Prognostic Modelling* (pea700). It develops numerical methods for the equations that govern the atmosphere, the land surface and the solid Earth, from ordinary differential equations to coupled nonlinear partial differential equations.

The script, the slides and the exercises are generated from one source. All code is written in Python and is shown next to the results it produces.

Syllabus

Module	pea700 Prognostic Modelling (Prognostische Modellierung)
Programme	MSc Physics of the Earth and Atmosphere
Credits	6 LP, 180 h
Format	Lecture (2 SWS) and exercises (2 SWS), 14 weeks
Assessment	Project
Prerequisite for assessment	Successful completion of the exercises (50 % of total points)
Lecturer	Stefan Kollet

Learning goals

Students

- know common numerical methods for geophysical and meteorological prognostic models and their properties,
- can judge the accuracy, stability, cost and limits of a method and interpret its results,
- can implement, verify and document numerical models in Python following good programming practice.

Outline

Unit	Part	Topic
1	Foundations	What is prognostic modelling?
2	Foundations	Governing equations and their classification
3	Foundations	The numerical toolbox I: derivatives, integrals, roots

Unit	Part	Topic
4	Foundations	The numerical toolbox II: matrices and waves
5	Time integration of ODEs	One-step methods
6	Time integration of ODEs	Stability, consistency, convergence, stiffness
7	Time integration of ODEs	Oscillations and multistep methods
8	Time integration of ODEs	Chaos, predictability, stochastic ODEs
9	Elliptic problems	Finite differences and finite volumes for elliptic problems
10	Elliptic problems	Direct solvers
11	Elliptic problems	Iterative solvers
12	Parabolic problems	Diffusion
13	Hyperbolic problems	Linear advection
14	Hyperbolic problems	Conservative and monotone advection
15	Hyperbolic problems	Waves and hyperbolic systems
16	Hyperbolic problems	Boundaries, domains and geometry
17	Nonlinear and coupled problems	Nonlinear PDEs
18	Nonlinear and coupled problems	Operator splitting, coupling and parametrization
19	Other discretizations	Galerkin and finite element methods
20	Other discretizations	Spectral methods
21	Outlook	Data assimilation and inverse problems
22	Outlook	High-performance computing and machine learning

Part I

Foundations

1 What is prognostic modelling?

Learning goals

After this unit you can

- explain what a prognostic model is and how prognostic and diagnostic equations differ,
- name the steps that lead from a physical law to a simulation result,
- list the sources of error in a numerical model and say which of them you control,
- explain round-off error and recognise loss of significance in a computation,
- distinguish verification from validation.

1.1 Models of the Earth system

In this lecture we deal with prognostic modelling of the physics of the Earth and the atmosphere. Before we write down any equation, we clarify what models are for.

Think first: what are models good for?

- **Prediction.** A model simulates the past and present state of a system and predicts its future state. Weather and climate models predict the future state of the atmosphere and the ocean. This is what we call prognostic modelling.
- **Filling gaps.** Measurements of temperature, rain or water levels are never continuous in space and time, not even from satellites. Models interpolate between locations and between measurement times. Nobody can measure the future, so looking ahead always needs a model that extrapolates.
- **Seeing the unobservable.** Models give access to variables for which no sensor exists, or which cannot be measured with the required accuracy or coverage, for example the water stored in the deep subsurface.
- **Virtual experiments.** Models allow experiments that are impossible, too expensive or too dangerous in reality: doubling atmospheric CO_2, removing a mountain range, moving an earthquake source.
- **Understanding.** Models help to understand complex systems and their feedbacks, because every process can be switched on and off.

Think first: what types of models exist, and what are their strengths?

Type	Strength	Weakness
Analog (laboratory experiment)	It is real	Needs a laboratory and often much money; valid only in the range of the experiment; scaling to the real system is difficult
Statistical, including machine learning	Works without knowing the governing laws; fast once trained	Needs much data; based on assumptions that are hard to check; unreliable outside the range of the training data
Mathematical (physics-based)	Based on conservation laws, therefore able to extrapolate	Based on simplifying assumptions; parameters are often unknown at the scale of the model; needs applied mathematics and computing

This lecture treats mathematical models. The three types are increasingly combined, for example when a machine-learning model is trained on the output of a physics-based model (Chapter 22).

1.1.1 A first mathematical model

A very familiar physical model is Newton's second law, $ma = F$. Written as a differential equation it reads

$$m\frac{\mathrm{d}^2u}{\mathrm{d}t^2} = F\left(t, u, \frac{\mathrm{d}u}{\mathrm{d}t}\right), \tag{1.1}$$

where $u(t)$ is the position of a particle at time t and F is the force acting on it. Many problems in science and engineering have this form: an equation that contains one or more derivatives of an unknown function. To model the motion of the particle we have to find the function $u(t)$ that satisfies Equation 1.1.

Throughout the lecture u denotes the unknown. Depending on the problem it is a position, a temperature, a pressure, a water depth or a vector of all of these.

1.1.2 Prognostic and diagnostic equations

A **prognostic equation** contains a time derivative of the unknown. It states how fast the state changes, and therefore allows us to step from the state now to the state a moment later:

$$\frac{\partial u}{\partial t} = \mathcal{F}(u, t). \tag{1.2}$$

A **diagnostic equation** contains no time derivative. It relates variables at the same instant. The ideal gas law $p = \rho RT$ and hydrostatic balance $\partial p/\partial z = -\rho g$ are diagnostic equations of atmospheric models.

A prognostic model consists of prognostic equations for the variables that carry the memory of the system, the **state**, together with diagnostic equations for everything that follows from the state. The mathematical task is an **initial value problem**: given the state at $t = 0$, find the state at later times.

Model	Prognostic variables (examples)	Equations
Numerical weather prediction, climate	wind, temperature, pressure or density, humidity	conservation of momentum, energy and mass in a rotating, stratified fluid
Ocean circulation	currents, temperature, salinity, sea surface height	the same, for sea water
Groundwater and soil hydrology	pressure head or water content	conservation of mass with Darcy's law
Seismic wave propagation	particle velocity, stress	conservation of momentum with a stress–strain relation

All these equations are derived in the same way, from the conservation of mass, momentum and energy (Chapter 2), and they are solved with the same small set of numerical methods. That is why meteorologists and geophysicists can share this lecture.

1.2 From physical law to computer code

Almost no prognostic model of a real system can be solved with pencil and paper. We have to solve it numerically, which means that a simulation result is separated from reality by a chain of steps:

1. **Physical system.** The atmosphere, an aquifer, the crust.

2. **Conceptual model.** Which processes matter, which are neglected, where the boundaries of the domain are.
3. **Mathematical model.** Differential equations with initial and boundary conditions and parameters.
4. **Discretization.** The continuous equations are replaced by algebraic equations for a finite number of unknowns.
5. **Solution algorithm.** The algebraic equations are solved, often only approximately.
6. **Implementation.** The algorithm is written as a program.
7. **Simulation and evaluation.** The program is run and its result is compared with observations.

Every step introduces errors. The lecture is mainly about steps 4 to 6, but you will only be able to judge a result if you keep the whole chain in mind.

1.2.1 The central idea: replace derivatives by differences

A computer can add, subtract, multiply and divide. It cannot differentiate. The simplest way to translate a differential equation into arithmetic is the **finite difference method**, which replaces a derivative by a difference quotient, for example

$$\frac{\mathrm{d}u}{\mathrm{d}t} \approx \frac{u(t+\Delta t) - u(t)}{\Delta t}, \tag{1.3}$$

where Δt is the time step. The derivative is the limit of the right hand side for $\Delta t \to 0$. On a computer Δt stays finite, and this is the origin of the discretization error.

As an example take the decay equation

$$\frac{\mathrm{d}u}{\mathrm{d}t} = -ku, \qquad u(0) = u_0, \tag{1.4}$$

which describes radioactive decay, the cooling of a body or the emptying of a linear reservoir. Its exact solution is $u(t) = u_0 e^{-kt}$. We insert Equation 1.3, write u_n for the approximation of $u(n\Delta t)$ and solve for the new value:

$$u_{n+1} = u_n - k\,\Delta t\, u_n. \tag{1.5}$$

This is a complete prognostic model. Starting from u_0 it produces the state at all later times using only multiplication and subtraction.

```python
import numpy as np
import matplotlib.pyplot as plt

k, u0, t_end = 1.0, 1.0, 4.0

fig, ax = plt.subplots(figsize=(6, 3.5))
t_exact = np.linspace(0, t_end, 200)
ax.plot(t_exact, u0 * np.exp(-k * t_exact), "k", label="exact")

for dt in [1.0, 0.5, 0.1]:
    n_steps = int(round(t_end / dt))
    u = np.empty(n_steps + 1)
    u[0] = u0
    for n in range(n_steps):
        u[n + 1] = u[n] - k * dt * u[n]
    ax.plot(dt * np.arange(n_steps + 1), u, "o-", ms=3, label=rf"$\Delta t$ = {dt}")
    error = abs(u[-1] - u0 * np.exp(-k * t_end))
    print(f"dt = {dt:4.1f}: error at t = {t_end:.0f} is {error:.4f}")

ax.set_xlabel("$t$")
ax.set_ylabel("$u$")
ax.legend()
plt.show()
```

```
dt =  1.0: error at t = 4 is 0.0183
dt =  0.5: error at t = 4 is 0.0144
dt =  0.1: error at t = 4 is 0.0035
```

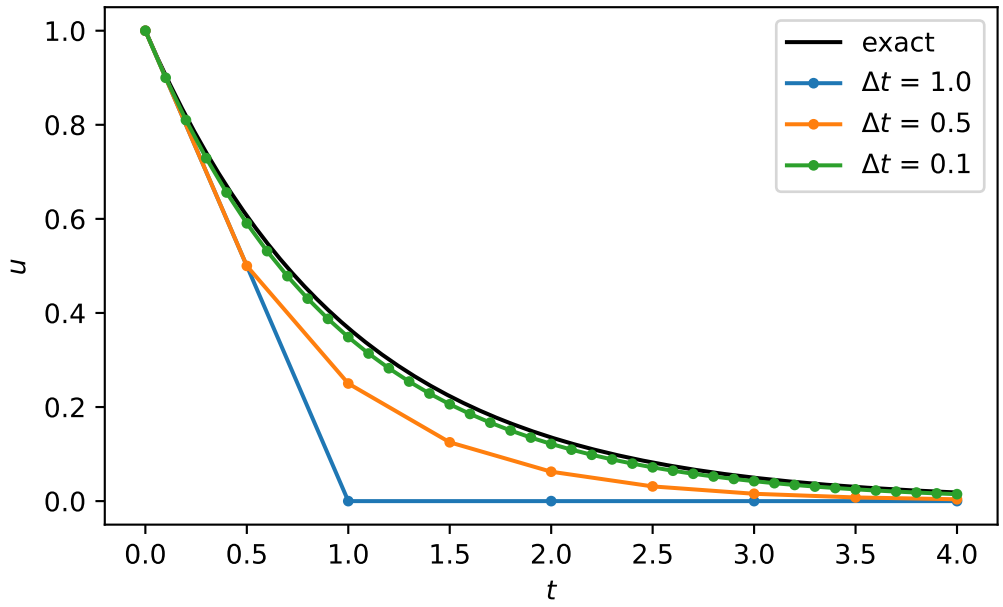


Figure 1.1: Numerical solution of the decay equation with the forward difference for three time steps, compared with the exact solution.

Figure 1.1 shows the behaviour we hope for: the smaller the time step, the closer the numerical solution is to the exact one. It also raises the questions that occupy us for the rest of the semester.

- How fast does the error decrease when Δt decreases? (Accuracy)
- With $\Delta t = 1$ the numerical solution drops to zero in one step. What happens for $\Delta t = 3$? (Stability)
- Is there a better formula than Equation 1.5 for the same cost? (Efficiency)

1.2.2 A first analysis

For this simple model the first two questions can be answered on paper. Equation 1.5 multiplies the solution by the same factor in every step, $u_{n+1} = (1 - k\Delta t)\, u_n$. Applying it n times,

$$u_1 = (1 - k\Delta t)\, u_0, \quad u_2 = (1 - k\Delta t)^2\, u_0, \quad \ldots, \quad u_n = (1 - k\Delta t)^n\, u_0. \tag{1.6}$$

The exact solution is multiplied in every step by $e^{-k\Delta t}$, since $u(t + \Delta t) = u_0 e^{-k(t+\Delta t)} = e^{-k\Delta t}\, u(t)$. The Taylor series of this factor is

$$e^{-k\Delta t} = 1 - k\Delta t + \frac{(k\Delta t)^2}{2} - \dots$$

The numerical factor $1 - k\Delta t$ consists of its first two terms.

Accuracy. In one step the numerical and the exact factor differ by $(k\Delta t)^2/2$. To reach a fixed time t we need $n = t/\Delta t$ steps, so the errors add up to something proportional to $n\,(k\Delta t)^2 \propto \Delta t$. The error at a fixed time is proportional to the time step: halving Δt halves the error.

Stability. The behaviour of $u_n = (1 - k\Delta t)^n u_0$ depends only on the size of the factor.

Time step	Factor $1 - k\Delta t$	Numerical solution
$k\Delta t < 1$	between 0 and 1	decays monotonically, as the exact solution
$k\Delta t = 1$	0	zero after one step
$1 < k\Delta t < 2$	between -1 and 0	decays, but changes sign in every step
$k\Delta t > 2$	below -1	oscillates and grows without bound

For $k\Delta t > 2$ the numerical solution has nothing in common with the exact one. The scheme is unstable. The time step of this scheme is therefore limited by the fastest process in the model, not by the accuracy we want. This observation, generalised to systems and to partial differential equations, is one of the main themes of the lecture (Chapter 6).

1.3 Sources of error and uncertainty

Think first: which errors occur when a numerical model is developed and applied?

Collect your own list before reading on, and sort it by the step of the modelling chain at which the error arises.

The following table sorts the errors by their origin (after Wick 2020).

Origin	Error	Description
Conceptual and mathematical model	Model error	Processes are missing or simplified, for example a linear equation replaces a nonlinear one. Found only by comparison with observations and by careful reasoning.
	Homogenization error	A model at full resolution is replaced by an averaged one; processes smaller than the grid are represented by parameters (parametrization, Chapter 18).
	Regularization error	An ill-posed problem is modified so that it becomes well-posed.
Data	Data error	Initial conditions, boundary conditions and parameters come from measurements and are uncertain and incomplete.
	Reference error	The observations used for evaluation are themselves inaccurate.
Discretization	Discretization error	Continuous functions are represented by a finite number of values.
	Interpolation error	Values between grid points are obtained from simpler functions.
Solution algorithm	Iteration error	Iterative algorithms are stopped after a finite number of steps.
Computer	Round-off error	A computer represents only a finite set of numbers.
Implementation	Bugs	Some make the program abort, which is the harmless case. Others produce results that look plausible and are wrong.

Two remarks help to keep this list in proportion.

First, the errors are of very different size, and the total error is dominated by the largest contribution. If the parameters of a groundwater model are known only within a factor of ten, it is pointless to reduce the discretization error to 0.01 %. Deciding which error dominates is a central task of numerical modelling.

Second, errors in the data are of two kinds. **Epistemic** uncertainty comes from lack of knowledge and can in principle be reduced by more measurements. **Aleatoric** uncertainty comes from variability that is random for all practical purposes. Both are treated by running the model many times (Chapter 8).

Of all these errors, the discretization, iteration and round-off errors and the bugs are the ones that numerical methods control. The rest of this unit treats round-off; the discretization error is the subject of most of the following units.

1.4 Floating point arithmetic and round-off

1.4.1 How a computer stores real numbers

A computer stores a real number as a **floating point number**

$$x = \pm\, m \cdot 2^{e},$$

with a significand m and an exponent e that each occupy a fixed number of bits. Python and NumPy use by default the *double precision* format of the IEEE 754 standard with 64 bits, of which 53 hold the significand. This has three consequences (Goldberg 1991).

1. **Finite precision.** A double carries about 16 significant decimal digits. The distance between 1 and the next larger floating point number is the **machine epsilon**, $\varepsilon = 2^{-52} \approx 2.2 \times 10^{-16}$.
2. **Finite range.** The largest double is about 1.8×10^{308}. Larger results overflow to `inf`.
3. **Most decimal fractions are not representable.** In base 2 the number 0.1 has infinitely many digits, as 1/3 has in base 10, and is rounded when it is stored.

```
import numpy as np

print("machine epsilon:        ", np.finfo(float).eps)
print("1 + eps/2 == 1:         ", 1.0 + np.finfo(float).eps / 2 == 1.0)
print("0.1 + 0.2 == 0.3:       ", 0.1 + 0.2 == 0.3)
print("0.1 + 0.2 =             ", repr(0.1 + 0.2))
print("largest double:         ", np.finfo(float).max)
print("single precision epsilon:", np.finfo(np.float32).eps)
```

```
machine epsilon:          2.220446049250313e-16
1 + eps/2 == 1:           True
0.1 + 0.2 == 0.3:         False
0.1 + 0.2 =               0.30000000000000004
largest double:           1.7976931348623157e+308
single precision epsilon: 1.1920929e-07
```

Every arithmetic operation therefore returns the exact result rounded to the nearest floating point number, with a relative error of at most $\varepsilon/2$. Two rules follow immediately.

- Never test floating point numbers for equality. Test whether their difference is small, for example with `np.isclose`.
- Single precision has only about 7 significant digits. Weather and climate models nevertheless increasingly use it, because it halves memory and data transfer. Whether this is safe has to be shown for each algorithm.

1.4.2 Loss of significance

A relative error of 10^{-16} sounds harmless, and for a single operation it is. It becomes dangerous when two nearly equal numbers are subtracted. The leading digits cancel, and the result consists mostly of the rounding errors of the operands. Consider

$$f(x) = \frac{1 - \cos x}{x^2} \;\to\; \frac{1}{2} \quad \text{for } x \to 0.$$

```python
def f_naive(x):
    return (1 - np.cos(x)) / x**2

def f_stable(x):
    return 2 * np.sin(x / 2) ** 2 / x**2

for x in [1e-2, 1e-4, 1e-6, 1e-8]:
    print(f"x = {x:.0e}:  naive {f_naive(x):.10f}   stable {f_stable(x):.10f}")
```

```
x = 1e-02:  naive 0.4999958333   stable 0.4999958333
x = 1e-04:  naive 0.4999999970   stable 0.4999999996
x = 1e-06:  naive 0.5000444503   stable 0.5000000000
x = 1e-08:  naive 0.0000000000   stable 0.5000000000
```

For $x = 10^{-8}$ the first formula returns 0 instead of 0.5: $\cos x$ differs from 1 by 5×10^{-17}, which is less than the machine epsilon, so it is stored as exactly 1. The second formula uses the identity $1 - \cos x = 2\sin^2(x/2)$, is mathematically identical and avoids the subtraction.

This has a direct consequence for our central idea. The difference quotient Equation 1.3 subtracts two numbers that become more and more equal as the step decreases. Making the step ever smaller therefore does not make a finite difference ever more accurate. We quantify this in Chapter 3.

1.4.3 Conditioning

Loss of significance is an example of a more general notion. A problem is **well-conditioned** if small relative changes of the input cause small relative changes of the result, and **ill-conditioned** if they cause large ones. The amplification factor is the **condition number** κ.

For the evaluation of a function $f(x)$ it follows from the Taylor series. A small change δx of the input changes the result by $\delta f \approx f'(x)\,\delta x$. We divide by f and expand with x to obtain relative changes on both sides:

$$\frac{\delta f}{f} \approx \frac{f'(x)\,\delta x}{f(x)} = \underbrace{\frac{x\,f'(x)}{f(x)}}_{\pm\kappa}\,\frac{\delta x}{x}, \qquad \kappa = \left|\frac{x\,f'(x)}{f(x)}\right|.$$

The problem. We apply this to the example above, $f(x) = (1-\cos x)/x^2$. For small x the Taylor series of the cosine gives

$$f(x) = \frac{1}{2} - \frac{x^2}{24} + \ldots, \qquad f'(x) \approx -\frac{x}{12},$$

and therefore

$$\kappa_f = \left|\frac{x\,f'(x)}{f(x)}\right| \approx \frac{x^2/12}{1/2} = \frac{x^2}{6}.$$

For $x = 10^{-8}$ this is about 10^{-17}. The problem is extremely well-conditioned: a small relative error in x has almost no effect on f. The failure of `f_naive` therefore cannot be blamed on the problem.

The algorithm. `f_naive` computes f in steps, and each step is a problem of its own with its own condition number. First it computes $c = \cos x$. The result is rounded, so c carries a relative error of up to ε. Then it computes $h(c) = 1 - c$. With the same formula, now for the input c,

$$\kappa_h = \left|\frac{c\,h'(c)}{h(c)}\right| = \frac{|c|}{|1-c|}.$$

For small x the Taylor series $\cos x = 1 - x^2/2 + x^4/24 - \dots$ gives $c \approx 1$ and $1 - c = x^2/2 - x^4/24 + \dots \approx x^2/2$, hence

$$\kappa_h \approx \frac{1}{x^2/2} = \frac{2}{x^2}.$$

For $x = 10^{-8}$ this is 2×10^{16}. The rounding error $\varepsilon \approx 10^{-16}$ of c is amplified to a relative error of order one: no correct digit remains. `f_stable` computes $\sin(x/2)$, squares it and divides by x^2. These steps have condition numbers of about 1 and 2, so the rounding errors stay at the level of ε.

```
eps = np.finfo(float).eps
print("     x   kappa problem   kappa step 1-cos x   predicted error   actual error")
for x in [1e-2, 1e-4, 1e-6, 1e-8]:
    kappa_f = x**2 / 6                                    # the problem f
    c = np.cos(x)
    kappa_h = abs(c) / abs(1 - c) if c != 1 else np.inf    # the step 1 - c in
↪   f_naive
    predicted = kappa_h * eps                             # bound for the relative
↪   error of f_naive
    actual = abs(f_naive(x) - f_stable(x)) / f_stable(x)   # f_stable serves as the
↪   exact value
    print(f"{x:6.0e}   {kappa_f:13.1e}   {kappa_h:18.1e}   {predicted:15.1e}
    ↪  {actual:12.1e}")
```

```
     x   kappa problem   kappa step 1-cos x   predicted error   actual error
 1e-02         1.7e-05              2.0e+04           4.4e-12        2.9e-13
 1e-04         1.7e-09              2.0e+08           4.4e-08        5.2e-09
 1e-06         1.7e-13              2.0e+12           4.4e-04        8.9e-05
 1e-08         1.7e-17                  inf               inf        1.0e+00
```

The last two columns compare the error predicted by the condition number of the step $1 - c$, $\kappa_h\,\varepsilon$, with the actual relative error of `f_naive`, for which the result of `f_stable` serves as the exact value. The prediction is an upper bound, since the rounding error of $\cos x$ is usually smaller than ε; it lies within a factor of about 10 of the actual error. Above all, it explains why the error grows by a factor of 10^4 each time x shrinks by a factor of 100. For $x = 10^{-8}$ the stored value of $\cos x$ is exactly 1: the step $1 - c$ has lost all information, which the infinite condition number expresses, and the relative error is 100 %.

Conditioning and stability. Conditioning is a property of the problem. It must be distinguished from the **stability** of an algorithm. An algorithm is unstable if it amplifies errors more than the problem itself does, typically because it contains an ill-conditioned intermediate step that the problem does not need. Here the problem is well-conditioned, `f_naive` is unstable

because of the step $1 - \cos x$, and `f_stable` is stable. For an ill-conditioned problem, in contrast, no algorithm helps; the problem itself has to be reformulated.

We meet the condition number again for linear systems of equations, where the same definition leads to the condition number of a matrix (Chapter 4, Chapter 11), and ill-conditioning in its most famous geophysical form, the limited predictability of weather (Chapter 8).

1.5 Concepts that return in every unit

A few concepts apply to all numerical methods of this lecture. We introduce them informally here and make them precise in Chapter 6.

Approximation The exact solution u is not available. We compute an approximation $\tilde{u}$ and need to know how far it is from u.

Consistency A method is consistent if the discrete equations approach the differential equation when the grid spacing and the time step go to zero.

Stability A method is stable if errors, for example round-off errors, do not grow without bound during the computation. Unstable methods typically show oscillations that grow in time. It is therefore important to know from the physics whether oscillations are to be expected.

Convergence A method converges if the numerical solution approaches the exact solution when the grid spacing and the time step go to zero. The rate at which it does so is the **order** of the method.

Efficiency The accuracy reached for a given amount of computing time, memory and energy. A method of higher order is often, but not always, more efficient.

Well-posedness A property of the mathematical problem: a solution exists, it is unique, and it depends continuously on the data (Chapter 2). A numerical method cannot repair a problem that is not well-posed.

1.6 Verification and validation

Two questions must be asked of every model (Oberkampf and Roy 2010).

Verification: are we solving the equations right? This is a mathematical question. It is answered by comparing the numerical solution with exact solutions and by checking that the error decreases at the expected rate when the grid is refined. Verification concerns discretization, algorithm and code, and it can be completed.

Validation: are we solving the right equations? This is a physical question. It is answered by comparing the simulation with observations. Validation concerns the conceptual and mathematical model and the data, and it is never complete, because a model can only be shown to agree with the observations available so far.

The order matters. A model that has not been verified cannot be validated, because agreement with observations may result from errors that compensate each other. In this lecture you verify every program you write.

1.7 Working practice

Numerical modelling is to a large part work on paper. Classifying the problem, deriving the discrete equations, and writing down the grid, the indices and the boundary conditions come first. Programming, debugging and plotting are the smaller part, and they go quickly when the paper work is complete.

The following habits save most of the time that is usually lost.

- **Simplify.** Start with the lowest resolution, the simplest boundary and initial conditions and constant coefficients. Add one complication at a time.
- **Know the answer in advance.** Test first on a problem with a known solution. For a linear system with five unknowns you can check every number by hand.
- **Write functions and test them.** A function that does one thing can be tested once and then trusted: on a problem with a known answer, and again after every change. In the exercises you collect your functions in your own module `toolbox.py` and reuse them in later sheets and in the project.
- **Use version control.** With `git` every working state of a program can be recovered, and every result can be attributed to the version of the code that produced it.
- **Make results reproducible.** A figure should be produced by a script from the raw output, without manual steps. All figures in this script are produced by the code shown next to them.

1.8 Summary

- A prognostic model predicts the future state of a system from its present state with equations that contain time derivatives. Diagnostic equations relate variables at the same time.
- A simulation result is separated from reality by a chain of steps: conceptual model, mathematical model, discretization, algorithm, implementation. Each contributes errors, and the largest contribution dominates.
- The basic idea of discretization is to replace derivatives by difference quotients. The resulting discretization error decreases with the step.
- Floating point numbers have about 16 significant digits. Subtracting nearly equal numbers destroys most of them.
- Verification checks the solution of the equations, validation checks the equations. Verification comes first.

1.9 Further reading

Durran (2010), chapter 1, for the view from atmospheric science; Goldberg (1991) for floating point arithmetic; Oberkampf and Roy (2010) for verification and validation.

2 Governing equations and their classification

Learning goals

After this unit you can

- compute and interpret gradient, divergence, curl and Laplacian,
- derive a conservation law from a balance over a control volume,
- classify a differential equation by order, linearity and type,
- recognise the prototype equations for diffusion, advection and waves in the equations of atmospheric, hydrological and seismic models,
- choose initial and boundary conditions that make a problem well-posed.

Numerical modelling starts on paper with the classification of the problem. The type of an equation decides which numerical methods work and which fail. This unit provides the language: the operators of vector calculus, the conservation laws built from them, the classification of the resulting equations, and the equations of this lecture with their types.

2.1 Vector calculus recap

2.1.1 Derivatives

If the unknown depends on a single variable, for example $u(t)$, its rate of change is the ordinary derivative $\mathrm{d}u/\mathrm{d}t$. If it depends on several variables, for example $u(x,t)$, the partial derivatives $\partial u/\partial t$ and $\partial u/\partial x$ give the rate of change with respect to one variable while the others are held fixed. Derivatives of order n are written $\mathrm{d}^n u/\mathrm{d}t^n$ and $\partial^n u/\partial x^n$. We also use the short forms u_t, u_x, u_{xx}.

2.1.2 Vectors

A vector in the plane is written $\vec{v} = v_1\vec{i} + v_2\vec{j}$, in space $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$, where $\vec{i}, \vec{j}, \vec{k}$ are the unit vectors along the coordinate axes. Its magnitude is

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2},$$

and the unit vector in the direction of $\vec{v}$ is $\vec{v}/\|\vec{v}\|$.

2.1.3 The gradient

The gradient of a scalar field $f(x, y, z)$ is the vector

$$\operatorname{grad} f = \nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}.$$

It has two geometric properties, shown in Figure 2.1.

- ∇f points in the direction in which f increases fastest, and is perpendicular to the contour of f through the point.
- $\|\nabla f\|$ is the rate of change in that direction. It is large where the contours are close together.

In physics, gradients drive fluxes: heat flows down the temperature gradient, groundwater flows down the gradient of hydraulic head, and the pressure gradient accelerates the wind.

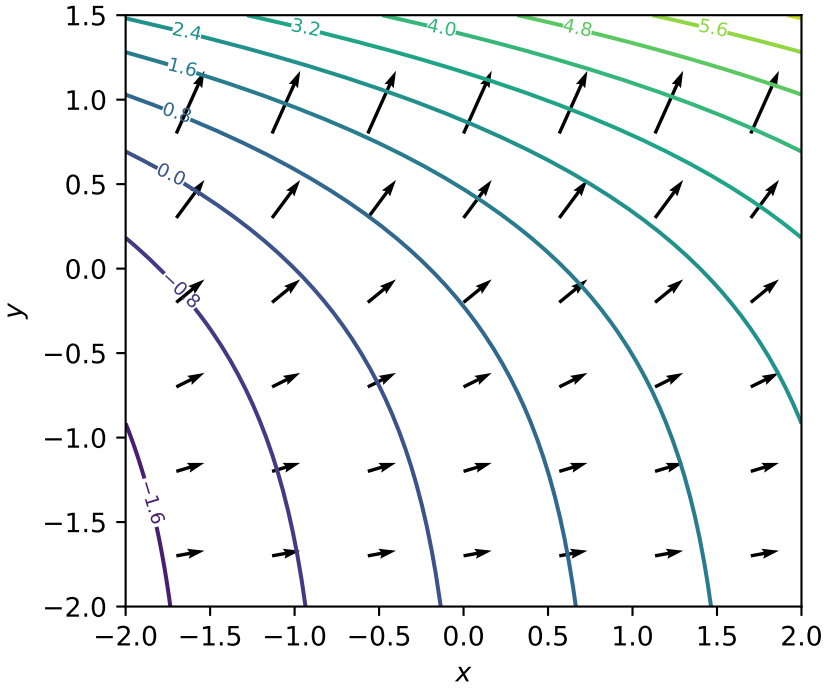


Figure 2.1: Contours of $f(x, y) = x + e^y$ and its gradient vectors. The vectors are perpendicular to the contours and longer where the contours are closer.

Exercise: gradient of $f(x, y) = x + e^y$ at the point $(1, 1)$

$\nabla f = \vec{i} + e^y \vec{j}$, so $\nabla f(1, 1) = \vec{i} + e\,\vec{j}$. The maximum rate of change at this point is $\|\nabla f\| = \sqrt{1 + e^2} \approx 2.90$.

2.1.4 The divergence

The divergence of a vector field $\vec{F}$ measures the outflow per unit volume at a point. Geometrically, it is the flux out of a small closed surface S around the point, divided by the enclosed volume V, in the limit of vanishing volume:

$$\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \lim_{V \to 0} \frac{1}{V} \oint_S \vec{F} \cdot \mathrm{d}\vec{A}. \tag{2.1}$$

Here $\mathrm{d}\vec{A}$ is the surface element with the direction of the outward normal. In Cartesian coordinates the divergence is

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}. \tag{2.2}$$

Why do both definitions agree? Consider a small box with one corner at (x_0, y_0, z_0) and edge lengths $\Delta x, \Delta y, \Delta z$ (Figure 2.2). On the face at $x = x_0$ the outward normal is $-\vec{i}$. If $\vec{F}$ is approximately constant over this small face, the flux through it is

$$\int_{S_1} \vec{F} \cdot \mathrm{d}\vec{A} \approx -F_1(x_0, y_0, z_0)\, \Delta y\, \Delta z.$$

On the opposite face at $x = x_0 + \Delta x$ the outward normal is $+\vec{i}$, and the flux is $F_1(x_0 + \Delta x, y_0, z_0)\, \Delta y\, \Delta z$. The sum of the two is

$$\frac{F_1(x_0 + \Delta x, y_0, z_0) - F_1(x_0, y_0, z_0)}{\Delta x}\, \Delta x\, \Delta y\, \Delta z \approx \frac{\partial F_1}{\partial x}\, \Delta x\, \Delta y\, \Delta z.$$

The other two pairs of faces contribute in the same way. Dividing the total flux by the volume $\Delta x\, \Delta y\, \Delta z$ gives Equation 2.2.

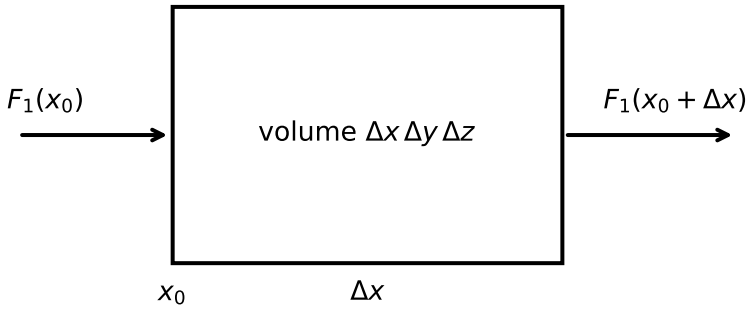


Figure 2.2: Fluxes in the x direction through a control volume. The divergence is the net outflow per unit volume.

This derivation is more than a proof. Summing fluxes over the faces of a small box is exactly what the finite volume method does (Chapter 9).

Two results that we use constantly:

Product rule. For a scalar field g and a vector field $\vec{F}$,

$$\nabla \cdot (g\vec{F}) = g\,\nabla \cdot \vec{F} + \vec{F} \cdot \nabla g. \tag{2.3}$$

Divergence theorem (Gauss). For a volume V with closed surface S,

$$\int_V \nabla \cdot \vec{F}\,\mathrm{d}V = \oint_S \vec{F} \cdot \mathrm{d}\vec{A}. \tag{2.4}$$

What flows out through the surface equals the sum of all sources inside.

2.1.5 Curl and Laplacian

The **curl** $\nabla \times \vec{F}$ measures the local rotation of a vector field. For a horizontal wind field $\vec{v} = u\vec{i} + v\vec{j}$ its vertical component is the relative vorticity

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y},$$

a central quantity of dynamic meteorology. A small paddle wheel placed in the flow and carried along with it spins with the angular velocity $\zeta/2$: anticlockwise for $\zeta > 0$, clockwise for $\zeta < 0$. Figure 2.3 shows two examples. In solid-body rotation, $u = -y$, $v = x$, we get $\zeta = 1 - (-1) = 2$.

In the shear flow $u = y$, $v = 0$, we get $\zeta = 0 - 1 = -1$. The streamlines of the shear flow are straight lines, yet the flow has vorticity: the faster flow on the upper side of the wheel pushes it clockwise. Curl measures local rotation, not curved streamlines.

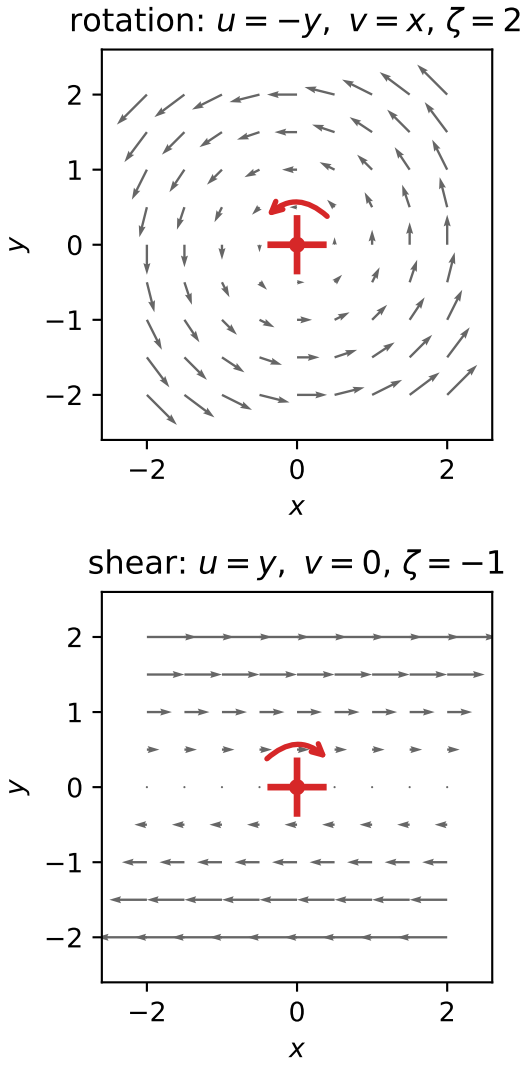


Figure 2.3: Two velocity fields with vorticity, each with a paddle wheel (red) and its sense of rotation. Top: solid-body rotation, $\zeta = 2$, anticlockwise. Bottom: shear flow with straight streamlines, $\zeta = -1$, clockwise.

The **Laplacian** is the divergence of the gradient:

$$\nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

It compares the value of f at a point with the average over its neighbourhood. To see this, start in one dimension and take the two neighbours of x at distance h. Their Taylor expansions are

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{6} f'''(x) + O(h^4),$$
$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{6} f'''(x) + O(h^4).$$

Adding the two lines, the odd-order terms cancel:

$$f(x+h) + f(x-h) = 2f(x) + h^2 f''(x) + O(h^4).$$

Dividing by 2 and subtracting $f(x)$ gives the difference between the average of the neighbours and the value at the point:

$$\underbrace{\frac{f(x+h) + f(x-h)}{2}}_{\text{neighbour average}} - f(x) = \frac{h^2}{2} f''(x) + O(h^4).$$

In two dimensions we do the same in the x and the y direction, with the four neighbours $(x \pm h, y)$ and $(x, y \pm h)$. Each pair gives its own second derivative, exactly as in one dimension:

$$f(x+h, y) + f(x-h, y) = 2f(x, y) + h^2 f_{xx} + O(h^4),$$
$$f(x, y+h) + f(x, y-h) = 2f(x, y) + h^2 f_{yy} + O(h^4).$$

We call the average over the four neighbours $\bar{f}$. Adding the two lines and dividing by 4 gives

$$\bar{f} = f(x, y) + \frac{h^2}{4} (f_{xx} + f_{yy}) + O(h^4),$$

and hence

$$\bar{f} - f(x, y) = \frac{h^2}{4} \nabla^2 f + O(h^4).$$

In three dimensions, with six neighbours, the factor is $h^2/6$. So, up to a positive factor, the Laplacian is the amount by which the neighbourhood average exceeds the value at the point:

- $\nabla^2 f < 0$: f is larger than its surroundings, as at a local maximum (a peak);
- $\nabla^2 f > 0$: f is smaller than its surroundings, as at a local minimum (a trough);
- $\nabla^2 f = 0$: f equals the average of its surroundings. Such functions are called harmonic; they have no interior maxima or minima.

This is why the Laplacian appears in every equation that describes smoothing. In the diffusion equation $\partial u/\partial t = D\nabla^2 u$ with $D > 0$, u decreases where it lies above the average of its neighbourhood and increases where it lies below. Peaks are eroded, troughs are filled, and the field approaches its local average. Rearranged, the formula above is also the finite difference approximation

$$\nabla^2 f \approx \frac{f(x+h,y) + f(x-h,y) + f(x,y+h) + f(x,y-h) - 4f(x,y)}{h^2}$$

that we will use for elliptic and parabolic problems.

2.2 Conservation laws

Nearly all equations of this lecture state that a quantity is conserved: mass, momentum or energy. They are all derived with the same argument.

Let u be the density of the conserved quantity (amount per volume) and $\vec{F}$ its flux (amount per area and time). The amount inside a fixed volume V changes only by flux through the surface and by sources s (amount per volume and time) inside:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V u \,\mathrm{d}V = -\oint_S \vec{F}\cdot \mathrm{d}\vec{A} + \int_V s \,\mathrm{d}V. \tag{2.5}$$

The minus sign appears because $\mathrm{d}\vec{A}$ points outward. We turn this balance into a differential equation in three steps.

1. The volume is fixed in space, so the time derivative can be taken inside the integral: $\frac{\mathrm{d}}{\mathrm{d}t}\int_V u \,\mathrm{d}V = \int_V \frac{\partial u}{\partial t}\,\mathrm{d}V$.
2. The divergence theorem Equation 2.4 converts the surface integral into a volume integral: $\oint_S \vec{F}\cdot\mathrm{d}\vec{A} = \int_V \nabla\cdot\vec{F}\,\mathrm{d}V$.
3. All terms are now integrals over the same volume and can be collected:

$$\int_V \left(\frac{\partial u}{\partial t} + \nabla\cdot\vec{F} - s\right)\mathrm{d}V = 0.$$

This holds for every volume V, however small and wherever it is placed. That is only possible if the integrand itself vanishes everywhere, which gives the differential form

$$\frac{\partial u}{\partial t} + \nabla \cdot \vec{F} = s. \tag{2.6}$$

Equation 2.6 is a prognostic equation for u, but it is not yet closed, because it contains the unknown flux. A **flux law** (constitutive relation) expresses $\vec{F}$ in terms of u. Two flux laws cover most of this lecture.

Advective flux. The quantity is carried along by a velocity field $\vec{v}$:

$$\vec{F} = \vec{v}\, u.$$

Diffusive flux. The quantity flows down its own gradient:

$$\vec{F} = -D\, \nabla u.$$

The second law has many names, depending on the quantity.

Law	Flux	Driven by	Coefficient
Fourier	heat	temperature gradient	thermal conductivity
Fick	dissolved substance	concentration gradient	diffusion coefficient
Darcy	groundwater	gradient of hydraulic head	hydraulic conductivity

We insert the total flux $\vec{F} = \vec{v}\, u - D\, \nabla u$ into Equation 2.6 and keep track of every sign. The divergence is linear, so it acts on the two parts separately:

$$\nabla \cdot \vec{F} = \nabla \cdot (\vec{v}\, u) + \nabla \cdot (-D\, \nabla u) = \nabla \cdot (\vec{v}\, u) - \nabla \cdot (D\, \nabla u).$$

Inserted into Equation 2.6:

$$\frac{\partial u}{\partial t} + \nabla \cdot (\vec{v}\, u) - \nabla \cdot (D\, \nabla u) = s.$$

Moving the diffusion term to the right-hand side changes its sign once more, which gives the **advection–diffusion equation**

$$\frac{\partial u}{\partial t} + \nabla \cdot (\vec{v}\, u) = \nabla \cdot (D\, \nabla u) + s. \tag{2.7}$$

The diffusion term now carries a plus sign. Two minus signs have cancelled: the one from the outward normal in Equation 2.5 (flux out of the volume reduces u) and the one from the flux law (the flux points down the gradient).

Keep track of the signs

Lost or doubled minus signs are among the most frequent and most time-consuming errors in numerical modelling, in course projects as well as in research codes. A code with a sign error usually runs; it just gives wrong results, and the error is often found only after days of debugging. The signs depend on conventions that differ between textbooks, disciplines and codes:

- **Direction of the normal**: outward (as here) or inward.
- **Flux law**: $\vec{F} = -D\nabla u$, or a flux defined positive in the direction of decreasing u.
- **Vertical coordinate**: z pointing up (height) or down (depth, common in hydrology and seismology). Gravity terms change sign with it.
- **Operator**: many solvers are written for $-\nabla^2 u = f$, because the matrix of $-\nabla^2$ is positive definite (Chapter 4); others for $\nabla^2 u = f$.
- **Source term**: $s > 0$ as a source or as a sink (e.g. extraction from a well, or evaporation).

Three habits prevent most of these errors:

1. **Write the convention down** at the top of the derivation and in the code (which direction is positive, what the flux means).
2. **Check the physics of the result.** In $\partial u/\partial t = D\nabla^2 u$, at a maximum $\nabla^2 u < 0$ (see the Laplacian above), so u must decrease there. With the wrong sign the maximum grows; this is anti-diffusion, and it is unstable.
3. **Test the code on a case with a known answer**: a Gaussian must spread and flatten, the total amount $\int u \, \mathrm{d}V$ must stay constant without sources, and water must flow from high to low hydraulic head.

Flux form and advective form

With the product rule Equation 2.3 the advection term becomes $\nabla \cdot (\vec{v}u) = \vec{v} \cdot \nabla u + u \, \nabla \cdot \vec{v}$. For a divergence-free (incompressible) flow the last term vanishes and the equation can be written in **advective form**, $\partial u/\partial t + \vec{v} \cdot \nabla u = \ldots$

The two forms are mathematically equivalent. Their discretizations are not. A discretization of the **flux form** conserves the total amount of u exactly, because what leaves one grid cell enters its neighbour. A discretization of the advective form in general does not (Chapter 14).

2.3 Classification of differential equations

2.3.1 Ordinary and partial differential equations

If the unknown depends on one independent variable, the equation contains only ordinary derivatives and is an **ordinary differential equation** (ODE). An example is the equation for the charge $Q(t)$ on a capacitor in a circuit with inductance L, resistance R and capacitance C:

$$L\frac{\mathrm{d}^2Q}{\mathrm{d}t^2} + R\frac{\mathrm{d}Q}{\mathrm{d}t} + \frac{1}{C}Q = E(t).$$

If the unknown depends on several independent variables, the equation contains partial derivatives and is a **partial differential equation** (PDE). The advection–diffusion equation Equation 2.7 is a PDE.

2.3.2 Systems

Several unknowns coupled by several equations form a system. The predator–prey equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = ax - \alpha xy, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = -cy + \gamma xy$$

for the number of prey x and predators y are a system of two ODEs. The models of the atmosphere and the ocean are systems of PDEs, one for each prognostic variable.

2.3.3 Order

The order of a differential equation is the order of the highest derivative in it. It determines how many initial and boundary conditions are needed.

Every ODE of order n can be rewritten as a system of n first-order ODEs. For Newton's law Equation 1.1 we introduce the velocity $v = \mathrm{d}u/\mathrm{d}t$ as a second unknown:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = v, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{F}{m}.$$

This is important in practice: numerical methods and software for ODEs are written for first-order systems. PDEs of second order in time can be treated in the same way; for seismic waves we will see that the wave equation is equivalent to a system of two first-order equations.

2.3.4 Linear and nonlinear

An equation is **linear** if the unknown and its derivatives appear only to the first power and are not multiplied with each other. The coefficients may depend on the independent variables. The general linear ODE of order n is

$$a_0(t)\,y^{(n)} + a_1(t)\,y^{(n-1)} + \cdots + a_n(t)\,y = g(t).$$

It is **homogeneous** if $g(t) = 0$.

Equation	Order	Linear?
$y' - 2xy = e^x$	1	yes
$y'' + 4y' + 4y = 0$	2	yes, homogeneous
$y' = x/y$	1	no
$y'' + \sin y = x^2$	2	no
$y''' + 2e^t y'' + yy' = t^4$	3	no
$u_t + c(x)\,u_x = 0$	1	yes
$u_t + u\,u_x = 0$	1	no

The last two rows deserve attention. Advection by a *given* velocity field $c(x)$ is linear, even though the coefficient varies. Advection of the velocity by itself, as in the momentum equations of atmosphere and ocean, is nonlinear.

The pendulum equation

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{L}\sin\theta = 0$$

is nonlinear. For small angles $\sin\theta \approx \theta$, and it becomes the linear equation of the harmonic oscillator. This **linearization** is a standard tool: the stability of numerical methods is analysed for linear equations, and the results are then applied to the nonlinear problem with care.

Linearity has a great practical advantage. Solutions of a linear homogeneous equation can be added (superposition), so that the solution can be analysed wave by wave (Chapter 4). Discretizing a linear PDE gives a linear system of algebraic equations; discretizing a nonlinear PDE gives a nonlinear system that has to be solved iteratively (Chapter 17).

2.3.5 Elliptic, parabolic, hyperbolic

The most important classification of PDEs concerns their type. Linear second-order PDEs in two independent variables have the general form

$$a\,u_{xx} + 2b\,u_{xy} + c\,u_{yy} + \text{lower-order terms} = f, \tag{2.8}$$

where the second variable y may be a space coordinate or time. The type is determined by the coefficients of the second derivatives alone:

Type	Condition	Prototype	Eigenvalues of A
Elliptic	$b^2 - ac < 0$	Laplace, Poisson: $u_{xx} + u_{yy} = f$	same sign
Parabolic	$b^2 - ac = 0$	Diffusion: $u_t = Du_{xx}$	one is zero
Hyperbolic	$b^2 - ac > 0$	Wave: $u_{tt} = c^2 u_{xx}$	opposite signs

The names come from the quadratic form $q(x, y) = ax^2 + 2bxy + cy^2$, whose contour $q = 1$ is an ellipse or a hyperbola under the first and third condition. In matrix notation,

$$q(x,y) = \begin{pmatrix} x & y \end{pmatrix} A \begin{pmatrix} x \\ y \end{pmatrix}, \qquad A = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

and the condition can be read off the signs of the eigenvalues of A. To see why, we compute the eigenvalues. They are the roots of the characteristic polynomial

$$\det(A - \lambda I) = \begin{vmatrix} a-\lambda & b \\ b & c-\lambda \end{vmatrix} = (a-\lambda)(c-\lambda) - b^2 = \lambda^2 - (a+c)\,\lambda + (ac - b^2).$$

A quadratic polynomial with roots λ_1 and λ_2 can be written as $(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$. Comparing the coefficients gives

$$\lambda_1 + \lambda_2 = a + c, \qquad \lambda_1\lambda_2 = ac - b^2 = \det A.$$

The eigenvalues are real, because A is symmetric. The discriminant of the characteristic polynomial confirms this: $(a+c)^2 - 4(ac - b^2) = (a-c)^2 + 4b^2 \geq 0$. The sign of the product $\lambda_1\lambda_2$ therefore decides the type, and it is the opposite of the sign of $b^2 - ac$:

- $b^2 - ac < 0$: $\lambda_1\lambda_2 > 0$, both eigenvalues have the same sign (elliptic);
- $b^2 - ac = 0$: $\lambda_1\lambda_2 = 0$, one eigenvalue is zero (parabolic);
- $b^2 - ac > 0$: $\lambda_1\lambda_2 < 0$, the eigenvalues have opposite signs (hyperbolic).

The geometric meaning follows from the eigenvectors. Because A is symmetric, its eigenvectors are orthogonal, and in coordinates (ξ, η) along the eigenvectors the mixed term disappears:

$$q = \lambda_1 \xi^2 + \lambda_2 \eta^2.$$

For $\lambda_1, \lambda_2 > 0$ the contour $q = 1$ is an ellipse with semi-axes $1/\sqrt{\lambda_1}$ and $1/\sqrt{\lambda_2}$. For eigenvalues of opposite sign it is a hyperbola. If $\lambda_2 = 0$, $q = \lambda_1 \xi^2$ does not depend on η at all, and $q = 1$ degenerates into two parallel lines, the limiting case between ellipse and hyperbola. The same rotation of coordinates applied to the PDE removes the mixed derivative and gives its **canonical form** $\lambda_1 u_{\xi\xi} + \lambda_2 u_{\eta\eta} + \text{lower-order terms} = f$. Elliptic equations then look like the Poisson equation, hyperbolic ones like the wave equation, and parabolic ones contain only one second derivative, like the diffusion equation.

As an example, take $u_{xx} + 4u_{xy} + u_{yy} = 0$, that is $a = c = 1$ and $b = 2$. The condition gives $b^2 - ac = 4 - 1 = 3 > 0$: hyperbolic. The eigenvalues confirm this: $\lambda_1 + \lambda_2 = 2$ and $\lambda_1 \lambda_2 = 1 - 4 = -3$, so $\lambda_1 = 3$ and $\lambda_2 = -1$, of opposite sign. The mixed derivative hides the type: both pure second derivatives have the same sign, yet the equation is a wave equation in rotated coordinates.

The eigenvalue formulation carries over to more than two variables, where the condition $b^2 - ac$ no longer exists (Wick 2020). With n independent variables, A is a symmetric $n \times n$ matrix. The equation is elliptic if all eigenvalues have the same sign, parabolic if one eigenvalue is zero and the others have the same sign, and hyperbolic if one eigenvalue has the opposite sign of the others. The heat equation $u_t = D(u_{xx} + u_{yy})$ in the variables (x, y, t), for example, has the eigenvalues D, D and 0 (up to an overall sign): parabolic. We meet quadratic forms again when we solve linear systems by minimisation (Chapter 11).

Check: classify the diffusion equation with this rule

Write $u_t - Du_{xx} = 0$ with the variables (x, t) in the form Equation 2.8. The only second derivative is u_{xx} with $a = -D$, and $b = c = 0$. Hence $b^2 - ac = 0$: parabolic. For the wave equation $u_{tt} - c_0^2 u_{xx} = 0$ we have $a = -c_0^2$, $b = 0$, $c = 1$ and $b^2 - ac = c_0^2 > 0$: hyperbolic.

What the three types mean physically is shown by their prototypes in the next section.

2.4 Prototype equations

Three special cases of conservation laws in one space dimension serve as prototypes, one for each type of the classification. Each has its own physical behaviour, and each requires its own numerical treatment.

Diffusion equation (heat equation), from Equation 2.7 with $\vec{v} = 0$ and constant D; parabolic:

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}. \tag{2.9}$$

Advection equation, with constant velocity c and $D = 0$; of first order and hyperbolic:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0. \tag{2.10}$$

Wave equation, with wave speed c; hyperbolic:

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}. \tag{2.11}$$

On an unbounded domain all three have exact solutions for the initial condition $u(x, 0) = f(x)$.

Advection. The solution is $u(x, t) = f(x - ct)$: the initial shape moves with velocity c and does not change. Information travels along the lines $x - ct = \text{const}$, the **characteristics** of the equation, with the finite speed c; first-order equations with this property are called hyperbolic, like the wave equation. To verify the solution, set $\xi = x - ct$ and apply the chain rule:

$$\frac{\partial u}{\partial t} = f'(\xi)\frac{\partial \xi}{\partial t} = -c\, f'(\xi), \qquad \frac{\partial u}{\partial x} = f'(\xi)\frac{\partial \xi}{\partial x} = f'(\xi) \quad \Longrightarrow \quad \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0.$$

This verifies the solution but does not tell us where it comes from. The box derives it.

Derivation: the method of characteristics

We look at u along a curve $x(t)$ in the (x, t) plane, that is at the function $U(t) = u(x(t), t)$. Its rate of change follows from the chain rule for a function of two variables, both of which depend on t:

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial u}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}t} = \frac{\partial u}{\partial t} + \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\partial u}{\partial x}.$$

The right-hand side looks like the left-hand side of Equation 2.10. We choose the curve such that the two agree, that is with

$$\frac{\mathrm{d}x}{\mathrm{d}t} = c.$$

Integrating from $t = 0$, where the curve starts at the point x_0, gives the straight line $x(t) = x_0 + ct$. Along this line the PDE becomes an ODE:

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0.$$

So U does not change, and u keeps the value it had at the starting point:

$$u(x_0 + ct, t) = U(t) = U(0) = u(x_0, 0) = f(x_0).$$

Finally we express x_0 by the point (x, t) we are interested in. The characteristic through (x, t) started at $x_0 = x - ct$, hence

$$u(x, t) = f(x - ct).$$

The derivation shows more than the verification:

- **Uniqueness.** Every point (x, t) lies on exactly one characteristic, and the value there is fixed by the initial condition. There is no other solution.
- **Domain of dependence.** $u(x, t)$ depends only on the initial value at the single point $x - ct$, not on f anywhere else. Information travels along the characteristics with speed c.
- **Generalisation.** The same steps work for a variable velocity $c(x, t)$: the characteristics are the solutions of the ODE $\mathrm{d}x/\mathrm{d}t = c(x, t)$, are no longer straight, and u is still constant along them. We return to this view of advection in the semi-Lagrangian methods (Chapter 14).

Wave equation. Differentiating once more gives $\partial^2 u/\partial t^2 = c^2 f''(\xi)$ and $\partial^2 u/\partial x^2 = f''(\xi)$, so $f(x - ct)$ also solves the wave equation. The sign of c drops out in the second derivative, so $f(x + ct)$ is a solution as well. For zero initial velocity the solution is $u(x, t) = \frac{1}{2}[f(x - ct) + f(x + ct)]$: the initial shape splits into two halves that travel in opposite directions. At $t = 0$ the two halves add up to $f(x)$, and their time derivatives, $\mp\frac{1}{2}cf'(x)$, cancel.

Diffusion. For the Gaussian initial condition $f(x) = e^{-x^2/w^2}$ the solution remains a Gaussian that widens and flattens,

$$u(x, t) = \frac{w}{\sqrt{w^2 + 4Dt}} \exp\left(-\frac{x^2}{w^2 + 4Dt}\right).$$

The squared width grows linearly in time, $w^2 + 4Dt$, and the amplitude decreases such that the integral of u, the total amount, stays constant.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-10, 10, 801)
w = 1.0                          # width of the initial Gaussian
f = lambda x: np.exp(-(x / w) ** 2)
D, c = 1.0, 1.0

def diffusion(x, t):
    return w / np.sqrt(w**2 + 4 * D * t) * np.exp(-x**2 / (w**2 + 4 * D * t))
```

```
solutions = {
    "Diffusion": diffusion,
    "Advection": lambda x, t: f(x - c * t),
    "Wave": lambda x, t: 0.5 * (f(x - c * t) + f(x + c * t)),
}

fig, axes = plt.subplots(1, 3, figsize=(9, 2.8), sharey=True)
for ax, (name, u) in zip(axes, solutions.items()):
    ax.plot(x, f(x), "k", label="$t = 0$")
    for t in [2, 4, 6]:
        ax.plot(x, u(x, t), label=f"$t = {t}$")
    ax.set_title(name)
    ax.set_xlabel("$x$")
axes[0].set_ylabel("$u$")
axes[0].legend(fontsize=8)
plt.tight_layout()
plt.show()
```

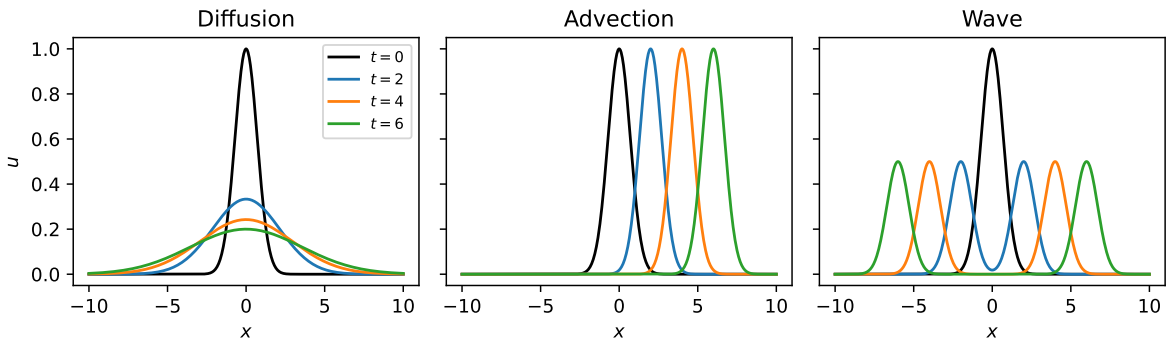


Figure 2.4: Exact solutions of the three prototype equations for the same Gaussian initial condition (black) at three later times.

Figure 2.4 contains the most important physical distinction of this lecture. Diffusion smooths and has no direction. Advection and waves transport information with a finite speed and preserve the shape. A good numerical method must reproduce this behaviour: it must not move a diffusing profile, and it must not smear or distort a travelling one. The second requirement turns out to be the harder one (Chapter 13).

A fourth prototype, the elliptic one, has no time derivative. In steady state, $\partial u/\partial t = 0$, the diffusion equation with a source becomes the **Poisson equation**

$$-\nabla^2 u = f, \tag{2.12}$$

and for $f = 0$ the **Laplace equation**. It describes equilibria: steady temperature and groundwater fields, the gravitational potential, the pressure in an incompressible fluid.

Figure 2.4 and the Poisson equation show why the classification matters: the three types behave differently.

	Elliptic	Parabolic	Hyperbolic
Describes	equilibrium	smoothing, relaxation to equilibrium	propagation
A disturbance is felt	everywhere at once	everywhere at once, but damped	only where the wave has arrived; finite speed
Solution is	smoother than the data	smoother than the initial condition	as rough as the initial condition
Needs	boundary conditions on the whole boundary	initial condition and boundary conditions	initial conditions; boundary conditions only where information enters
Numerical task	solve one large linear system	time stepping, often implicit	time stepping with attention to the propagation speed

2.5 The equations of this lecture

The equations of real models are combinations of the prototypes. We list the ones that appear in later units, so that you recognise the building blocks.

Shallow water equations. A layer of fluid of depth h and velocity u over a flat bottom, in one dimension and without rotation:

$$\begin{aligned} \frac{\partial h}{\partial t} + \frac{\partial (hu)}{\partial x} &= 0, \\ \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + g\frac{\partial h}{\partial x} &= 0. \end{aligned} \tag{2.13}$$

The first equation is conservation of mass in flux form, the second is conservation of momentum in advective form. The system is nonlinear.

To see what it describes we **linearize** it about a state of rest with depth H. We write $h = H + \eta$ and assume that the surface elevation η and the velocity u are small, so that products of small quantities can be neglected: $hu \approx Hu$ and $u\,\partial u/\partial x \approx 0$. What remains is

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0, \qquad \frac{\partial u}{\partial t} + g\frac{\partial \eta}{\partial x} = 0. \tag{2.14}$$

These are two first-order equations for two unknowns. We combine them into one second-order equation for η alone by eliminating u. The two equations are coupled through derivatives of u: the first contains $\partial u/\partial x$, the second $\partial u/\partial t$. Differentiating the first equation with respect to t and the second with respect to x produces the same mixed derivative of u in both. Since H and g are constants, they stay in front of the derivatives:

$$\frac{\partial^2 \eta}{\partial t^2} + H\frac{\partial^2 u}{\partial t\,\partial x} = 0, \qquad \frac{\partial^2 u}{\partial x\,\partial t} + g\frac{\partial^2 \eta}{\partial x^2} = 0.$$

For smooth functions the order of differentiation does not matter (Schwarz's theorem), so $\partial^2 u/\partial t\,\partial x = \partial^2 u/\partial x\,\partial t$. The second equation gives this mixed derivative in terms of η:

$$\frac{\partial^2 u}{\partial x\,\partial t} = -g\frac{\partial^2 \eta}{\partial x^2}.$$

Inserting it into the first equation,

$$\frac{\partial^2 \eta}{\partial t^2} + H\left(-g\frac{\partial^2 \eta}{\partial x^2}\right) = 0,$$

and moving the second term to the right-hand side gives

$$\frac{\partial^2 \eta}{\partial t^2} = gH\frac{\partial^2 \eta}{\partial x^2},$$

the wave equation Equation 2.11 with $c = \sqrt{gH}$. The shallow water equations describe **gravity waves**; they are a hyperbolic system. Note how the signs work out: the minus sign from solving the second equation for the mixed derivative and the one from moving the term to the right-hand side combine to $+gH$. Since $gH > 0$ the result is a wave equation; with $-gH$ it would be elliptic, and there would be no waves. Eliminating η instead, by differentiating the first equation with respect to x and the second with respect to t, gives the same wave equation for u. For an ocean of depth 4 km, $c = \sqrt{9.81 \cdot 4000}$ m/s ≈ 200 m/s, the speed of a tsunami. With rotation added the system contains much of the dynamics of the atmosphere and the ocean, which makes it the standard test bed for the numerical methods of weather and climate models (Chapter 15).

What makes a system of first-order equations hyperbolic? A first-order system

$$\frac{\partial \vec{q}}{\partial t} + B\frac{\partial \vec{q}}{\partial x} = 0$$

is hyperbolic if the matrix B has real eigenvalues and a full set of eigenvectors. It can then be decomposed into advection equations, one for each eigenvalue, and the eigenvalues are the speeds.

For the linearized shallow water equations Equation 2.14, $\vec{q} = (\eta, u)^T$ and

$$B = \begin{pmatrix} 0 & H \\ g & 0 \end{pmatrix}, \qquad \det(B - \lambda I) = \lambda^2 - gH = 0 \quad \Longrightarrow \quad \lambda = \pm\sqrt{gH}.$$

The two eigenvalues are the speeds of the gravity waves travelling to the right and to the left. For the full equations Equation 2.13 the same calculation gives $u \pm \sqrt{gh}$: the waves are in addition carried by the flow.

Primitive equations. The equations of atmospheric and ocean models are the conservation laws for momentum, mass and energy of a stratified fluid on the rotating sphere, with the hydrostatic approximation in the vertical. Each is an advection equation for one prognostic variable with source terms: pressure gradient and Coriolis force for momentum, heating for temperature, phase changes for humidity. Their dynamics are dominated by advection and waves, that is, by hyperbolic behaviour; diffusion terms represent the mixing by turbulence.

Groundwater flow. Conservation of mass with Darcy's law gives a diffusion equation for the hydraulic head h,

$$S_s \frac{\partial h}{\partial t} = \nabla \cdot (K \nabla h) + q, \tag{2.15}$$

with specific storage S_s, hydraulic conductivity K and sources q. It is parabolic, and in steady state elliptic. K varies in space over many orders of magnitude, which is the main numerical difficulty (Chapter 9).

Richards equation. In the unsaturated soil, water content θ and hydraulic conductivity K depend on the pressure head ψ:

$$\frac{\partial \theta(\psi)}{\partial t} = \nabla \cdot [K(\psi) \nabla(\psi + z)]\,. \tag{2.16}$$

The equation is a parabolic diffusion equation whose coefficients depend strongly on the solution. It is the prototype of a nonlinear problem in this lecture (Chapter 17).

Seismic waves. Conservation of momentum with Hooke's law gives, for a shear wave propagating in the x direction with particle velocity v and shear stress σ,

$$\rho \frac{\partial v}{\partial t} = \frac{\partial \sigma}{\partial x}, \qquad \frac{\partial \sigma}{\partial t} = \mu \frac{\partial v}{\partial x}, \tag{2.17}$$

with density ρ and shear modulus μ. For constant ρ and μ we differentiate the first equation with respect to t and insert the second:

$$\rho \frac{\partial^2 v}{\partial t^2} = \frac{\partial}{\partial x} \frac{\partial \sigma}{\partial t} = \mu \frac{\partial^2 v}{\partial x^2} \quad \Longrightarrow \quad \frac{\partial^2 v}{\partial t^2} = \frac{\mu}{\rho} \frac{\partial^2 v}{\partial x^2}.$$

This is the wave equation Equation 2.11 with the shear wave speed $c = \sqrt{\mu/\rho}$, and Equation 2.17 is a hyperbolic system. The structure is the same as in Equation 2.14: two first-order equations for two variables that drive each other.

Real models mix the types. The advection–diffusion equation Equation 2.7 is formally parabolic, but behaves like a hyperbolic equation when advection dominates. The same holds for the Navier–Stokes equations at high Reynolds number. Numerical models therefore often treat the terms of one equation with different methods (Chapter 18).

2.6 Initial and boundary conditions

A differential equation has many solutions. Initial and boundary conditions select one.

Initial conditions prescribe the state at $t = 0$. Their number equals the order of the time derivative: one for the diffusion equation, $u(x, 0) = u_0(x)$, and two for the wave equation, $u(x, 0)$ and $u_t(x, 0)$.

Boundary conditions prescribe the behaviour on the boundary $\partial\Omega$ of the domain Ω. For second-order equations three types are used, where $\partial u/\partial n$ is the derivative in the direction of the outward normal.

Type	Condition on $\partial\Omega$	Physical meaning	Example
Dirichlet	$u = g$	the value is prescribed	fixed temperature; water level of a lake bounding an aquifer
Neumann	$\partial u/\partial n = g$	the flux is prescribed	insulated wall or impermeable layer ($g = 0$); prescribed heat flux
Robin	$\partial u/\partial n + \alpha u = g$	the flux depends on the value	heat exchange with the surrounding air

A condition with $g = 0$ is called homogeneous. On each part of the boundary exactly one condition is prescribed.

A fourth kind is common in atmospheric modelling: **periodic** boundary conditions, $u(0,t) = u(L,t)$. On a latitude circle they are exact. Limited-area models face the opposite situation: their lateral boundaries are artificial, and waves must be able to leave the domain without reflection. Such open boundaries are a subject of their own (Chapter 16).

2.7 Well-posed problems

A problem is **well-posed** in the sense of Hadamard[1] if

1. a solution exists,
2. the solution is unique,
3. the solution depends continuously on the data, that is on initial conditions, boundary conditions and parameters.

If one of the conditions is violated, a numerical solution is meaningless or impossible. Three examples show that this is of practical relevance.

Too few conditions. The Poisson equation with Neumann conditions on the whole boundary prescribes only fluxes. If u is a solution, so is u plus any constant, which violates condition 2. Moreover a solution exists only if the prescribed boundary fluxes balance the sources, which concerns condition 1. In the discrete problem this shows up as a singular matrix.

The wrong direction of time. The diffusion equation smooths. Solving it backward in time, in order to find the initial state that led to an observed state, must therefore roughen: small-scale errors in the observed state are amplified without bound, which violates condition 3. Problems of this kind, where causes are inferred from effects, are the **inverse problems** of geophysics and the **data assimilation** problem of meteorology. They need additional information to become well-posed (Chapter 21).

The wrong kind of conditions. Hadamard's own example shows that the conditions must match the type of the equation. Take the Laplace equation in the upper half plane $y > 0$ and prescribe, as for the wave equation, the value and the normal derivative on the line $y = 0$:

$$u_{xx} + u_{yy} = 0, \qquad u(x,0) = 0, \qquad u_y(x,0) = \frac{1}{n}\sin(nx).$$

We look for a solution of the form $u = \sin(nx)\,Y(y)$. Inserting it, $u_{xx} = -n^2 \sin(nx)\,Y$ and $u_{yy} = \sin(nx)\,Y''$, so the PDE becomes the ODE $Y'' = n^2 Y$ with the general solution $Y = A\cosh(ny) + B\sinh(ny)$. The first condition gives $Y(0) = A = 0$. The second gives $Y'(0) = nB = 1/n$, so $B = 1/n^2$ and

[1] Jacques Hadamard (1865–1963), French mathematician, who formulated these conditions in 1902 and also proved the prime number theorem.

$$u(x,y) = \frac{1}{n^2}\sin(nx)\sinh(ny).$$

Now let $n \to \infty$. The data tend to zero, $|u_y(x,0)| \le 1/n$, and for zero data the solution is $u = 0$. But at any fixed height $y > 0$ the solution grows without bound, since $\sinh(ny) \approx \frac{1}{2}e^{ny}$: at $y = 1$ and $n = 20$, data of size 0.05 produce a solution of size $\frac{1}{400}\sinh(20) \approx 6 \cdot 10^5$. Condition 3 is violated.

The same data are harmless for the wave equation $u_{tt} = u_{xx}$ with t in place of y. There the ODE is $Y'' = -n^2 Y$, the hyperbolic functions become trigonometric ones, and the solution $u = \frac{1}{n^2}\sin(nx)\sin(nt)$ stays below $1/n^2$. The only difference is the sign in front of n^2, and that sign is the type of the equation. An elliptic equation needs one condition on the whole boundary, not two conditions on part of it (see the table in Section 2.4). In geophysics this example appears as the **downward continuation** of gravity and magnetic fields: computing the field at depth from measurements at the surface amplifies the short wavelengths exponentially.

Condition 3 for the mathematical problem has a counterpart for the numerical method: stability (Chapter 6). And well-posed does not mean predictable for all times. The equations of the atmosphere are well-posed, yet small errors in the initial state grow so fast that the forecast loses its value after about two weeks (Chapter 8).

2.8 Summary

- Gradient, divergence, curl and Laplacian are the building blocks of all equations in this lecture. The divergence is the net outflow per volume.
- A conservation law states that the amount in a volume changes only by the flux through its surface and by sources: $\partial u/\partial t + \nabla \cdot \vec{F} = s$. Advective and diffusive fluxes close the equation.
- Classify before you compute: ODE or PDE, order, linear or nonlinear, elliptic, parabolic or hyperbolic. The type decides the numerical method.
- The prototypes show what the types mean: diffusion smooths; advection and waves propagate with finite speed; the Poisson equation describes equilibria.
- The equations of atmospheric, hydrological and seismic models are combinations of the prototypes, and each can be classified the same way.
- A well-posed problem has the right number of initial and boundary conditions, and its solution depends continuously on the data.

2.9 Further reading

Boyce and DiPrima (2009), chapter 1, for the classification of ODEs; Wick (2020) for the classification of PDEs and well-posedness; Durran (2010), chapter 1, for the equations of

geophysical fluid dynamics.

3 The numerical toolbox I: derivatives, integrals, roots

Learning goals

After this unit you can

- derive finite difference formulas and their truncation error from the Taylor series,
- measure the order of convergence of a numerical method,
- explain why a finite difference cannot be made arbitrarily accurate, and estimate the best step,
- interpolate and integrate numerically and derive the error of the basic rules,
- find roots with bisection and Newton's method and explain their rates of convergence.

Numerical models are built from a small number of tools. This unit introduces the tools that come from calculus: approximating derivatives, interpolating, integrating and finding roots. All of them rest on the Taylor series. The next unit adds the tools from linear algebra and Fourier analysis.

About the code in this script

From this unit on, each computation is shown in two tabs.

- **Step by step** is written the way you would write it yourself after the first exercise sheets: explicit loops, one operation per line, a comment on each line. Read this version first. New Python is explained below the code where it first appears.
- **Compact** does the same in the style that experienced Python programmers use: whole NumPy arrays instead of loops, and functions from libraries such as SciPy. It is shorter and faster but hides the individual steps.

Both versions give the same results. In the PDF the two versions follow each other.

3.1 Taylor series and finite differences

3.1.1 The Taylor series

The Taylor series expresses the value of a smooth function at a neighbouring point by its value and its derivatives at x:

$$u(x+h) = u(x) + h\,u'(x) + \frac{h^2}{2}u''(x) + \frac{h^3}{6}u'''(x) + \frac{h^4}{24}u''''(x) + \dots \tag{3.1}$$

The general term is $\frac{h^k}{k!}u^{(k)}(x)$, with $2! = 2$, $3! = 6$, $4! = 24$. If the series is stopped after the term with h^n, the rest can be written as a single term,

$$u(x+h) = \sum_{k=0}^{n} \frac{h^k}{k!}u^{(k)}(x) + \frac{h^{n+1}}{(n+1)!}u^{(n+1)}(\xi)$$

with some ξ (the Greek letter xi) between x and $x+h$ (Lagrange form of the remainder). This is what makes the dots precise: the neglected part behaves like the first neglected term, provided $u^{(n+1)}$ is bounded. "Smooth" means that the derivatives that appear exist and are bounded.

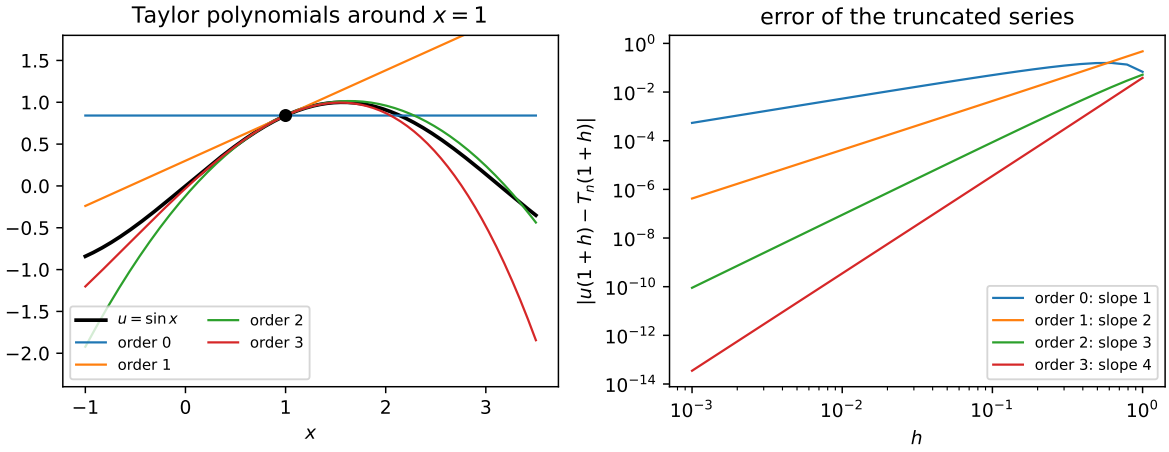


Figure 3.1: Left: Taylor polynomials of $\sin x$ around $x = 1$, stopped after the term with h^n. Right: their error at $x = 1 + h$; it falls with slope $n + 1$.

Figure 3.1 shows the series at work for $u = \sin x$ around $x = 1$. Each additional term makes the polynomial follow the function over a longer distance from x. The right panel shows the error of the polynomial stopped after the term with h^n, evaluated at $x + h$. On logarithmic

axes it is a straight line with slope $n+1$, as the remainder term $\frac{h^{n+1}}{(n+1)!}u^{(n+1)}(\xi)$ predicts. Every truncation error in this unit arises in this way.

Replacing h by $-h$ changes the sign of the odd terms, because $(-h)^k = -h^k$ for odd k:

$$u(x-h) = u(x) - h\,u'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u'''(x) + \frac{h^4}{24}u''''(x) - \dots \tag{3.2}$$

Every finite difference formula is a combination of such series, chosen so that unwanted terms cancel.

Notation. We write $e = O(h^p)$, "e is of order h^p", if $|e| \le Ch^p$ for all sufficiently small h, with a constant C that does not depend on h. A method whose error is $O(h^p)$ is **of order** p. Halving h then divides the error by about 2^p.

3.1.2 Forward and backward difference

We solve Equation 3.1 for $u'(x)$. Subtracting $u(x)$ on both sides gives

$$u(x+h) - u(x) = h\,u'(x) + \frac{h^2}{2}u''(x) + \frac{h^3}{6}u'''(x) + \dots,$$

and dividing by h

$$\frac{u(x+h)-u(x)}{h} = u'(x) + \frac{h}{2}u''(x) + \frac{h^2}{6}u'''(x) + \dots.$$

Bringing everything except $u'(x)$ to the other side,

$$u'(x) = \underbrace{\frac{u(x+h)-u(x)}{h}}_{\text{forward difference}} \underbrace{- \frac{h}{2}u''(x) - \frac{h^2}{6}u'''(x) - \dots}_{\text{truncation error}} \tag{3.3}$$

The terms that are dropped when only the difference quotient is kept form the **truncation error**. Its leading term is $-\frac{h}{2}u''(x)$, proportional to h. The forward difference is of first order: its error is $O(h)$.

The same steps applied to Equation 3.2 give

$$u(x) - u(x-h) = h\,u'(x) - \frac{h^2}{2}u''(x) + \frac{h^3}{6}u'''(x) - \dots,$$

and after division by h the **backward difference**

$$u'(x) = \frac{u(x) - u(x-h)}{h} + \frac{h}{2}u''(x) - \frac{h^2}{6}u'''(x) + \ldots, \tag{3.4}$$

which is also of first order, with the opposite sign of the leading error.

3.1.3 Central difference

The errors of the two one-sided formulas have opposite signs, which suggests combining them. We subtract Equation 3.2 from Equation 3.1 term by term:

Term	in $u(x+h)$	in $u(x-h)$	difference
u	u	u	0
u'	$h\,u'$	$-h\,u'$	$2h\,u'$
u''	$\frac{h^2}{2}u''$	$\frac{h^2}{2}u''$	0
u'''	$\frac{h^3}{6}u'''$	$-\frac{h^3}{6}u'''$	$\frac{h^3}{3}u'''$
u''''	$\frac{h^4}{24}u''''$	$\frac{h^4}{24}u''''$	0

All even terms cancel:

$$u(x+h) - u(x-h) = 2h\,u'(x) + \frac{h^3}{3}u'''(x) + \ldots$$

Dividing by $2h$ gives $\frac{u(x+h)-u(x-h)}{2h} = u'(x) + \frac{h^2}{6}u'''(x) + \ldots$, and solving for $u'(x)$ the **central difference**

$$u'(x) = \frac{u(x+h) - u(x-h)}{2h} - \frac{h^2}{6}u'''(x) - \ldots \tag{3.5}$$

Its error is $O(h^2)$: second order. Halving h halves the error of the one-sided differences but quarters the error of the central difference, at the same cost of two function values.

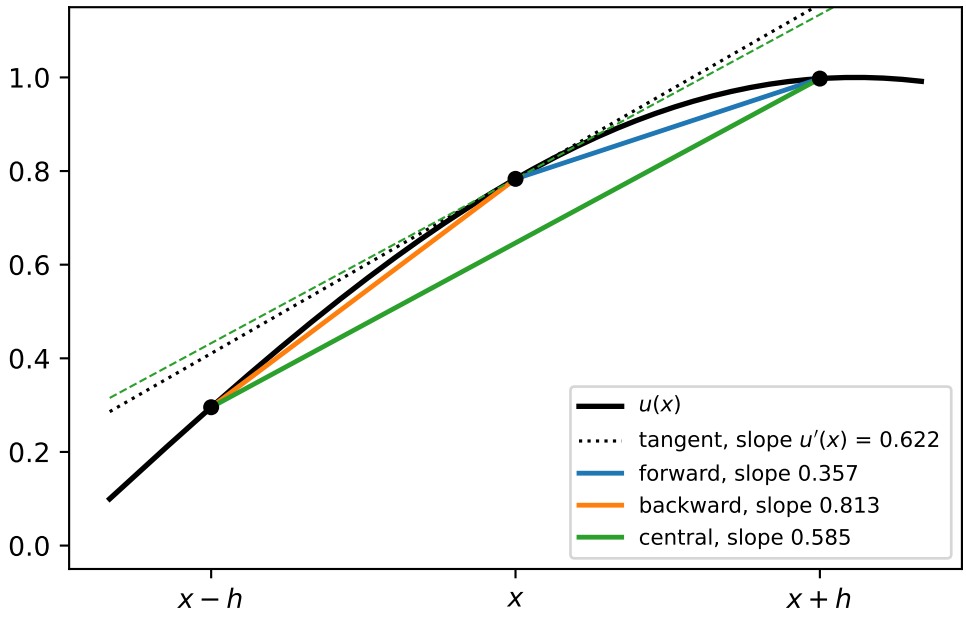


Figure 3.2: Difference quotients are slopes of chords. Forward and backward chord lie on one side of x; the central chord (moved to x: thin dashed line) is nearly parallel to the tangent. Here $u = \sin x$, $x = 0.9$, $h = 0.6$.

Geometrically, each difference quotient is the slope of a chord (Figure 3.2). The forward chord connects x and $x + h$, the backward chord $x - h$ and x. Where the function curves, one of them is too flat and the other too steep, by about $\frac{h}{2}|u''|$ each, with opposite signs: in the figure $u'' = -\sin 0.9 = -0.78$ and $\frac{h}{2}|u''| = 0.23$, while the slopes differ from the tangent slope 0.622 by -0.27 and $+0.19$ (the difference from 0.23 comes from the higher terms). The central chord spans the symmetric interval $[x - h, x + h]$, and the curvature tilts it in neither direction. For a parabola, where $u''' = 0$, the error term $-\frac{h^2}{6}u'''$ vanishes and the central chord is exactly parallel to the tangent.

3.1.4 Second derivative

Adding the two series instead cancels the odd terms (last column of the table, with the signs of $u(x - h)$ reversed):

$$u(x + h) + u(x - h) = 2u(x) + h^2 u''(x) + \frac{h^4}{12}u''''(x) + \dots,$$

where $\frac{h^4}{12} = 2 \cdot \frac{h^4}{24}$. Subtracting $2u(x)$ gives $u(x+h) - 2u(x) + u(x-h) = h^2 u''(x) + \frac{h^4}{12} u''''(x) + \ldots$, and dividing by h^2 and solving for $u''(x)$ gives the standard approximation of the second derivative,

$$u''(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} - \frac{h^2}{12} u''''(x) - \ldots, \tag{3.6}$$

which is also of second order. This formula is the core of every finite difference model of diffusion.

There is a second way to obtain it, which becomes important when the diffusion coefficient varies in space. The second derivative is the derivative of the first derivative. We approximate the first derivative half-way between the grid points,

$$u'\Big(x + \tfrac{h}{2}\Big) \approx \frac{u(x+h) - u(x)}{h}, \qquad u'\Big(x - \tfrac{h}{2}\Big) \approx \frac{u(x) - u(x-h)}{h}.$$

Seen from the point $x + \frac{h}{2}$, the first quotient uses the two neighbours at distance $\frac{h}{2}$ on either side: it is a central difference with step $h/2$ and therefore of second order. The same holds for the second quotient at $x - \frac{h}{2}$. We then take the central difference of these two values, again with step $h/2$:

$$u''(x) \approx \frac{u'\Big(x + \frac{h}{2}\Big) - u'\Big(x - \frac{h}{2}\Big)}{h} = \frac{1}{h}\left[\frac{u(x+h) - u(x)}{h} - \frac{u(x) - u(x-h)}{h}\right] = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}.$$

In this form the formula is a difference of two fluxes, as in the derivation of the divergence in Chapter 2.

3.1.5 Measuring the order of convergence

If the error behaves as $e(h) \approx Ch^p$, then taking the logarithm gives

$$\log e = \log C + p \log h.$$

In a plot with logarithmic axes the error is a straight line with slope p. To compute p from two step sizes h_1 and h_2 we divide the two errors. The unknown constant C drops out,

$$\frac{e(h_1)}{e(h_2)} = \frac{Ch_1^p}{Ch_2^p} = \left(\frac{h_1}{h_2}\right)^p,$$

and taking the logarithm of both sides, $\log\left(e(h_1)/e(h_2)\right) = p\log(h_1/h_2)$, gives the **observed order**

$$p = \frac{\log\left(e(h_1)/e(h_2)\right)}{\log\left(h_1/h_2\right)}. \tag{3.7}$$

If the step is halved, $h_1/h_2 = 2$ and $p = \log_2\left(e(h_1)/e(h_2)\right)$: an error that falls by a factor of 4 means second order.

Comparing the observed order with the theoretical one is the most important test of a numerical code. A bug almost always reduces the order. We test the forward and central difference with $u = \sin$, whose derivative cos is known, at $x = 1$.

Step by step

```
import numpy as np                    # NumPy: np.sin, np.cos, np.log, ...

x = 1.0                               # point where we approximate u'(x)
exact = np.cos(x)                     # exact derivative of sin at x

h_list = [0.1, 0.05, 0.025, 0.0125] # step sizes, each half the previous one
error_forward = []                    # empty lists, filled in the loop
error_central = []

for h in h_list:                      # h takes the values 0.1, 0.05, ... in turn
    forward = (np.sin(x + h) - np.sin(x)) / h
    central = (np.sin(x + h) - np.sin(x - h)) / (2 * h)
    error_forward.append(abs(forward - exact))   # add the error to the list
    error_central.append(abs(central - exact))

print("     h    error forward  order  error central  order")
for i in range(1, len(h_list)):       # i = 1, 2, 3; compare step i with step i-1
    log_ratio_h = np.log(h_list[i - 1] / h_list[i])
    p_forward = np.log(error_forward[i - 1] / error_forward[i]) / log_ratio_h
    p_central = np.log(error_central[i - 1] / error_central[i]) / log_ratio_h
    print(f"{h_list[i]:7.4f} {error_forward[i]:12.2e} {p_forward:6.2f}"
          f" {error_central[i]:14.2e} {p_central:6.2f}")
```

```
     h    error forward  order  error central  order
 0.0500      2.13e-02   1.01       2.25e-04   2.00
 0.0250      1.06e-02   1.01       5.63e-05   2.00
 0.0125      5.27e-03   1.00       1.41e-05   2.00
```

Compact

```
import numpy as np

x = 1.0
h = np.array([0.1, 0.05, 0.025, 0.0125])      # all step sizes at once
error_forward = abs((np.sin(x + h) - np.sin(x)) / h - np.cos(x))
error_central = abs((np.sin(x + h) - np.sin(x - h)) / (2 * h) - np.cos(x))

# e[:-1] is the array without its last, e[1:] without its first element
def observed_order(e, h):
    return np.log(e[:-1] / e[1:]) / np.log(h[:-1] / h[1:])

print("forward, observed order:", observed_order(error_forward, h).round(3))
print("central, observed order:", observed_order(error_central, h).round(3))
```

```
forward, observed order: [1.014 1.007 1.004]
central, observed order: [1.999 2.    2.   ]
```

The observed orders are 1 and 2, as derived. Moreover the forward error is close to $\frac{h}{2}|u''(1)| = \frac{h}{2}\sin 1 \approx 0.42\,h$, the leading term of Equation 3.3.

New Python in the step-by-step version.

- `[0.1, 0.05, ...]` is a **list**, `[]` an empty list. `error_forward.append(value)` adds a value at the end of the list.
- `for h in h_list:` executes the indented lines once for each element of the list, with `h` equal to that element. It replaces the counter of a `while` loop.
- `range(1, len(h_list))` produces the whole numbers $1, 2, \ldots$ up to `len(h_list) - 1`; `len` is the number of elements. `h_list[i]` is element number `i`, counted from **0**: `h_list[0]` is 0.1.
- In `f"{h:7.4f}"` the part after the colon sets the format: 7 characters wide with 4 digits after the decimal point. `e` instead of `f` gives the exponential notation, `1.23e-04`.

3.1.6 Why the step cannot be made arbitrarily small

According to Equation 3.3 the error vanishes for $h \to 0$. On a computer it does not, and we can estimate why.

The computer does not hold $u(x)$ and $u(x+h)$ but rounded values $u(x)(1+\delta_1)$ and $u(x+h)(1+\delta_2)$ with $|\delta_{1,2}| \le \varepsilon$, the machine epsilon (Chapter 1). The computed numerator is

$$u(x+h)(1+\delta_2) - u(x)(1+\delta_1) = \big[u(x+h) - u(x)\big] + u(x+h)\,\delta_2 - u(x)\,\delta_1,$$

so it differs from the exact numerator by

$$|u(x+h)\,\delta_2 - u(x)\,\delta_1| \le \varepsilon\,\big(|u(x+h)| + |u(x)|\big) \approx 2\varepsilon|u|,$$

where we used $u(x+h) \approx u(x)$ for small h and wrote $|u|$ for $|u(x)|$. The division by h is itself accurate to ε, but it turns this absolute error of the numerator into an error of $2\varepsilon|u|/h$ in the difference quotient: the smaller h, the larger the error. Together with the leading truncation error from Equation 3.3 the total error of the forward difference is bounded by

$$e(h) = \underbrace{\frac{h}{2}|u''|}_{\text{truncation}} + \underbrace{\frac{2\varepsilon|u|}{h}}_{\text{round-off}}. \tag{3.8}$$

The first term decreases with h, the second increases. The minimum is where the derivative with respect to h vanishes:

$$\frac{\mathrm{d}e}{\mathrm{d}h} = \frac{|u''|}{2} - \frac{2\varepsilon|u|}{h^2} = 0 \quad\Longrightarrow\quad h^2 = \frac{4\varepsilon|u|}{|u''|} \quad\Longrightarrow\quad h_{\text{opt}} = 2\sqrt{\frac{\varepsilon|u|}{|u''|}}.$$

Inserting h_{opt} into Equation 3.8, both terms become equal,

$$\frac{h_{\text{opt}}}{2}|u''| = \sqrt{\frac{\varepsilon|u|}{|u''|}}\,|u''| = \sqrt{\varepsilon|u||u''|}, \qquad \frac{2\varepsilon|u|}{h_{\text{opt}}} = \varepsilon|u|\sqrt{\frac{|u''|}{\varepsilon|u|}} = \sqrt{\varepsilon|u||u''|},$$

and the smallest error is their sum,

$$e(h_{\text{opt}}) = 2\sqrt{\varepsilon\,|u|\,|u''|}.$$

If u and u'' are of order one, the best step is of the order $\sqrt{\varepsilon} \approx 10^{-8}$, and the smallest error that can be reached is also about 10^{-8}: half of the 16 digits are lost.

For the central difference the numerator has the same round-off error $2\varepsilon|u|$, but it is divided by $2h$, which gives $\varepsilon|u|/h$. With the truncation error $\frac{h^2}{6}|u'''|$ from Equation 3.5,

$$e(h) = \frac{h^2}{6}|u'''| + \frac{\varepsilon|u|}{h}, \qquad \frac{\mathrm{d}e}{\mathrm{d}h} = \frac{h}{3}|u'''| - \frac{\varepsilon|u|}{h^2} = 0 \quad\Longrightarrow\quad h_{\text{opt}} = \left(\frac{3\varepsilon|u|}{|u'''|}\right)^{1/3}.$$

At h_{opt} we have $\varepsilon|u| = \frac{1}{3}h_{\text{opt}}^3|u'''|$, so the round-off term is $\frac{1}{3}h_{\text{opt}}^2|u'''|$, twice the truncation term, and

$$e(h_{\text{opt}}) = \frac{h_{\text{opt}}^2}{6}|u'''| + \frac{h_{\text{opt}}^2}{3}|u'''| = \frac{h_{\text{opt}}^2}{2}|u'''|.$$

For u and u''' of order one, $h_{\text{opt}} \sim \varepsilon^{1/3} \approx 10^{-5}$ and $e(h_{\text{opt}}) \sim \varepsilon^{2/3} \approx 10^{-11}$.

Step by step

```
import matplotlib.pyplot as plt     # the plotting library

x = 1.0
exact = np.cos(x)
h_values = []                       # step sizes
error_forward = []                  # errors of the two formulas
error_central = []
slope_1 = []                        # reference lines h/2 and h^2/6
slope_2 = []

for k in range(60):                 # k = 0, 1, ..., 59
    h = 10.0 ** (-15 + 0.25 * k)    # h = 10^-15, 10^-14.75, ..., 10^-0.25
    forward = (np.sin(x + h) - np.sin(x)) / h
    central = (np.sin(x + h) - np.sin(x - h)) / (2 * h)
    h_values.append(h)
    error_forward.append(abs(forward - exact))
    error_central.append(abs(central - exact))
    slope_1.append(h / 2)
    slope_2.append(h**2 / 6)

fig, ax = plt.subplots(figsize=(6, 3.8))     # a figure with one set of axes
ax.loglog(h_values, error_forward, ".", label="forward difference")
ax.loglog(h_values, error_central, ".", label="central difference")
ax.loglog(h_values, slope_1, "k--", lw=0.8, label="slope 1 and 2")
ax.loglog(h_values, slope_2, "k--", lw=0.8)
ax.set_ylim(1e-13, 1)               # show errors between 1e-13 and 1
ax.set_xlabel("$h$")
ax.set_ylabel("error")
ax.legend()                         # the box with the labels
plt.show()
```

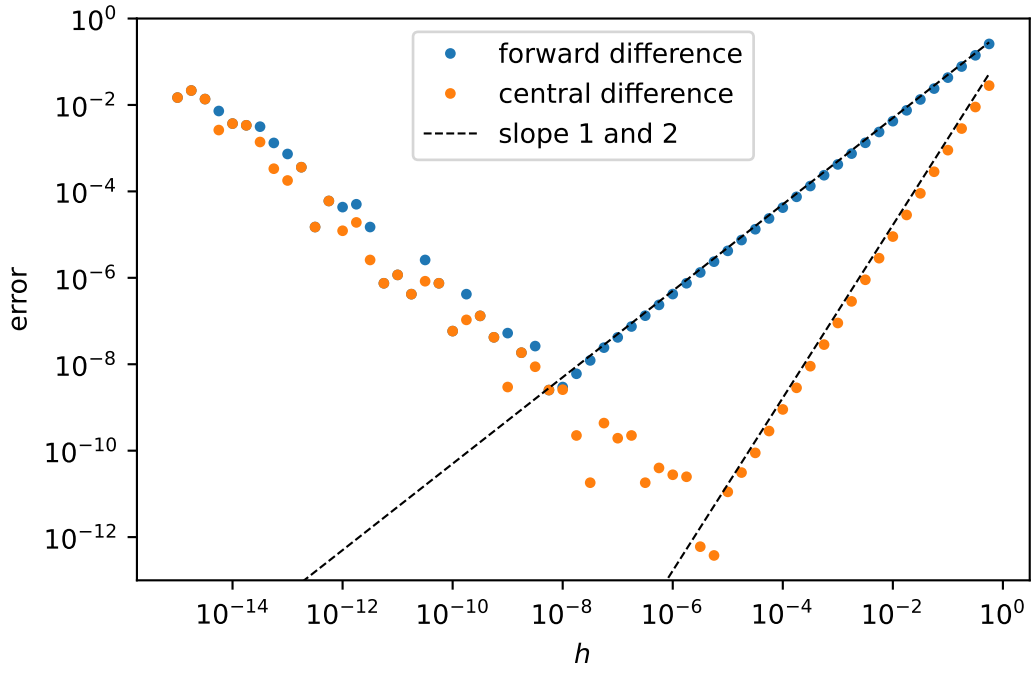


Figure 3.3: Error of the forward and central difference for the derivative of $\sin x$ at $x = 1$. The error decreases with the theoretical slope until round-off takes over.

Compact

```
import matplotlib.pyplot as plt

h = 10.0 ** np.arange(-15, 0, 0.25)          # all 60 step sizes as one array
forward = abs((np.sin(x + h) - np.sin(x)) / h - np.cos(x))
central = abs((np.sin(x + h) - np.sin(x - h)) / (2 * h) - np.cos(x))

fig, ax = plt.subplots(figsize=(6, 3.8))
ax.loglog(h, forward, ".", label="forward difference")
ax.loglog(h, central, ".", label="central difference")
ax.loglog(h, h / 2, "k--", lw=0.8, label="slope 1 and 2")
ax.loglog(h, h**2 / 6, "k--", lw=0.8)
ax.set(ylim=(1e-13, 1), xlabel="$h$", ylabel="error")
ax.legend()
plt.show()
```

The exponents are computed from the integer counter `k`, as in the time loop of Exercise 1, instead of adding 0.25 repeatedly. `ax.loglog` plots with logarithmic axes; `"."` draws dots, `"k--"` a black (k) dashed line, `lw` is the line width.

Figure 3.3 confirms the estimates: the error of the forward difference stops decreasing near $h = 10^{-8}$, at about 10^{-8}, that of the central difference near $h = 10^{-5}$, at about 10^{-11}. Single points can lie lower, where the rounding errors happen to cancel; the estimates are bounds for the typical error. Left of the optimum the error grows like $1/h$ and scatters, because the rounding errors δ_1, δ_2 vary irregularly from one h to the next. The figure has two lessons. First, a higher order gives a smaller error for the same step and is less affected by round-off. Second, in a convergence test the step must be neither so large that higher-order terms disturb the slope nor so small that round-off dominates.

3.1.7 Stencils

The points used by a difference formula form its **stencil**. Figure 3.4 collects the stencils of this unit, together with two that are not derived here. All of them follow from the same idea: combine Taylor series so that the unwanted terms cancel. For a given stencil and derivative, the weights are the solution of a small linear system (Fornberg 1988).

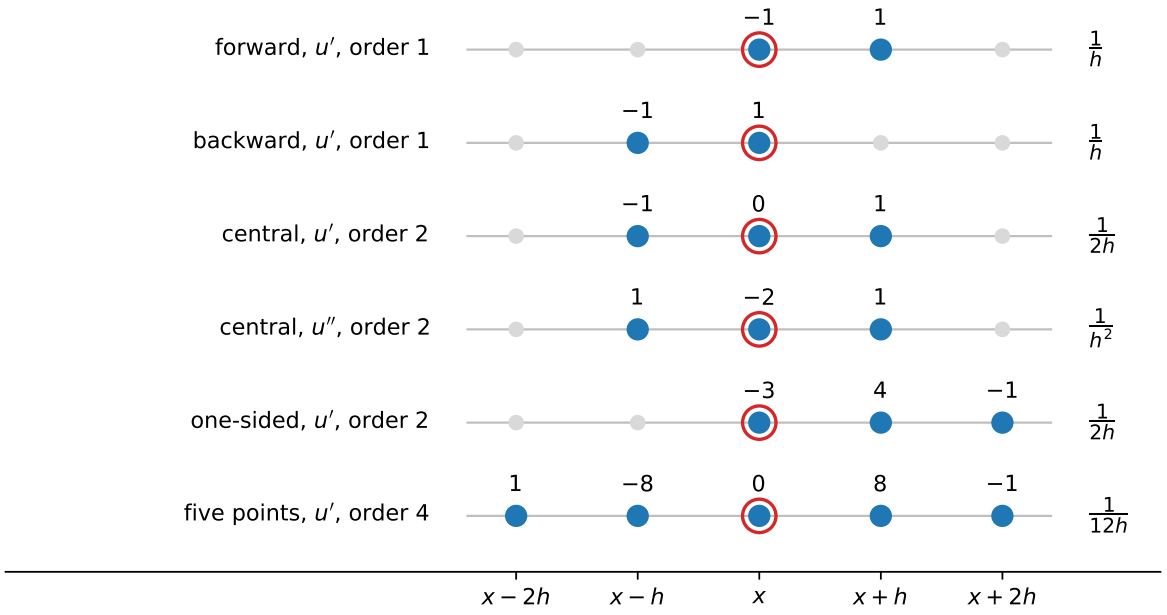


Figure 3.4: Stencils of common difference formulas. The weights are written above the points used (blue), the common factor on the right; the red circle marks the point x at which the derivative is approximated.

The **one-sided** formula is needed at a boundary. There the central difference would use a point outside the domain, where the model computes no value: below the ground there is no air

temperature. The derivative at the boundary is needed for fluxes through it, such as the heat flux $-k\,\partial T/\partial z$ into the ground, and for Neumann boundary conditions. The forward difference uses only points inside, but it is of first order and can reduce the accuracy of an otherwise second-order model. One more point, $x+2h$, gives second order,

$$u'(x) = \frac{-3u(x) + 4u(x+h) - u(x+2h)}{2h} + \frac{h^2}{3}u'''(x) + \ldots,$$

with an error constant twice that of the central difference. Problem 2 of exercise sheet 3 derives it. Models often use instead a **ghost point** outside the domain, whose value is set by the boundary condition (Chapter 9). The **five-point** formula cancels one more pair of terms than the central difference and is of fourth order; it returns in Chapter 4.

In the next unit the stencil is written down at every grid point. Each stencil then becomes one row of a matrix, and the $1, -2, 1$ stencil of the second derivative becomes the most important matrix of this lecture.

3.2 Interpolation

Interpolation constructs a function that passes through given data points (x_j, u_j). Models need it all the time: to obtain a value between grid points, to transfer fields between different grids, to find the value at the departure point of an air parcel (Chapter 14).

3.2.1 Linear interpolation

Between two neighbours x_j and $x_{j+1} = x_j + h$ the straight line through the two data points is

$$p(x) = u_j + \frac{u_{j+1} - u_j}{h}(x - x_j). \tag{3.9}$$

At $x = x_j$ it gives u_j, at $x = x_{j+1}$ it gives $u_j + (u_{j+1} - u_j) = u_{j+1}$. The special case at the midpoint, $x - x_j = h/2$, is the average $\frac{1}{2}(u_j + u_{j+1})$, which is used whenever a value is needed half-way between grid points.

How large is the error? For a polynomial p of degree n through $n+1$ points $x_0, \ldots, x_n$ the interpolation error is

$$u(x) - p(x) = \frac{u^{(n+1)}(\xi)}{(n+1)!}\prod_{j=0}^{n}(x - x_j) \tag{3.10}$$

for some ξ in the interval that contains x and the data points. The symbol $\prod$ (capital pi) denotes a product, in the same way as $\sum$ denotes a sum:

$$\prod_{j=0}^{n}(x - x_j) = (x - x_0)(x - x_1)\cdots(x - x_n),$$

with one factor for each data point; for $n = 1$ it is $(x - x_0)(x - x_1)$. The Greek letter ξ (xi) stands for a point whose existence the derivation guarantees but whose position it does not determine, as in the Lagrange remainder of the Taylor series. It depends on x. In estimates we therefore replace $|u^{(n+1)}(\xi)|$ by its largest value on the interval, $\max|u^{(n+1)}|$.

Derivation of the interpolation error

At a data point both sides are zero. Fix an x that is not a data point and write $w(t) = \prod_{j=0}^{n}(t - x_j)$, a polynomial of degree $n + 1$ with leading term t^{n+1}. The function

$$g(t) = u(t) - p(t) - \frac{u(x) - p(x)}{w(x)}\, w(t)$$

is zero at the $n + 1$ data points, because there $u = p$ and $w = 0$, and also at $t = x$, where the last term equals $u(x) - p(x)$. It has $n + 2$ zeros. Between two zeros of a smooth function its derivative has a zero (Rolle's theorem). Hence g' has at least $n+1$ zeros, g'' at least n, and $g^{(n+1)}$ at least one, at some point ξ. Now $p^{(n+1)} = 0$, because p has degree n, and $w^{(n+1)} = (n + 1)!$, because only the leading term t^{n+1} survives $n + 1$ differentiations. Therefore

$$0 = g^{(n+1)}(\xi) = u^{(n+1)}(\xi) - \frac{u(x) - p(x)}{w(x)}\,(n + 1)!,$$

which, solved for $u(x) - p(x)$, is Equation 3.10.

For the straight line, $n = 1$, the error is $\frac{1}{2}u''(\xi)\,(x - x_j)(x - x_{j+1})$. Writing $t = x - x_j$, the product is $t\,(t - h)$ for $0 \le t \le h$. Its derivative $2t - h$ vanishes at $t = h/2$, the midpoint, where the product equals $\frac{h}{2}\cdot\left(-\frac{h}{2}\right) = -\frac{h^2}{4}$. Hence

$$|u(x) - p(x)| \le \frac{1}{2}\cdot\frac{h^2}{4}\max|u''| = \frac{h^2}{8}\max|u''|.$$

Linear interpolation is of second order, and it is exact where the function has no curvature.

3.2.2 Polynomials of high degree and the Runge phenomenon

Polynomial interpolation passes a single polynomial of degree n through $n + 1$ points. It can be written down directly in the Lagrange form

$$p(x) = \sum_{j=0}^{n} u_j \, L_j(x), \qquad L_j(x) = \prod_{i \neq j} \frac{x - x_i}{x_j - x_i}.$$

Each L_j is a product of n linear factors, a polynomial of degree n. At $x = x_j$ every factor equals one, so $L_j(x_j) = 1$. At any other data point x_i the factor $(x - x_i)$ is zero, so $L_j(x_i) = 0$. In the sum for $p(x_i)$ therefore only the term $j = i$ survives, and $p(x_i) = u_i$. For $n = 1$ the Lagrange form is $p(x) = u_0 \frac{x - x_1}{x_0 - x_1} + u_1 \frac{x - x_0}{x_1 - x_0}$, which is Equation 3.9 written differently.

From Equation 3.10 one would expect the accuracy to improve with n, because of the factor $1/(n+1)!$. On equidistant points it does not. The product $\prod(x - x_j)$ is much larger near the ends of the interval than in the middle, and for many functions the derivatives $u^{(n+1)}$ grow faster than the factorial. The polynomial oscillates near the ends, and the oscillations grow with n. This is the **Runge phenomenon**.

There are two remedies. **Piecewise interpolation** uses polynomials of low degree between neighbouring points. **Cubic splines** are piecewise cubic polynomials that in addition make the first and second derivative continuous at the data points. Alternatively, the data points can be placed more densely near the ends, for example at the Chebyshev points $x_j = \cos(j\pi/n)$; this keeps the product $\prod(x - x_j)$ small everywhere. Models rarely have this freedom: the grid is given.

Step by step

```
from scipy.interpolate import CubicSpline   # cubic splines from SciPy

def runge(x):
    return 1 / (1 + 25 * x**2)

def lagrange(x_data, u_data, x):
    # value at x of the polynomial through the points (x_data, u_data)
    n_points = len(x_data)
    p = 0.0
    for j in range(n_points):
        L = 1.0                     # L_j(x), built up factor by factor
        for i in range(n_points):
            if i != j:              # != means "not equal"
                L = L * (x - x_data[i]) / (x_data[j] - x_data[i])
        p = p + u_data[j] * L
    return p

x_data = []                         # 11 equidistant data points in [-1, 1]
u_data = []
for j in range(11):
    x_data.append(-1 + 0.2 * j)
```

```
    u_data.append(runge(-1 + 0.2 * j))

spline = CubicSpline(x_data, u_data)        # the spline through the data

x_fine = []                             # 401 points for drawing the curves
u_exact = []
u_polynomial = []
u_spline = []
for k in range(401):
    x = -1 + 0.005 * k
    x_fine.append(x)
    u_exact.append(runge(x))
    u_polynomial.append(lagrange(x_data, u_data, x))
    u_spline.append(spline(x))

fig, ax = plt.subplots(figsize=(6, 3.5))
ax.plot(x_fine, u_exact, "k", label="function")
ax.plot(x_fine, u_polynomial, label="polynomial, degree 10")
ax.plot(x_fine, u_spline, "--", label="cubic spline")
ax.plot(x_data, u_data, "ko", ms=4)         # data points as black circles
ax.set_xlabel("$x$")
ax.legend()
plt.show()
```

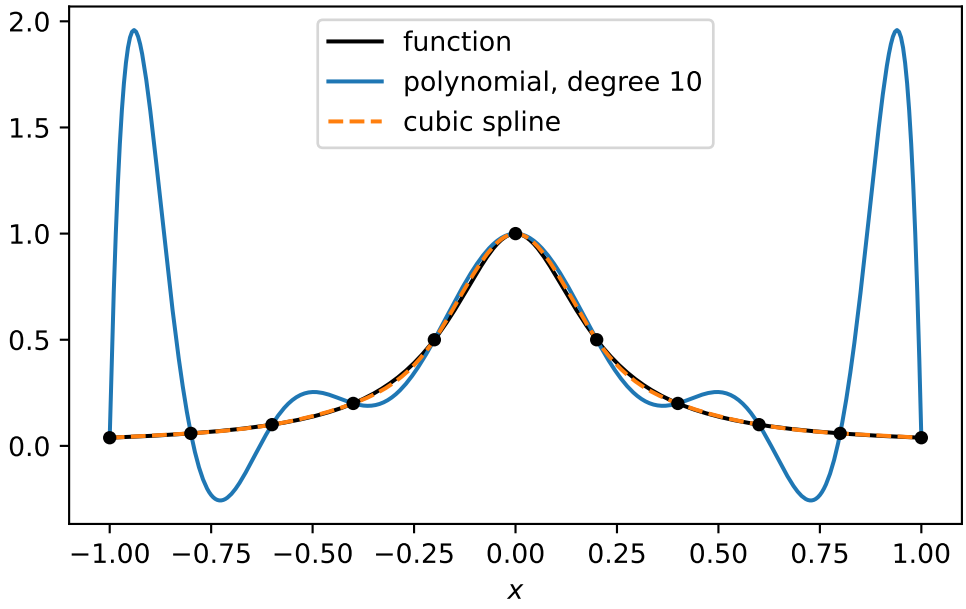


Figure 3.5: Interpolation of $1/(1+25x^2)$ through 11 equidistant points. The polynomial of degree 10 oscillates near the ends; the cubic spline does not.

Compact

```
from scipy.interpolate import BarycentricInterpolator, CubicSpline

runge = lambda x: 1 / (1 + 25 * x**2)       # a function in one line
x_data = np.linspace(-1, 1, 11)             # 11 equidistant points
x_fine = np.linspace(-1, 1, 401)
polynomial = BarycentricInterpolator(x_data, runge(x_data))
spline = CubicSpline(x_data, runge(x_data))

fig, ax = plt.subplots(figsize=(6, 3.5))
ax.plot(x_fine, runge(x_fine), "k", label="function")
ax.plot(x_fine, polynomial(x_fine), label="polynomial, degree 10")
ax.plot(x_fine, spline(x_fine), "--", label="cubic spline")
ax.plot(x_data, runge(x_data), "ko", ms=4)
ax.set_xlabel("$x$")
ax.legend()
plt.show()
```

The step-by-step function `lagrange` is a direct transcription of the Lagrange form.

New Python. `def name(arguments):` defines a **function**: the indented lines are executed when the function is called, and `return` hands back the result. The two `for` loops in `lagrange` are **nested**: for each `j` the inner loop runs through all `i`. The cubic spline requires solving a linear system for the second derivatives at the data points; we take it from the library SciPy. `CubicSpline(x_data, u_data)` returns an object that can be called like a function. The compact version uses `BarycentricInterpolator`, a numerically more stable way to evaluate the same polynomial.

The lesson reaches beyond interpolation. Accuracy is improved reliably by refining the grid with a method of fixed, moderate order. Raising the order on a fixed grid works only under special conditions (Chapter 20).

3.3 Quadrature

Quadrature approximates an integral by a weighted sum of function values. The interval $[a, b]$ is divided into N subintervals of width $h = (b - a)/N$, and on each the integrand is replaced by a polynomial.

3.3.1 Midpoint and trapezoidal rule

Consider one subinterval of width h with midpoint m. We expand the integrand around the midpoint, $f(m+s) = f(m) + sf'(m) + \frac{1}{2}s^2 f''(m) + \frac{1}{6}s^3 f'''(m) + \ldots$, and integrate term by term

from $s=-h/2$ to $s=h/2$. The terms with odd powers of s are odd functions and integrate to zero over the symmetric interval. For the others,

$$\int_{-h/2}^{h/2} 1\,\mathrm{d}s = h, \qquad \int_{-h/2}^{h/2} s^2\,\mathrm{d}s = \left[\frac{s^3}{3}\right]_{-h/2}^{h/2} = \frac{2}{3}\left(\frac{h}{2}\right)^3 = \frac{h^3}{12}.$$

Hence

$$\int_{m-h/2}^{m+h/2} f(x)\,\mathrm{d}x = h\,f(m) + \frac{f''(m)}{2}\cdot\frac{h^3}{12} + O(h^5) = \underbrace{h\,f(m)}_{\text{midpoint rule}} + \frac{h^3}{24}f''(m) + O(h^5). \quad (3.11)$$

The **trapezoidal rule** uses the two end points instead. From the same expansion at $s=\pm h/2$, $f(m\pm\frac{h}{2}) = f(m)\pm\frac{h}{2}f'(m)+\frac{h^2}{8}f''(m)\pm\ldots$, so that in the sum of the two the f' terms cancel:

$$f\left(m-\tfrac{h}{2}\right) + f\left(m+\tfrac{h}{2}\right) = 2f(m) + \frac{h^2}{4}f''(m) + O(h^4),$$

and

$$\frac{h}{2}\left[f\left(m-\tfrac{h}{2}\right) + f\left(m+\tfrac{h}{2}\right)\right] = h\,f(m) + \frac{h^3}{8}f''(m) + O(h^5).$$

We replace $hf(m)$ with Equation 3.11, $hf(m) = \int f\,\mathrm{d}x - \frac{h^3}{24}f''(m)$:

$$\frac{h}{2}[\ldots] = \int f\,\mathrm{d}x - \frac{h^3}{24}f''(m) + \frac{h^3}{8}f''(m) = \int f\,\mathrm{d}x + \frac{h^3}{12}f''(m),$$

since $\frac{1}{8}-\frac{1}{24} = \frac{3-1}{24} = \frac{1}{12}$. Solved for the integral,

$$\int_{m-h/2}^{m+h/2} f(x)\,\mathrm{d}x = \underbrace{\frac{h}{2}\left[f\left(m-\tfrac{h}{2}\right) + f\left(m+\tfrac{h}{2}\right)\right]}_{\text{trapezoidal rule}} - \frac{h^3}{12}f''(m) + O(h^5). \quad (3.12)$$

Composite rules. On the whole interval we sum over the N subintervals with midpoints $m_i = a + (i+\frac{1}{2})h$, $i=0,\ldots,N-1$, and end points $x_i = a+ih$, $i=0,\ldots,N$. The composite midpoint rule is $h\sum_i f(m_i)$. In the composite trapezoidal rule every interior point $x_1,\ldots,x_{N-1}$ is the end point of two subintervals and appears twice with the weight $h/2$:

$$T_N = h\left[\tfrac{1}{2}f(x_0) + f(x_1) + \cdots + f(x_{N-1}) + \tfrac{1}{2}f(x_N)\right].$$

The local errors add up. For the midpoint rule

$$\sum_{i=0}^{N-1} \frac{h^3}{24} f''(m_i) = \frac{h^2}{24} \sum_{i=0}^{N-1} h\, f''(m_i) \approx \frac{h^2}{24} \int_a^b f''(x)\,\mathrm{d}x = \frac{h^2}{24}\left[f'(b) - f'(a)\right],$$

because $\sum_i h f''(m_i)$ is itself the midpoint rule for $\int f''$. The same argument for the trapezoidal rule gives $-\frac{h^2}{12}\left[f'(b) - f'(a)\right]$. Each subinterval contributes $O(h^3)$, but there are $N = (b-a)/h$ of them, so the composite error is $O(h^2)$: both composite rules are of **second order**. The error of the trapezoidal rule is twice that of the midpoint rule and has the opposite sign.

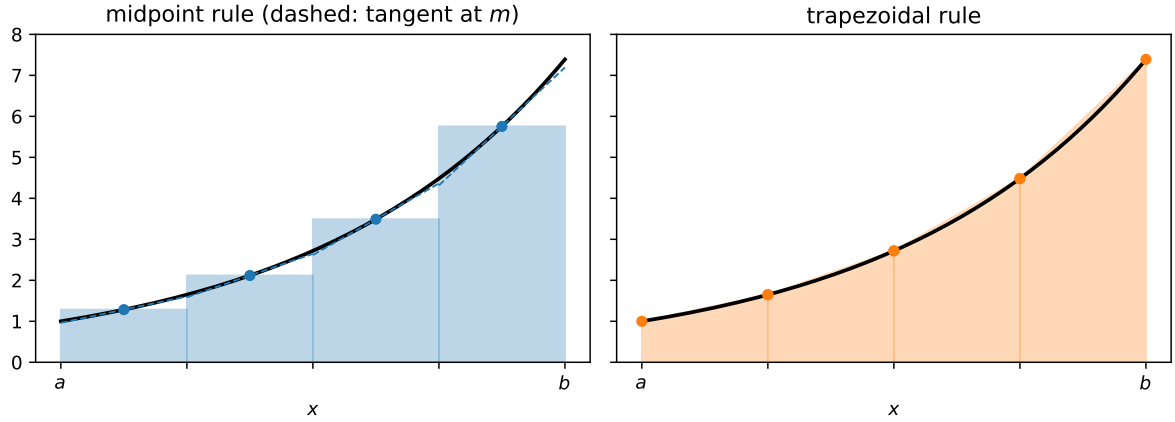


Figure 3.6: Composite midpoint and trapezoidal rule for $f = e^x$ with $N = 4$. For a convex function the trapezoids lie above the curve; the rectangles have the same area as the trapezoids under the tangents (dashed), which lie below the curve.

Figure 3.6 shows the signs of the two errors for a convex function, $f'' > 0$. The trapezoids lie above the curve, because a chord of a convex function lies above it: the trapezoidal rule is too large, in agreement with $\int f\,\mathrm{d}x - \text{trapezoidal} = -\frac{h^3}{12} f'' < 0$. For the midpoint rule, consider the tangent at the midpoint, $t(m+s) = f(m) + s\, f'(m)$. Its integral over the subinterval is

$$\int_{-h/2}^{h/2} \left(f(m) + s\, f'(m)\right) \mathrm{d}s = h\, f(m) + f'(m) \left[\frac{s^2}{2}\right]_{-h/2}^{h/2} = h\, f(m) + f'(m) \left(\frac{h^2}{8} - \frac{h^2}{8}\right) = h\, f(m)\,:$$

the rectangle has the same area as the trapezoid under the tangent. A tangent of a convex function lies below it, so the midpoint rule is too small, in agreement with $\int f\,\mathrm{d}x - h f(m) = +\frac{h^3}{24} f'' > 0$.

The composite error depends only on the derivative of f at the two end points. If f is periodic and the interval is one period, $f'(b) = f'(a)$ and the h^2 term vanishes. A more complete analysis (the Euler–Maclaurin formula) shows that the same holds for all higher terms, which contain $f'''(b) - f'''(a)$, $f^{(5)}(b) - f^{(5)}(a)$, and so on. For smooth periodic functions the trapezoidal rule is therefore extremely accurate.

3.3.2 Simpson's rule

Opposite signs again invite a combination. Taking two thirds of Equation 3.11 and one third of Equation 3.12,

$$\tfrac{2}{3}\left(\int f - \tfrac{h^3}{24} f''\right) + \tfrac{1}{3}\left(\int f + \tfrac{h^3}{12} f''\right) = \int f + \left(-\tfrac{2}{72} + \tfrac{1}{36}\right) h^3 f'' = \int f,$$

the h^3 terms cancel. In function values, the combination is

$$\tfrac{2}{3} h\, f(m) + \tfrac{1}{3} \cdot \tfrac{h}{2}\left[f\left(m - \tfrac{h}{2}\right) + f\left(m + \tfrac{h}{2}\right)\right] = \frac{h}{6}\left[f\left(m - \tfrac{h}{2}\right) + 4f(m) + f\left(m + \tfrac{h}{2}\right)\right],$$

which is **Simpson's rule**:

$$\int_{m-h/2}^{m+h/2} f(x)\,\mathrm{d}x \approx \frac{h}{6}\left[f\left(m - \tfrac{h}{2}\right) + 4f(m) + f\left(m + \tfrac{h}{2}\right)\right].$$

The next terms of both rules are odd powers of s, which vanish, and terms with h^5. The local error of Simpson's rule is therefore $O(h^5)$, and its composite form is of **fourth order**. Many textbooks write Simpson's rule with the distance $\hat{h} = h/2$ between neighbouring points, $\frac{\hat{h}}{3}[f_0 + 4f_1 + f_2]$; this is the same formula.

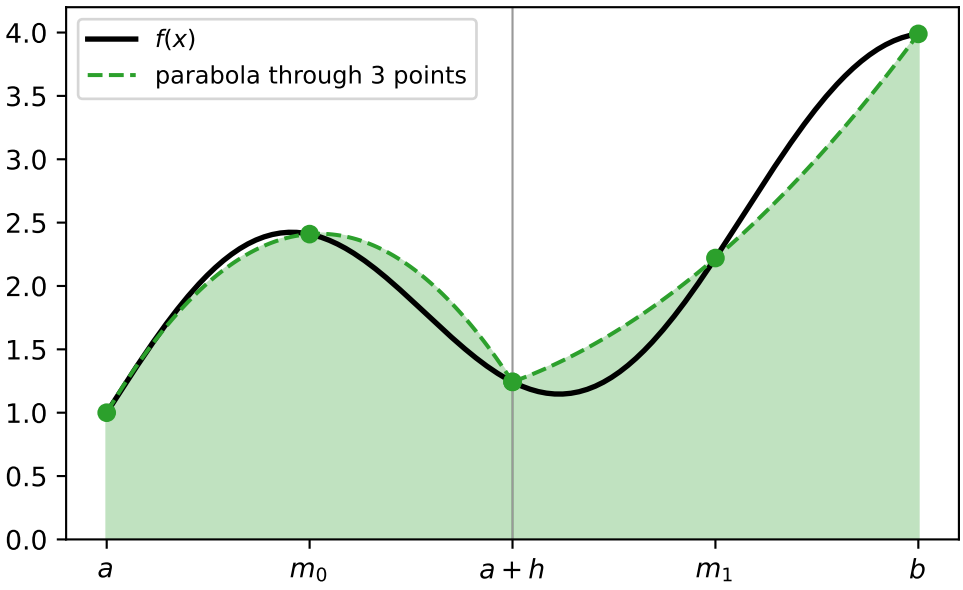


Figure 3.7: Simpson's rule with $N = 2$ subintervals: on each it integrates the parabola (dashed) through the end points and the midpoint exactly.

Simpson's rule also has a geometric meaning (Figure 3.7): on each subinterval it is the exact integral of the parabola p through the three points $m - \frac{h}{2}$, m, $m + \frac{h}{2}$. To see this, we use what the derivation above shows. Its error terms contain only f'''' and higher derivatives, because the terms with f'' cancel and those with odd derivatives integrate to zero. For a polynomial of degree at most three all these derivatives are zero, so Simpson's rule integrates such polynomials exactly. The parabola p has degree two, hence Simpson's rule applied to p gives $\int p\,\mathrm{d}x$. Simpson's rule uses only the three values $f(m - \frac{h}{2})$, $f(m)$ and $f(m + \frac{h}{2})$, and there $p = f$. Applied to f it therefore gives the same number as applied to p, namely $\int p\,\mathrm{d}x$. In the figure the parabola follows f closely on the first subinterval and differs visibly on the second. Smaller subintervals reduce the difference.

Rule	On one subinterval	Error of the composite rule
Midpoint	$h\,f(m)$	$O(h^2)$
Trapezoidal	$\frac{h}{2}\left[f(m-\frac{h}{2}) + f(m+\frac{h}{2})\right]$	$O(h^2)$
Simpson	$\frac{h}{6}\left[f(m-\frac{h}{2}) + 4f(m) + f(m+\frac{h}{2})\right]$	$O(h^4)$

We test the three rules on $\int_0^1 e^x\,\mathrm{d}x = e - 1$ with $N = 4, 8, 16, 32$. Since N doubles, h halves, and the observed order is $\log_2$ of the ratio of successive errors.

Step by step

```
def midpoint(f, a, b, n):
    h = (b - a) / n                 # width of a subinterval
    total = 0.0
    for i in range(n):              # subintervals i = 0, ..., n-1
        m = a + (i + 0.5) * h       # midpoint of subinterval i
        total = total + f(m)
    return h * total

def trapezoidal(f, a, b, n):
    h = (b - a) / n
    total = 0.5 * f(a) + 0.5 * f(b) # the two end points count half
    for i in range(1, n):           # interior points i = 1, ..., n-1
        total = total + f(a + i * h)
    return h * total

def simpson(f, a, b, n):
    return (2 * midpoint(f, a, b, n) + trapezoidal(f, a, b, n)) / 3

exact = np.e - 1                    # integral of exp(x) from 0 to 1
n_list = [4, 8, 16, 32]
error_m = []
error_t = []
error_s = []
for n in n_list:
    error_m.append(abs(midpoint(np.exp, 0, 1, n) - exact))
    error_t.append(abs(trapezoidal(np.exp, 0, 1, n) - exact))
    error_s.append(abs(simpson(np.exp, 0, 1, n) - exact))

print(" N   midpoint  order   trapezoid  order   Simpson   order")
for i in range(1, len(n_list)):
    p_m = np.log2(error_m[i - 1] / error_m[i])  # h halved: order = log2(ratio)
    p_t = np.log2(error_t[i - 1] / error_t[i])
    p_s = np.log2(error_s[i - 1] / error_s[i])
    print(f"{n_list[i]:2d} {error_m[i]:9.2e} {p_m:5.2f}",
          f"{error_t[i]:9.2e} {p_t:5.2f}  {error_s[i]:9.2e} {p_s:5.2f}")
ratio = error_t[-1] / error_m[-1]
print("error trapezoidal / error midpoint for N = 32:", round(ratio, 3))
```

```
 N   midpoint  order   trapezoid  order   Simpson   order
 8  1.12e-03  2.00  2.24e-03  2.00   1.46e-07  4.00
16  2.80e-04  2.00  5.59e-04  2.00   9.10e-09  4.00
32  6.99e-05  2.00  1.40e-04  2.00   5.69e-10  4.00
error trapezoidal / error midpoint for N = 32: 2.0
```

Compact

```
def midpoint(f, a, b, n):
    h = (b - a) / n
    return h * np.sum(f(a + h * (np.arange(n) + 0.5)))   # all midpoints at once

def trapezoidal(f, a, b, n):
    x = np.linspace(a, b, n + 1)                          # all n+1 end points
    return (b - a) / n * (0.5 * f(x[0]) + np.sum(f(x[1:-1])) + 0.5 * f(x[-1]))

def simpson(f, a, b, n):
    return (2 * midpoint(f, a, b, n) + trapezoidal(f, a, b, n)) / 3

n = np.array([4, 8, 16, 32])
for rule in [midpoint, trapezoidal, simpson]:
    errors = np.array([abs(rule(np.exp, 0, 1, k) - np.e + 1) for k in n])
    print(f"{rule.__name__:12s} error for N = 32: {errors[-1]:.2e},"
          f" observed order: {np.log2(errors[:-1] / errors[1:]).round(2)}")
```

```
midpoint     error for N = 32: 6.99e-05, observed order: [2. 2. 2.]
trapezoidal  error for N = 32: 1.40e-04, observed order: [2. 2. 2.]
simpson      error for N = 32: 5.69e-10, observed order: [4. 4. 4.]
```

The output confirms the orders, and also that the trapezoidal error is twice the midpoint error. A function can be handed to another function like a number: `midpoint(np.exp, 0, 1, n)` integrates `np.exp`, and inside `midpoint` it is called as `f(m)`. `np.log2` is the logarithm to base 2, and `error_t[-1]` is the last element of the list.

3.3.3 Where quadrature enters prognostic models

- **Time stepping.** Integrating $\mathrm{d}u/\mathrm{d}t = f(t,u)$ over one time step gives $u(t+\Delta t) = u(t) + \int_t^{t+\Delta t} f\,\mathrm{d}t$. Every time stepping scheme is a quadrature rule for this integral. Evaluating f only at the start of the step, $\int \approx \Delta t\, f(t,u(t))$ (the left rectangle rule, of first order), gives the forward Euler scheme Equation 1.5; the trapezoidal rule gives a scheme of second order (Chapter 5).
- **Finite volumes.** The fluxes through the faces of a grid cell and the sources inside it are integrals (Chapter 9).
- **Finite elements.** The entries of the system matrix are integrals over elements, evaluated with Gauss quadrature, which chooses the positions of the points as well as the weights (Chapter 19).

3.4 Root finding

Many steps of a model require the solution of a nonlinear equation $f(x) = 0$: every step of an implicit time stepping scheme for a nonlinear problem, the saturation temperature of an air parcel, the water table in a soil column.

As an example we take the temperature of the ground surface on a sunny day. The surface absorbs the radiation Q, solar plus the longwave radiation from the atmosphere, and loses energy in two ways: it emits longwave radiation σT^4 (Stefan–Boltzmann law, for a black body), and it heats the air above it by the sensible heat flux $h(T - T_a)$, which is proportional to the difference between the surface temperature T and the air temperature T_a. The surface temperature adjusts until the energy going out equals the energy coming in:

$$\sigma T^4 + h\,(T - T_a) = Q.$$

We neglect the evaporation and the heat flux into the soil, as for dry, bare ground. With the Stefan–Boltzmann constant $\sigma = 5.67 \cdot 10^{-8}$ W m^{-2} K^{-4}, a transfer coefficient $h = 10$ W m^{-2} K^{-1} for a light wind, $T_a = 288$ K and $Q = 990$ W m^{-2} (800 W m^{-2} of sunshine of which 80 % are absorbed, plus 350 W m^{-2} from the atmosphere), we seek the root of

$$f(T) = \sigma T^4 + h\,(T - T_a) - Q = 0. \tag{3.13}$$

The equation is of fourth degree in T. A formula for the roots exists but is so long that nobody uses it, and in a land surface model the equation becomes more complicated as soon as evaporation and soil heat flux are added. Such equations are solved iteratively, in every grid cell and every time step.

A root finder needs to know where to look. The physics gives an interval that must contain the root:

- At $T = T_a$ the sensible heat flux is zero, and the emission $\sigma T_a^4 = 390$ W m^{-2} is less than Q: $f(T_a) = -600 < 0$. The surface takes in more than it gives off, so it is warmer than the air.
- At $T = T_a + Q/h = 387$ K the sensible heat flux alone equals Q, and the emission comes on top: $f(T_a + Q/h) = \sigma T^4 > 0$. Here the surface gives off too much.

Since f is continuous and changes sign, it has a root between 288 K and 387 K. Figure 3.8 shows both sides of the balance and f.

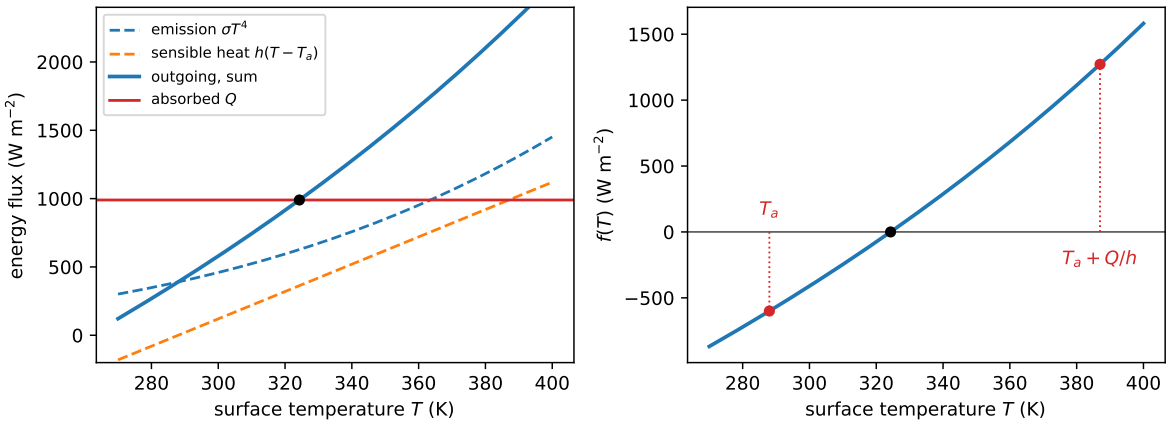


Figure 3.8: Left: the outgoing energy, emission plus sensible heat flux, grows with the surface temperature T and equals the absorbed radiation Q at the root (dot). Right: $f(T)$ is negative at T_a and positive at $T_a + Q/h$, so the root lies in between.

The root is at about 324 K, 51 °C, a realistic value for dry ground at noon. The left panel also shows that the root is unique: the outgoing energy grows with T, so it equals Q only once. Newton's method below also needs the derivative,

$$f'(T) = 4\sigma T^3 + h,$$

which is positive for every $T > 0$, as the picture suggests.

3.4.1 Bisection

If f is continuous and $f(a)$ and $f(b)$ have opposite signs, there is a root in $[a, b]$ (intermediate value theorem). Bisection evaluates f at the midpoint $c = (a + b)/2$. If $f(a)$ and $f(c)$ have opposite signs, the root lies in $[a, c]$, otherwise in $[c, b]$, and the method continues with that half.

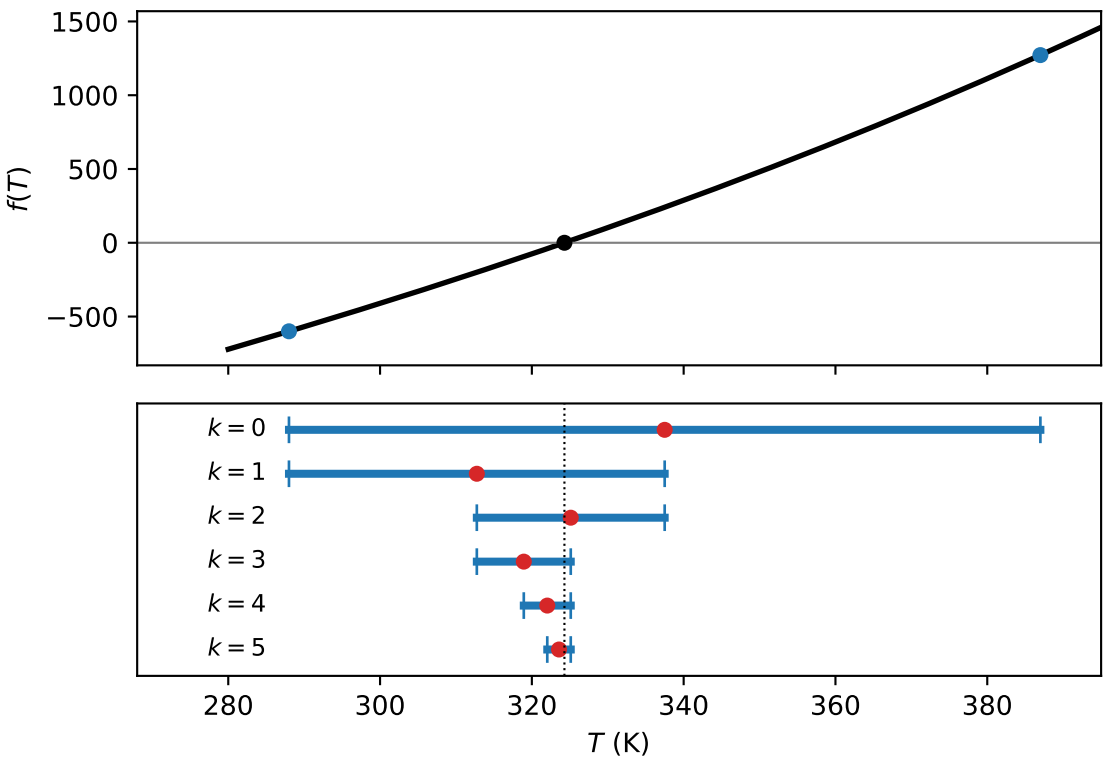


Figure 3.9: Bisection for the energy balance on $[T_a, T_a + Q/h] = [288, 387]$ K. Each row is the interval of iteration k with its midpoint (red); each interval is half of the previous one and contains the root (dotted).

Figure 3.9 shows the first six intervals for our example. In the first iteration $f(c) > 0$ at the midpoint $c = 337.5$ K, the same sign as $f(b)$, so the root lies in the left half $[288, 337.5]$. At the next midpoint, 312.75 K, $f < 0$, so the root lies in the right half; and so on.

After k iterations the interval that contains the root has the length $(b - a)/2^k$. Its midpoint, our estimate, is at most half of this length away from the root:

$$e_k \le \frac{b - a}{2^{k+1}}.$$

To reach a tolerance τ we need $(b - a)/2^{k+1} \le \tau$, that is $2^{k+1} \ge (b - a)/\tau$, or after taking the logarithm to base 2,

$$k \ge \log_2 \frac{b - a}{\tau} - 1.$$

For $b - a = 99$ K and $\tau = 10^{-12}$ K this is $k \geq 46.5 - 1$, that is 46 iterations. Each iteration gains one binary digit, and $\log_2 10 \approx 3.3$ iterations gain one decimal digit. This is **linear convergence**: the error bound is multiplied by a fixed factor, here $\frac{1}{2}$, in every iteration. The method cannot fail, but it is slow, and it does not extend to systems of equations.

"Cannot fail" assumes that f is continuous on $[a, b]$. A function with a pole also changes sign: $1/x$ is negative at $x = -1$ and positive at $x = 1$ but has no root in between, and bisection would converge to the pole at $x = 0$. Before bisecting, plot f or make sure in another way that it is continuous on the bracket.

3.4.2 Newton–Raphson

Newton's method uses the Taylor series Equation 3.1. At the current estimate x_k we keep the first two terms,

$$f(x_k + \delta) \approx f(x_k) + \delta\, f'(x_k),$$

which is the tangent to f at x_k, and choose δ such that the right hand side is zero: $\delta = -f(x_k)/f'(x_k)$. The next estimate is $x_{k+1} = x_k + \delta$:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{3.14}$$

Geometrically, x_{k+1} is the point where the tangent at x_k crosses zero.

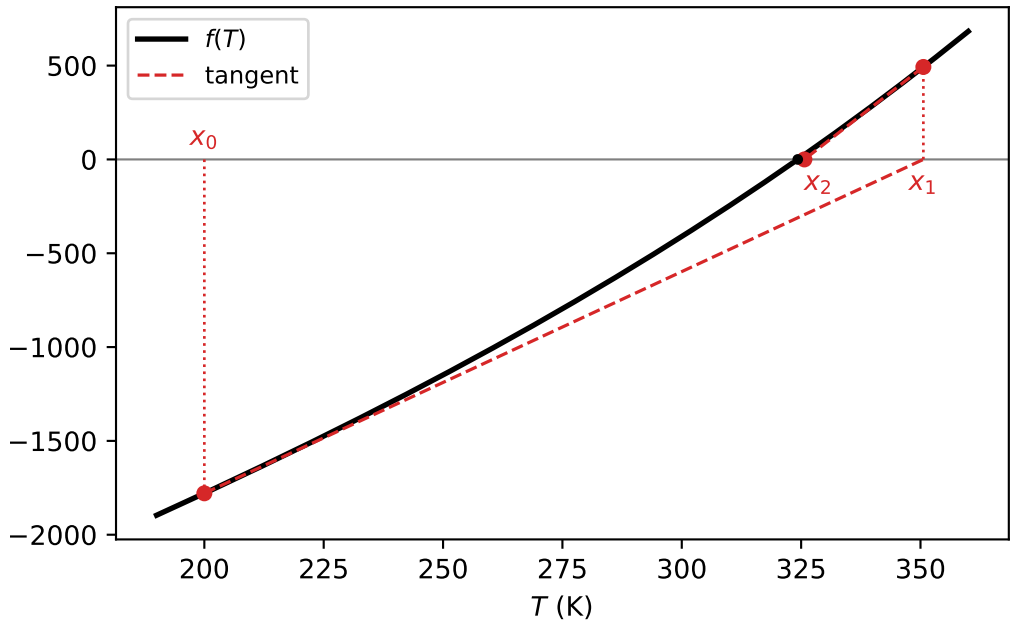


Figure 3.10: Newton steps for the energy balance from the poor start $x_0 = 200$ K. Each new estimate is where the tangent (dashed) crosses zero. Because f is convex, the tangent lies below the curve and x_1 lands to the right of the root (black dot).

Figure 3.10 shows the first steps from a deliberately poor start, $x_0 = 200$ K, so that they are visible; from T_a the tangent would lie almost on the curve. The tangent at x_0 crosses zero at $x_1 = 350.6$ K, 26 K to the right of the root. That it lands on the right is no accident: $f'' = 12\sigma T^2 > 0$, so f is convex, and the tangent of a convex function lies below the curve. It therefore reaches zero later than f does. From x_1 on, all estimates lie to the right of the root and approach it from above: $x_2 = 325.7$ K is within 1.4 K of the root at 324.29 K, and x_3 within 0.004 K.

How fast does it converge? Let r be the root and $e_k = x_k - r$ the error. We expand f around x_k and evaluate at the root, now keeping the next term in the Lagrange form. The step from x_k to r is $r - x_k = -e_k$:

$$0 = f(r) = f(x_k) - e_k\, f'(x_k) + \frac{e_k^2}{2} f''(\xi),$$

with ξ between x_k and r. Dividing by $f'(x_k)$,

$$0 = \frac{f(x_k)}{f'(x_k)} - e_k + \frac{f''(\xi)}{2f'(x_k)}\, e_k^2,$$

and rearranging,

$$e_k - \frac{f(x_k)}{f'(x_k)} = \frac{f''(\xi)}{2f'(x_k)}\, e_k^2.$$

The left hand side is $x_k - r - f(x_k)/f'(x_k) = x_{k+1} - r = e_{k+1}$ by Equation 3.14. Hence

$$e_{k+1} = \frac{f''(\xi)}{2f'(x_k)}\, e_k^2 \approx \frac{f''(r)}{2f'(r)}\, e_k^2. \tag{3.15}$$

The new error is proportional to the *square* of the old one: **quadratic convergence**. An error of 10^{-3} becomes 10^{-6} and then 10^{-12}; the number of correct digits doubles in every iteration. The argument requires $f'(r) \neq 0$, a simple root. At a double root, where f touches zero without crossing, $f'(r) = 0$ and Newton's method converges only linearly. The sign of the factor also confirms the picture: with $f'' > 0$ and $f' > 0$ it is positive, so $e_{k+1} > 0$, and every estimate after the first lies to the right of the root.

The code finds the root with both methods. It then repeats the Newton steps from $T_0 = T_a$ and prints the error and the ratio e_{k+1}/e_k^2, which according to Equation 3.15 should approach $f''(r)/(2f'(r))$, with $f''(T) = 12\sigma T^2$.

Step by step

```
sigma = 5.67e-8                     # Stefan-Boltzmann constant, W m^-2 K^-4
h = 10.0                            # transfer coefficient, W m^-2 K^-1
Ta = 288.0                          # air temperature, K
Q = 990.0                           # absorbed radiation, W m^-2

def f(T):                           # energy out minus energy in
    return sigma * T**4 + h * (T - Ta) - Q

def dfdT(T):                        # derivative of f
    return 4 * sigma * T**3 + h

def bisection(f, a, b, tol):
    fa = f(a)
    k = 0                           # counts the iterations
    while (b - a) / 2 > tol:        # midpoint not yet within tol of the root
        c = (a + b) / 2             # midpoint
        fc = f(c)
        k = k + 1
        if fa * fc <= 0:            # different signs: the root is in [a, c]
            b = c
        else:                       # same sign: the root is in [c, b]
```

```
            a = c
            fa = fc
    return (a + b) / 2, k           # return two values: the root and k

def newton(f, dfdx, x, tol):
    k = 0
    step = 2 * tol                  # any value > tol, so that the loop starts
    while abs(step) > tol and k < 50:   # stop after 50 iterations at most
        step = -f(x) / dfdx(x)      # Newton step
        x = x + step
        k = k + 1
    return x, k

root_b, k_b = bisection(f, Ta, Ta + Q / h, 1e-12)   # receive both values
root_n, k_n = newton(f, dfdT, Ta, 1e-12)
print(f"bisection: T = {root_b:.12f} K after {k_b} iterations")
print(f"Newton:    T = {root_n:.12f} K after {k_n} iterations")

r = root_n                          # the root, as reference
d2f = 12 * sigma * r**2             # second derivative f''(r)
C = d2f / (2 * dfdT(r))
T = Ta
error_old = abs(T - r)
for k in range(1, 4):
    T = T - f(T) / dfdT(T)
    error = abs(T - r)
    ratio = error / error_old**2
    print(f"iteration {k}: error = {error:.1e} K,  error / old error^2 =
    ↪  {ratio:.5f}")
    error_old = error
print(f"predicted ratio f''(r) / (2 f'(r)) = {C:.5f}")
```

```
bisection: T = 324.291588862769 K after 46 iterations
Newton:    T = 324.291588862769 K after 5 iterations
iteration 1: error = 2.6e+00 K,  error / old error^2 = 0.00199
iteration 2: error = 1.4e-02 K,  error / old error^2 = 0.00202
iteration 3: error = 3.9e-07 K,  error / old error^2 = 0.00202
predicted ratio f''(r) / (2 f'(r)) = 0.00202
```

Compact

```
from scipy.optimize import root_scalar

sigma, h, Ta, Q = 5.67e-8, 10.0, 288.0, 990.0
```

```
f = lambda T: sigma * T**4 + h * (T - Ta) - Q
dfdT = lambda T: 4 * sigma * T**3 + h

b = root_scalar(f, bracket=[Ta, Ta + Q / h], method="bisect", xtol=1e-12)
n = root_scalar(f, fprime=dfdT, x0=Ta, method="newton", xtol=1e-12)
print(f"bisection: T = {b.root:.12f} K after {b.iterations} iterations")
print(f"Newton:    T = {n.root:.12f} K after {n.iterations} iterations")
```

```
bisection: T = 324.291588862769 K after 47 iterations
Newton:    T = 324.291588862769 K after 5 iterations
```

The errors of the Newton iteration show the squaring predicted by Equation 3.15: 2.6 K, then 0.014 K, then $4 \cdot 10^{-7}$ K; the exponent roughly doubles from one iteration to the next, and the ratio e_{k+1}/e_k^2 agrees with the predicted value to three digits. The ratio is small, 0.002 K^{-1}, because the curvature of f is small compared with its slope. One more iteration reaches the round-off level, about 10^{-13} K for numbers near 300; the error can then not decrease further. SciPy's bisection counts one iteration more than ours because it uses a slightly different stopping test. `while` conditions can be combined with `and`: the loop continues only while both are true. The second condition protects against an endless loop if the iteration does not converge. A function can `return` several values separated by commas; they are received as `root_b, k_b = ...`.

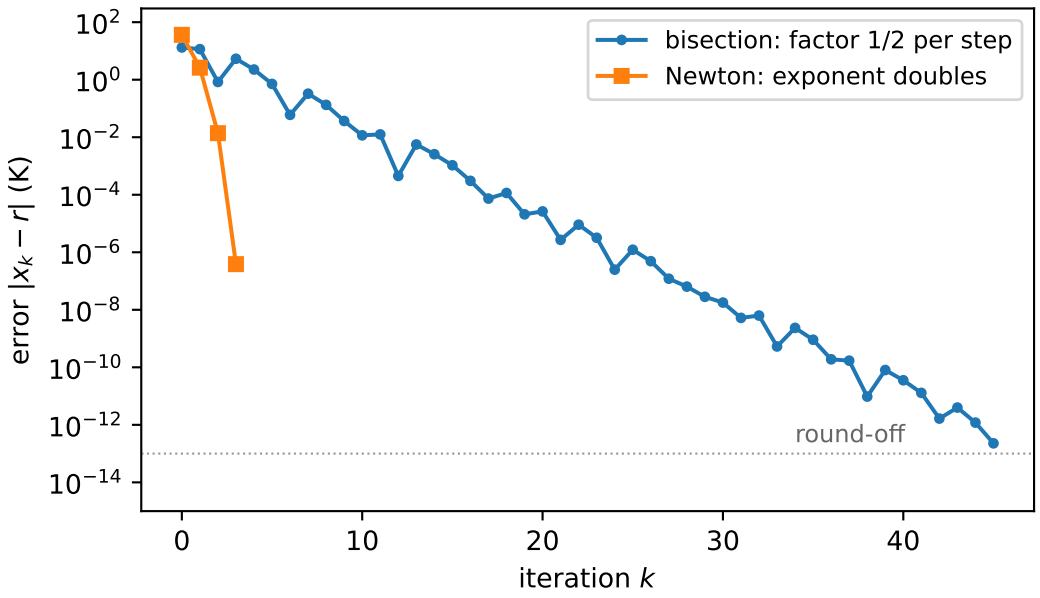


Figure 3.11: Error of bisection on $[T_a, T_a + Q/h]$ and of Newton's method from T_a, on a logarithmic axis.

Figure 3.11 compares the two rates. On a logarithmic axis the bisection error falls along a straight line, on average by a factor of 2 per iteration. It does not fall monotonically, because a midpoint can by chance lie closer to the root than the next one; the bound $(b-a)/2^{k+1}$ decreases monotonically. The Newton error curve bends downward ever more steeply, because the exponent doubles, and it reaches the round-off level after four iterations, where bisection needs about 45.

The speed has a price, which is also visible in Equation 3.15. Newton's method needs the derivative. And it needs a starting value that is close enough: the argument assumed that e_k is small, and where $f'(x_k) \approx 0$ the tangent is nearly horizontal and the next estimate is thrown far away. The energy balance is a friendly case, with $f' > 0$ and $f'' > 0$ everywhere; the box below shows less friendly ones. In practice the two methods are combined: bisection until the interval is small, then Newton. `scipy.optimize.root_scalar` offers such safeguarded methods, for example `method="brentq"`.

3.4.3 Newton's method for systems

Newton's method is important for a further reason: it generalises to systems. For N equations $\vec{F}(\vec{x}) = 0$ in N unknowns, the Taylor series of each component F_i to first order is

$$F_i(\vec{x}_k + \delta\vec{x}) \approx F_i(\vec{x}_k) + \sum_{j=1}^{N} \frac{\partial F_i}{\partial x_j}(\vec{x}_k)\,\delta x_j,$$

or, in matrix form,

$$\vec{F}(\vec{x}_k + \delta\vec{x}) \approx \vec{F}(\vec{x}_k) + J(\vec{x}_k)\,\delta\vec{x}, \qquad J_{ij} = \frac{\partial F_i}{\partial x_j},$$

with the **Jacobian matrix** J in the place of the derivative. Setting the right hand side to zero gives a linear system for the update:

$$J(\vec{x}_k)\,\delta\vec{x} = -\vec{F}(\vec{x}_k), \qquad \vec{x}_{k+1} = \vec{x}_k + \delta\vec{x}. \tag{3.16}$$

Example. The circle $x^2 + y^2 = 4$ and the hyperbola $xy = 1$ intersect in four points. With

$$\vec{F}(x,y) = \begin{pmatrix} x^2 + y^2 - 4 \\ xy - 1 \end{pmatrix}, \qquad J(x,y) = \begin{pmatrix} 2x & 2y \\ y & x \end{pmatrix}$$

and the starting point $(2, 0.5)$, Newton's method finds the intersection near it. The exact value is $x = \sqrt{2+\sqrt{3}} \approx 1.9319$, $y = 1/x \approx 0.5176$.

Step by step

```
x, y = 2.0, 0.5                     # starting point
for k in range(1, 6):
    F = np.array([x**2 + y**2 - 4,  # the two equations
                  x * y - 1])
    J = np.array([[2 * x, 2 * y],   # the Jacobian matrix, row by row
                  [y,     x    ]])
    delta = np.linalg.solve(J, -F)  # solve J delta = -F
    x = x + delta[0]
    y = y + delta[1]
    error = abs(x - np.sqrt(2 + np.sqrt(3)))   # distance from the exact x
    print(f"iteration {k}: x = {x:.12f}, y = {y:.12f}, error in x = {error:.1e}")
```

```
iteration 1: x = 1.933333333333, y = 0.516666666667, error in x = 1.5e-03
iteration 2: x = 1.931852741096, y = 0.517637054822, error in x = 1.1e-06
iteration 3: x = 1.931851652579, y = 0.517638090204, error in x = 8.0e-13
iteration 4: x = 1.931851652578, y = 0.517638090205, error in x = 2.2e-16
iteration 5: x = 1.931851652578, y = 0.517638090205, error in x = 0.0e+00
```

Compact

```
from scipy.optimize import root

F = lambda v: [v[0]**2 + v[1]**2 - 4, v[0] * v[1] - 1]
J = lambda v: [[2 * v[0], 2 * v[1]], [v[1], v[0]]]
print(root(F, [2.0, 0.5], jac=J).x)
```

```
[1.93185165 0.51763809]
```

Again the number of correct digits doubles in every iteration, until the round-off level is reached. `np.linalg.solve(J, -F)` solves the linear system $J\,\delta\vec{x} = -\vec{F}$; how this is done is the subject of the next unit.

A nonlinear problem is thus replaced by a sequence of linear ones. This is how nonlinear PDEs such as the Richards equation are solved (Chapter 17), and it is one reason why the solution of linear systems, the subject of the next unit, is so central.

3.5 Summary

- Finite difference formulas and their truncation errors follow from the Taylor series by combining series so that unwanted terms cancel. The forward and backward differences are of first order, the central differences for the first and second derivative of second order. For any stencil, the weights follow from a linear system.
- The order of a method is measured as the slope of the error in a log–log plot, or from the ratio of the errors for two step sizes. This is the basic verification test.
- Round-off limits the accuracy of a finite difference. The optimal step balances truncation and round-off error and is far larger than the machine epsilon: about $\sqrt{\varepsilon}$ for the forward and $\varepsilon^{1/3}$ for the central difference.
- Linear interpolation and the midpoint and trapezoidal rules are of second order. Polynomials of high degree on equidistant points oscillate (Runge). Combining two rules whose errors have opposite signs gives a rule of higher order, such as Simpson's rule.
- Bisection converges linearly and always. Newton's method converges quadratically near a simple root and generalises to systems, where each iteration solves a linear system with the Jacobian.

3.6 Further reading

LeVeque (2007), chapter 1, for finite difference approximations; Fornberg (1988) for stencils on arbitrary grids.

4 The numerical toolbox II: matrices and waves

Learning goals

After this unit you can

- set up the matrix of a finite difference discretization and describe its structure,
- derive the eigenvalues of the model matrix and its condition number,
- explain what the condition number says about the solution of a linear system,
- decompose a signal on a grid into waves with the discrete Fourier transform,
- explain the shortest resolvable wave, aliasing and the effective resolution of a grid.

Two more tools are needed before we turn to differential equations. Matrices describe what a discretization does to all grid points at once. Waves describe what it does to each spatial scale. The two views are closely related: the eigenvectors of the most important matrix of this lecture are waves.

4.1 From a differential equation to a matrix

4.1.1 A model problem

Consider the one-dimensional Poisson equation with homogeneous Dirichlet conditions,

$$-\frac{\mathrm{d}^2 u}{\mathrm{d}x^2} = f(x), \qquad u(0) = u(L) = 0. \tag{4.1}$$

We divide $[0, L]$ into $N + 1$ intervals of width $h = L/(N + 1)$ with grid points $x_j = jh$. The values $u_0 = u_{N+1} = 0$ are known; the unknowns are $u_1, \ldots, u_N$. At each interior point we replace the second derivative by the central difference Equation 3.6:

$$\frac{-u_{j-1} + 2u_j - u_{j+1}}{h^2} = f_j, \qquad j = 1, \ldots, N.$$

These are N linear equations for N unknowns. We write them out for $N = 4$, using $u_0 = u_5 = 0$ in the first and the last equation:

$$
\begin{aligned}
2u_1 - u_2 &= h^2 f_1 \\
-u_1 + 2u_2 - u_3 &= h^2 f_2 \\
-u_2 + 2u_3 - u_4 &= h^2 f_3 \\
-u_3 + 2u_4 &= h^2 f_4
\end{aligned}
\quad \Longleftrightarrow \quad
\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}
\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}
= h^2 \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix}.
$$

In general, discretizing a linear differential equation leads to a linear system

$$A\vec{x} = \vec{b}$$

with as many unknowns as grid points. Because we multiplied each equation by h^2, the factor h^2 now stands on the right hand side: here $\vec{x} = (u_1, \ldots, u_N)^T$, $\vec{b} = h^2(f_1, \ldots, f_N)^T$, and A is the **model matrix**

$$
A = \begin{pmatrix}
2 & -1 & & & \\
-1 & 2 & -1 & & \\
& \ddots & \ddots & \ddots & \\
& & -1 & 2 & -1 \\
& & & -1 & 2
\end{pmatrix}. \tag{4.2}
$$

Row j of the matrix is the stencil of the difference formula, centred on the diagonal. A accompanies us through the units on elliptic and parabolic problems. The rest of this section works out three of its properties, each of which matters later:

1. **Sparsity** (Section 4.1.2): how much memory and work a solver needs.
2. **Symmetry and definiteness** (Section 4.1.3): whether $A\vec{x} = \vec{b}$ has exactly one solution, and which solvers we may use.
3. **Eigenvalues and eigenvectors** (Section 4.1.4): explicit formulas for all N of them. They give the definiteness, the matrix norm and the condition number (Section 4.2), and they connect the matrix to waves (Section 4.3).

4.1.2 Sparsity

Each row has at most three non-zero entries, because each grid point is coupled only to its neighbours. Of the N^2 entries only $3N - 2$ are non-zero. Such a matrix is **sparse**. Sparse matrices are stored and processed without their zeros, for example with `scipy.sparse`. For a model with 10^8 grid points this is the difference between possible and impossible: the full matrix would have 10^{16} entries.

Result

A has $3N-2$ non-zero entries out of N^2. Memory and work grow like N, not like N^2, as long as a solver does not fill in the zeros (Chapter 10).

4.1.3 Symmetry and definiteness

The goal of this subsection: show that A is symmetric and positive definite, and see what this means. The definiteness tells us whether $A\vec{x} = \vec{b}$ has exactly one solution, and the fastest solvers for large systems only work for such matrices (Chapter 11).

A is **symmetric**, $A = A^T$: the entry in row i and column j equals the entry in row j and column i. The reason is that the stencil is symmetric: point j couples to its left neighbour with the same weight, -1, as to its right neighbour.

The **definiteness** of a symmetric matrix is the sign of its **quadratic form**, the number

$$q(\vec{x}) = \vec{x}^T A \vec{x} = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i \, a_{ij} \, x_j$$

that the matrix assigns to every vector $\vec{x}$. There are five cases:

$q(\vec{x})$ for all $\vec{x} \neq 0$	the matrix is
> 0	positive definite
≥ 0, and $= 0$ for some $\vec{x} \neq 0$	positive semidefinite
< 0	negative definite
≤ 0, and $= 0$ for some $\vec{x} \neq 0$	negative semidefinite
> 0 for some $\vec{x}$, < 0 for others	indefinite

For $N = 2$ the quadratic form is a surface over the (x_1, x_2) plane, and the cases can be seen (Figure 4.1). A positive definite matrix gives a bowl with its only minimum at the origin, a negative definite matrix an upside-down bowl, an indefinite matrix a saddle. A semidefinite matrix gives a trough: along one direction the surface stays at zero.

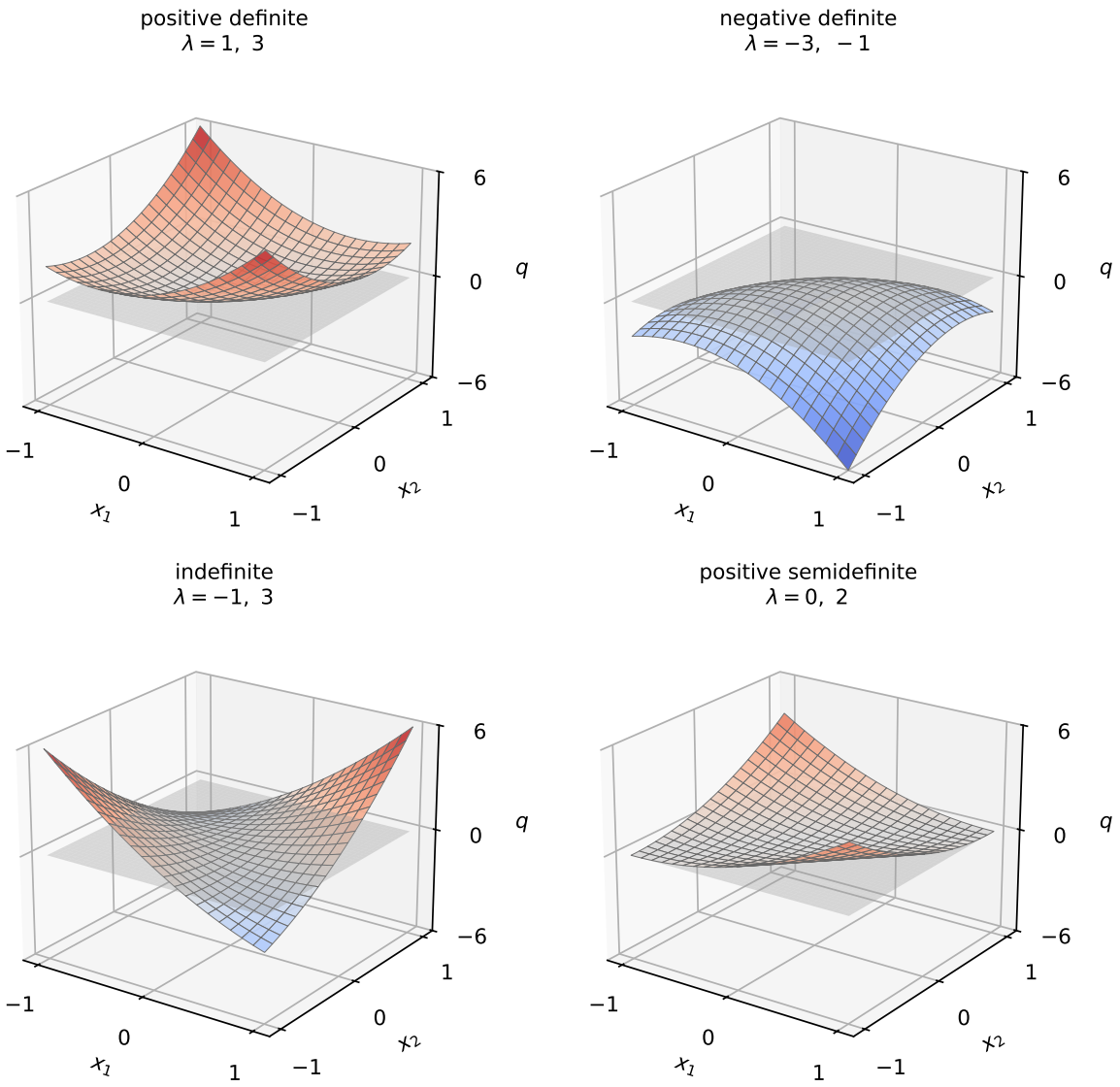


Figure 4.1: The quadratic form $q = \vec{x}^T A \vec{x}$ of four symmetric 2×2 matrices, with their eigenvalues λ; the grey plane is $q = 0$. Top left: the model matrix for $N = 2$. Bottom right: $q = (x_1 - x_2)^2$ is zero along the line $x_1 = x_2$.

How do we determine the definiteness? There are two ways.

By rewriting the quadratic form as a sum of squares. For the model matrix with $N = 3$ we first multiply out, row by row,

$$\vec{x}^T A\vec{x} = x_1(2x_1 - x_2) + x_2(-x_1 + 2x_2 - x_3) + x_3(-x_2 + 2x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3.$$

Now we split each $2x_j^2$ into $x_j^2 + x_j^2$ and group the terms so that each mixed term $-2x_jx_{j+1}$ completes a binomial, $x_j^2 - 2x_jx_{j+1} + x_{j+1}^2 = (x_j - x_{j+1})^2$:

$$\vec{x}^T A\vec{x} = x_1^2 + \underbrace{(x_1^2 - 2x_1x_2 + x_2^2)}_{(x_1-x_2)^2} + \underbrace{(x_2^2 - 2x_2x_3 + x_3^2)}_{(x_2-x_3)^2} + x_3^2.$$

The same works for every N:

$$\vec{x}^T A\vec{x} = x_1^2 + \sum_{j=1}^{N-1} (x_j - x_{j+1})^2 + x_N^2. \tag{4.3}$$

A sum of squares cannot be negative. It is zero only if every square is zero: $x_1 = 0$, then $x_2 = x_1 = 0$, and so on, so $\vec{x} = 0$. Hence A is positive definite.

Equation 4.3 also shows *why*. The squares are differences of neighbouring values, the discrete version of $\int (u')^2 \, \mathrm{d}x$. The two single squares x_1^2 and x_N^2 come from the first and last rows, where the Dirichlet values $u_0 = u_{N+1} = 0$ were used. Without them, as for Neumann or periodic boundaries, a constant vector $x_1 = \cdots = x_N$ gives $q = 0$: the matrix is then only positive semidefinite, like the trough in Figure 4.1.

By the eigenvalues. For a symmetric matrix the signs of the eigenvalues decide: positive definite means all eigenvalues are positive, negative definite all negative, semidefinite that some are zero, and indefinite that there are eigenvalues of both signs. Figure 4.1 gives the eigenvalues for each case. Why this works is shown in Section 4.1.4 (Equation 4.5), where we also confirm the result for A with its eigenvalues. For large matrices this is the practical test, because eigenvalues can be computed even when the quadratic form cannot be rearranged by hand.

Why we want positive definite. A positive definite matrix is invertible. If $A\vec{x} = \vec{0}$ for some $\vec{x} \neq \vec{0}$, then $\vec{x}^T A\vec{x} = 0$, which a positive definite matrix does not allow. So $A\vec{x} = \vec{b}$ has exactly one solution for every $\vec{b}$: the discrete problem is well-posed in the sense of Chapter 1. For Neumann or periodic boundaries the constant vector gives $A\vec{x} = \vec{0}$, and the solution is fixed only up to a constant.

The sign matters. The second difference itself, $(u_{j-1} - 2u_j + u_{j+1})/h^2$, has the matrix $-A/h^2$, which is negative definite. This is why we wrote the Poisson equation with a minus sign, $-u'' = f$: then the matrix is positive definite. Matrices from diffusion problems have this property, and the most efficient solvers rely on it (Chapter 11).

Result

The model matrix A is symmetric and positive definite (SPD): its quadratic form is the sum of squares Equation 4.3, and it is positive for every $\vec{x} \neq \vec{0}$. Therefore $A\vec{x} = \vec{b}$ has exactly one solution. Without Dirichlet values the matrix would be only positive semidefinite and singular.

4.1.4 Eigenvalues and eigenvectors

The goal of this subsection: formulas for all N eigenvalues λ_m and eigenvectors $\vec{v}_m$ of the model matrix. The result is Equation 4.6 at the end. We use it three times: to confirm that A is positive definite (below), to compute the norm and the condition number of A (Section 4.2), and to connect the matrix to waves (Section 4.3).

Eigenvalues of symmetric matrices. An **eigenvector** $\vec{v} \neq \vec{0}$ of a matrix A is a vector that A only stretches,

$$A\vec{v} = \lambda\vec{v},$$

with the **eigenvalue** λ. For a symmetric $N \times N$ matrix all eigenvalues are real, and there are N eigenvectors that are orthogonal to each other. We number them $m = 1, \ldots, N$ and scale them to length 1, so that

$$\vec{v}_m^T \vec{v}_n = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{otherwise.} \end{cases} \tag{4.4}$$

Such vectors are called **orthonormal**. Any vector can be written as a combination $\vec{x} = \sum_m c_m \vec{v}_m$, and by Equation 4.4 the coefficients are $c_m = \vec{v}_m^T \vec{x}$ and the length is $\|\vec{x}\|_2^2 = \vec{x}^T\vec{x} = \sum_m c_m^2$. Applying A multiplies each part by its eigenvalue, $A\vec{x} = \sum_m \lambda_m c_m \vec{v}_m$. This is the one property of eigenvectors that everything below uses.

The eigenvalue test for definiteness. In the product $\vec{x}^T A\vec{x}$ only the terms with $m = n$ survive, by Equation 4.4:

$$\vec{x}^T A\vec{x} = \sum_m \sum_n c_m c_n \lambda_n \, \vec{v}_m^T \vec{v}_n = \sum_m \lambda_m c_m^2. \tag{4.5}$$

If all $\lambda_m > 0$, this is positive for every $\vec{x} \neq \vec{0}$. If one $\lambda_m < 0$, the choice $\vec{x} = \vec{v}_m$ gives $q = \lambda_m < 0$. For example, $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has the eigenvalues 3 and -1: for $\vec{x} = (1, 1)$ the quadratic form is $1 + 4 + 1 = 6 > 0$, for $\vec{x} = (1, -1)$ it is $1 - 4 + 1 = -2 < 0$, the saddle in Figure 4.1.

Now to the eigenvalues of A itself. We find them in three steps.

Step 1: a guess from the continuous problem. Note that we now solve a different problem than Equation 4.1. The solution u of the Poisson equation depends on f; an eigenvector does not involve f at all. The continuous counterpart of $A\vec{v} = \lambda\vec{v}$ is the eigenvalue problem

$$-\frac{\mathrm{d}^2 v}{\mathrm{d}x^2} = \mu\, v, \qquad v(0) = v(L) = 0.$$

Its solutions are sines: for $v(x) = \sin(kx)$, differentiating twice gives $-v'' = k^2 \sin(kx)$, so $\mu = k^2$, and $v(L) = 0$ requires $k = m\pi/L$ with an integer m: m half waves fit into the domain. Since A is $-\mathrm{d}^2/\mathrm{d}x^2$ multiplied by h^2, we expect its eigenvectors to be the same sines sampled on the grid, and try

$$v_j = \sin(j\theta), \qquad j = 1, \ldots, N.$$

θ is an angle, the phase by which the sine advances from one grid point to the next. At the grid points $x_j = jh$ a sine wave $\sin(kx)$ takes the values $\sin(k\, jh) = \sin(j\theta)$, so

$$\theta = kh,$$

the wavenumber times the grid spacing, a number without unit. A long wave advances little from point to point, θ small; the shortest wave that the grid can show advances by almost π. Which values of θ are allowed is decided in step 3.

Step 2: the interior rows give the eigenvalue. Row j of $A\vec{v}$ is

$$(A\vec{v})_j = -\sin\big((j-1)\theta\big) + 2\sin(j\theta) - \sin\big((j+1)\theta\big).$$

The addition theorems $\sin(a \mp b) = \sin a \cos b \mp \cos a \sin b$ give, when added, $\sin(a-b) + \sin(a+b) = 2\sin a \cos b$. With $a = j\theta$ and $b = \theta$ the two outer terms combine to $-2\sin(j\theta)\cos\theta$, and

$$(A\vec{v})_j = 2\sin(j\theta) - 2\sin(j\theta)\cos\theta = (2 - 2\cos\theta)\, v_j.$$

Every row gives the same factor $2 - 2\cos\theta$, and this factor is the eigenvalue, for any θ so far. (Section 4.3.1 repeats this step in one line with the complex exponential.)

Step 3: the boundary rows fix θ. The first and the last row lack one neighbour. The formula of step 2 holds for them only if the missing values vanish, like the Dirichlet values $u_0 = u_{N+1} = 0$ at the boundary nodes $j = 0$ and $j = N+1$. At the left, $v_0 = \sin 0 = 0$ holds automatically. At the right, $v_{N+1} = \sin\big((N+1)\theta\big) = 0$ requires $(N+1)\theta = m\pi$ with an integer m, so $\theta = m\pi/(N+1)$. The values $m = 1, \ldots, N$ give N different eigenvectors; $m = 0$ and

$m = N + 1$ give $\vec{v} = \vec{0}$, and larger m repeat the same vectors. With N eigenvectors for an $N \times N$ matrix, we have found all of them.

Result: eigenvalues and eigenvectors of the model matrix

For $m = 1, 2, \ldots, N$,

$$\begin{aligned} \theta_m &= \frac{m\pi}{N+1}, \\ \lambda_m &= 2 - 2\cos\theta_m = 4\sin^2\frac{\theta_m}{2}, \\ (\vec{v}_m)_j &= \sqrt{\frac{2}{N+1}}\,\sin(j\theta_m), \qquad j = 1, \ldots, N. \end{aligned} \tag{4.6}$$

- m counts the **half waves** of eigenvector $\vec{v}_m$ across the domain (Figure 4.2). $m = 1$ is the longest wave, $m = N$ the shortest, which changes sign from grid point to grid point.
- $\theta_m = k_m h$ is the phase step per grid point of the sine with wavenumber $k_m = m\pi/L$.
- Since $0 < \theta_m < \pi$, the eigenvalues increase with m and lie between 0 and 4: $0 < \lambda_1 < \lambda_2 < \cdots < \lambda_N < 4$. The smallest is $\lambda_{\min} = \lambda_1$, the largest $\lambda_{\max} = \lambda_N$.
- All eigenvalues are positive, so A is positive definite by Equation 4.5, as found in Section 4.1.3 with the sum of squares.

Three remarks on the result. The second form of λ_m follows from $\cos\theta = 1 - 2\sin^2(\theta/2)$. The factor $\sqrt{2/(N+1)}$ scales the eigenvectors to length 1, because $\sum_{j=1}^{N} \sin^2(j\theta_m) = (N+1)/2$; any multiple of an eigenvector is also an eigenvector, so this factor does not change λ_m. And the result agrees with the continuous problem: there are $N + 1$ intervals, so $h = L/(N+1)$, and the sine with $k = m\pi/L$ has the phase step

$$\theta = kh = \frac{m\pi}{L} \cdot \frac{L}{N+1} = \frac{m\pi}{N+1} = \theta_m.$$

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.sparse import diags

N = 20
A = diags([-1.0, 2.0, -1.0], [-1, 0, 1], shape=(N, N))
eigenvalues, eigenvectors = np.linalg.eigh(A.toarray())

fig, ax = plt.subplots(figsize=(6.5, 3))
j = np.arange(1, N + 1)
for index in [0, 1, N - 1]:
    v = eigenvectors[:, index]
    ax.plot(j, v * np.sign(v[0]), "o-", ms=3, label=rf"$m = {index + 1}$, $\lambda =
↪   {eigenvalues[index]:.3f}$")
```

```
ax.set_xlabel("grid point $j$")
ax.set_xticks([1, 5, 10, 15, 20])
ax.legend(fontsize=8)
plt.show()
```

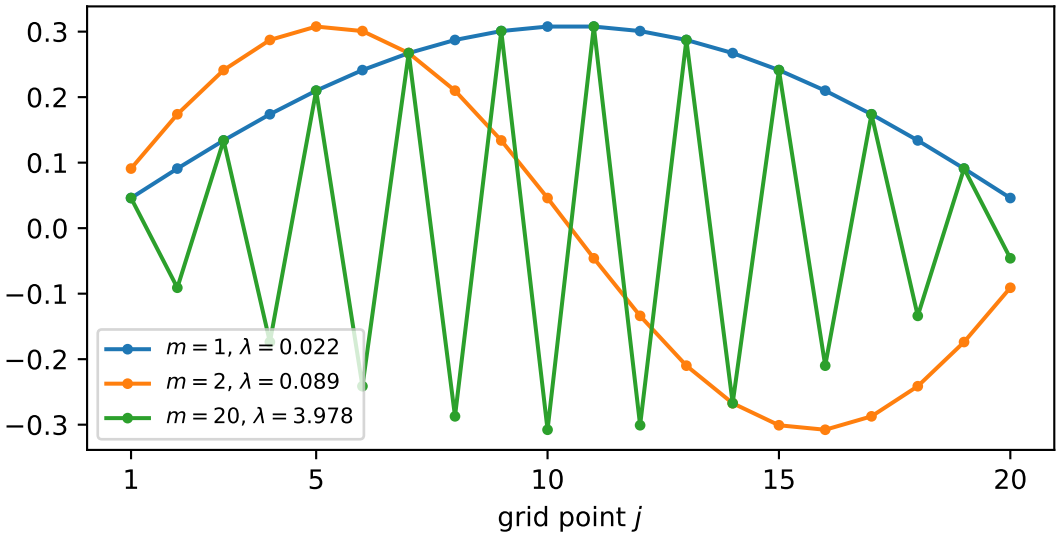


Figure 4.2: Eigenvectors of the model matrix for $N = 20$: the two longest waves and the shortest wave.

What the eigenvectors are good for. They solve Equation 4.1 without a solver. Write the right hand side in eigenvectors, $\vec{b} = \sum_m \beta_m \vec{v}_m$ with $\beta_m = \vec{v}_m^T \vec{b}$. Then

$$\vec{u} = \sum_{m=1}^{N} \frac{\beta_m}{\lambda_m} \vec{v}_m \tag{4.7}$$

solves $A\vec{u} = \vec{b}$, because A multiplies each $\vec{v}_m$ by λ_m and so restores β_m. The solution of the discrete Poisson problem is therefore a sum of the sine waves, and each wave of the right hand side is divided by its eigenvalue. This is the discrete form of solving the Poisson equation with a Fourier sine series: with $\vec{b} = h^2\vec{f}$ and, for long waves, $\lambda_m \approx \theta_m^2 = (kh)^2$, the factor is $h^2/\lambda_m \approx 1/k^2$, exactly the factor $1/\mu$ of the continuous problem.

Equation 4.7 also shows where errors are amplified. An error in the right hand side is divided by λ_m as well. In the longest wave, $m = 1$, it is divided by the smallest eigenvalue λ_1 and amplified most; in the shortest, $m = N$, by the largest λ_N. The ratio of the two, λ_N/λ_1, is the condition number of the next section.

4.2 Norms and the condition number

The question of this section: if the right hand side $\vec{b}$ contains an error, for example round-off or a measurement error, how large is the resulting error in the solution of $A\vec{x} = \vec{b}$? To answer it we need a measure for the size of vectors and of matrices (Section 4.2.1). With it we derive a bound for the error, the condition number (Section 4.2.2), and evaluate it for the model matrix (Section 4.2.3).

4.2.1 Norms

The size of a vector is measured by a norm, for example the Euclidean norm and the maximum norm,

$$\|\vec{x}\|_2 = \sqrt{\sum_j x_j^2}, \qquad \|\vec{x}\|_\infty = \max_j |x_j|.$$

Norms are what we mean by "the error" of a solution on a grid. The norm of a matrix is the largest factor by which it can stretch a vector,

$$\|A\| = \max_{\vec{x}\neq 0} \frac{\|A\vec{x}\|}{\|\vec{x}\|}, \qquad \text{so that} \qquad \|A\vec{x}\| \le \|A\|\,\|\vec{x}\|. \tag{4.8}$$

The 2-norm of a symmetric matrix. For the Euclidean norm, the matrix norm follows from the eigenvalues. We use the orthonormal eigenvectors of Section 4.1.4: write $\vec{x} = \sum_m c_m \vec{v}_m$, so that $\|\vec{x}\|_2^2 = \sum_m c_m^2$ and $A\vec{x} = \sum_m \lambda_m c_m \vec{v}_m$. Then

$$\|A\vec{x}\|_2^2 = \sum_m \lambda_m^2 c_m^2 \le \max_m \lambda_m^2 \sum_m c_m^2 = \max_m \lambda_m^2\, \|\vec{x}\|_2^2.$$

Each part of $\vec{x}$ is stretched by its own eigenvalue, so the stretch can be no larger than the largest $|\lambda_m|$. It reaches that value for $\vec{x}$ equal to the corresponding eigenvector. The inverse matrix has the same eigenvectors and the eigenvalues $1/\lambda_m$, because $A\vec{v}_m = \lambda_m \vec{v}_m$ gives $A^{-1}\vec{v}_m = \vec{v}_m/\lambda_m$; its largest stretch is $1/\min_m |\lambda_m|$.

Result

For a symmetric matrix,

$$\|A\|_2 = \max_m |\lambda_m|, \qquad \|A^{-1}\|_2 = \frac{1}{\min_m |\lambda_m|}. \tag{4.9}$$

For the model matrix all λ_m are positive, so $\|A\|_2 = \lambda_N$ and $\|A^{-1}\|_2 = 1/\lambda_1$, with λ_m from Equation 4.6.

4.2.2 The condition number

Let $\vec{b}$ be perturbed by $\delta\vec{b}$, and let $\delta\vec{x}$ be the resulting change of the solution. Subtracting $A\vec{x} = \vec{b}$ from $A(\vec{x} + \delta\vec{x}) = \vec{b} + \delta\vec{b}$ gives $A\,\delta\vec{x} = \delta\vec{b}$, hence

$$\delta\vec{x} = A^{-1}\delta\vec{b} \quad \Longrightarrow \quad \|\delta\vec{x}\| \leq \|A^{-1}\|\,\|\delta\vec{b}\|,$$

using Equation 4.8. The same inequality applied to $\vec{b} = A\vec{x}$ gives $\|\vec{b}\| \leq \|A\|\,\|\vec{x}\|$, or $1/\|\vec{x}\| \leq \|A\|/\|\vec{b}\|$. Multiplying the two inequalities,

$$\frac{\|\delta\vec{x}\|}{\|\vec{x}\|} \leq \underbrace{\|A\|\,\|A^{-1}\|}_{\kappa(A)}\,\frac{\|\delta\vec{b}\|}{\|\vec{b}\|}. \tag{4.10}$$

The factor $\kappa(A) = \|A\|\,\|A^{-1}\|$ is the **condition number** of the matrix. It bounds the amplification of relative errors, and is therefore the condition number of Chapter 1 for the problem "solve a linear system". The bound is reached: put the solution in the eigenvector with the largest eigenvalue, $\vec{x} = \vec{v}_{\max}$, so $\vec{b} = \lambda_{\max}\vec{x}$ is large, and the error in the one with the smallest, $\delta\vec{b} \propto \vec{v}_{\min}$, so $\delta\vec{x} = \delta\vec{b}/\lambda_{\min}$ is large.

Result

With Equation 4.9, a symmetric positive definite matrix has the condition number

$$\kappa(A) = \frac{\lambda_{\max}}{\lambda_{\min}}. \tag{4.11}$$

A relative error in $\vec{b}$ can grow by up to this factor in $\vec{x}$. Rule of thumb: with $\kappa = 10^p$ about p of the 16 digits of double precision are lost.

4.2.3 The condition number of the model matrix

We now insert the eigenvalues Equation 4.6 into Equation 4.11, for large N. The largest eigenvalue is λ_N. Its angle $\theta_N = N\pi/(N+1)$ is close to π, so $\cos\theta_N \approx -1$ and

$$\lambda_{\max} = \lambda_N = 2 - 2\cos\theta_N \approx 4.$$

The smallest eigenvalue is λ_1. Its angle $\theta_1 = \pi/(N+1)$ is small, and with the Taylor expansion $\cos\theta \approx 1 - \theta^2/2$,

$$\lambda_{\min} = \lambda_1 = 2 - 2\cos\theta_1 \approx 2 - 2\Big(1 - \frac{\theta_1^2}{2}\Big) = \theta_1^2 = \frac{\pi^2}{(N+1)^2}.$$

Result

The condition number of the model matrix is

$$\kappa(A) = \frac{\lambda_N}{\lambda_1} \approx \frac{4(N+1)^2}{\pi^2} \approx 0.4\, N^2. \tag{4.12}$$

It grows as N^2, that is as $1/h^2$. For $N = 1000$, $\kappa \approx 4 \times 10^5$: about 6 of 16 digits can be lost.

The computation confirms the formulas:

```
for N in [10, 100, 1000]:
    A = diags([-1.0, 2.0, -1.0], [-1, 0, 1], shape=(N, N))
    eigenvalues = np.linalg.eigvalsh(A.toarray())
    m = np.arange(1, N + 1)
    exact = 2 - 2 * np.cos(m * np.pi / (N + 1))
    print(f"N = {N:4d}: max. deviation from the formula {abs(eigenvalues -
    ↪  exact).max():.1e},"
          f" condition number {eigenvalues[-1] / eigenvalues[0]:9.1f},"
          f" estimate {4 * (N + 1)**2 / np.pi**2:9.1f},"
          f" non-zeros {A.nnz} of {N * N}")
```

```
N =   10: max. deviation from the formula 4.4e-16, condition number      48.4, estimate
N =  100: max. deviation from the formula 2.7e-15, condition number    4133.6, estimate    4
N = 1000: max. deviation from the formula 3.6e-15, condition number  406095.0, estimate  406
```

Refining the grid makes the discretization more accurate and the linear system harder to solve. This conflict is the reason why solvers for linear systems take up two units of this lecture (Chapter 10, Chapter 11).

4.3 Fourier analysis

The eigenvectors of the model matrix are sine waves (Equation 4.6). This is no accident, and this section turns it into a general tool. We want to answer three questions:

1. How do we split any grid function into waves? (Section 4.3.3)
2. Which waves can a grid represent at all? (Section 4.3.4)
3. What does a difference formula do to each wave? The answer connects back to the eigenvalues λ_m. (Section 4.3.5, Section 4.3.6)

The tool for all three is the complex exponential, which we introduce first (Section 4.3.1).

4.3.1 The complex exponential

Fourier analysis is written with the complex exponential $e^{i\varphi}$. It looks more difficult than sine and cosine, but it is only a compact way to handle both at once. Everything follows from one rule, $i^2 = -1$.

Euler's formula. A complex number $z = a + ib$ is a point in the plane, with the real part a on the horizontal and the imaginary part b on the vertical axis. The complex exponential is the point on the unit circle at angle φ (Figure 4.3):

$$e^{i\varphi} = \cos\varphi + i\sin\varphi. \tag{4.13}$$

To see why, insert $x = i\varphi$ into the Taylor series $e^x = 1 + x + x^2/2! + x^3/3! + \dots$ and use $i^2 = -1$, $i^3 = -i$, $i^4 = 1$:

$$\begin{aligned} e^{i\varphi} &= 1 + i\varphi - \frac{\varphi^2}{2!} - i\frac{\varphi^3}{3!} + \frac{\varphi^4}{4!} + \dots \\ &= \Big(1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \dots\Big) + i\Big(\varphi - \frac{\varphi^3}{3!} + \dots\Big) = \cos\varphi + i\sin\varphi. \end{aligned}$$

The even powers are real and form the series of the cosine, the odd powers are imaginary and form the series of the sine. Replacing φ by $-\varphi$ changes only the sign of the sine, $e^{-i\varphi} = \cos\varphi - i\sin\varphi$. Adding and subtracting the two gives

$$\cos\varphi = \frac{e^{i\varphi} + e^{-i\varphi}}{2}, \qquad \sin\varphi = \frac{e^{i\varphi} - e^{-i\varphi}}{2i}\ : \tag{4.14}$$

a real wave is the sum of two rotations in opposite directions.

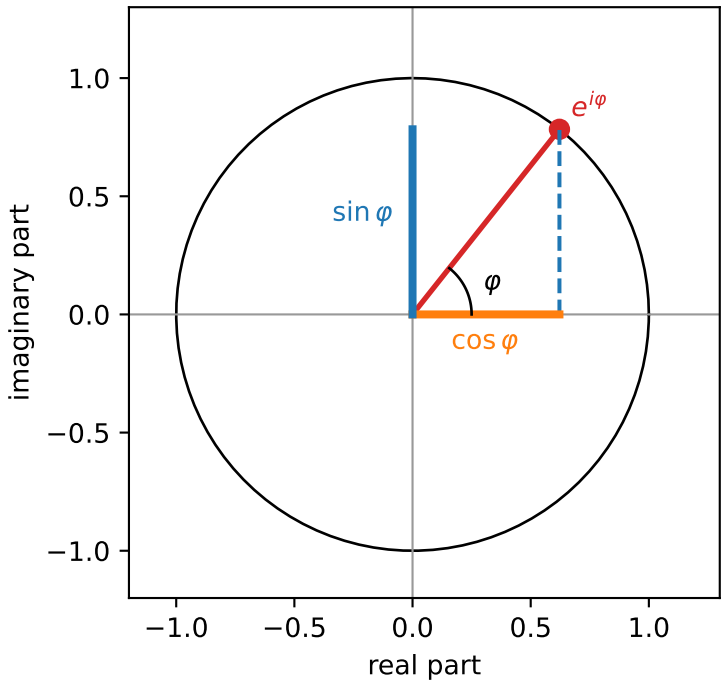


Figure 4.3: The complex exponential $e^{i\varphi}$ is the point on the unit circle at angle φ. Its real part is $\cos\varphi$, its imaginary part $\sin\varphi$.

Why it makes life easier. Three rules replace the whole collection of trigonometric identities:

- *Multiplying adds angles*: $e^{ia}\,e^{ib} = e^{i(a+b)}$. Written out with Equation 4.13, the real and imaginary parts of this one line are the addition theorems for $\cos(a+b)$ and $\sin(a+b)$.
- *A shift is a multiplication*: $e^{ik(x+\Delta x)} = e^{ikx}\,e^{ik\Delta x}$. Moving a wave by one grid point multiplies it by a fixed factor.
- *A derivative is a multiplication*: $\frac{\mathrm{d}}{\mathrm{d}x}e^{ikx} = ik\,e^{ikx}$ and $\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{ikx} = -k^2\,e^{ikx}$.

The last two rules are why waves are the natural test functions for difference formulas (Section 4.3.5): a difference formula is a combination of shifts, so it also just multiplies the wave by a factor.

The model matrix once more. As an example, we repeat step 2 of the eigenvalue computation in Section 4.1.4. Instead of the sine we try $v_j = e^{ij\theta} = \cos(j\theta) + i\sin(j\theta)$, which carries the cosine and the sine at the same time. This is a trick for the algebra, not a new eigenvector: the sine is its imaginary part. Row j of $A\vec{v}$ is

$$(A\vec{v})_j = -e^{i(j-1)\theta} + 2e^{ij\theta} - e^{i(j+1)\theta} = e^{ij\theta}\bigl(2 - e^{-i\theta} - e^{i\theta}\bigr) = (2 - 2\cos\theta)\,v_j,$$

using the shift rule and Equation 4.14. The eigenvalue appears without any addition theorem.

Why does this tell us anything about the sine? The entries of A are real, so A acts on the real and the imaginary part separately: $A\vec{v} = A\vec{c} + i\,A\vec{s}$ with $c_j = \cos(j\theta)$ and $s_j = \sin(j\theta)$. The right hand side is $(2 - 2\cos\theta)\,(\vec{c} + i\vec{s})$, with a real factor. Comparing real and imaginary parts gives $(A\vec{c})_j = (2 - 2\cos\theta)\,c_j$ and $(A\vec{s})_j = (2 - 2\cos\theta)\,s_j$: one computation with the complex exponential does the work of two, one for the cosine and one for the sine.

The boundary conditions come in at the end. The unknowns sit at the interior nodes $j = 1, \ldots, N$; the boundary nodes $j = 0$ (left) and $j = N + 1$ (right) carry the Dirichlet values $u_0 = u_{N+1} = 0$, so an eigenvector must vanish there. $e^{ij\theta}$ alone does not, its modulus is one at every node. But $e^{-ij\theta}$ has the same eigenvalue, because $\cos(-\theta) = \cos\theta$, and so has every combination of the two. By Equation 4.14 the combination $\sin(j\theta) = (e^{ij\theta} - e^{-ij\theta})/(2i)$ is zero at the left boundary node, $j = 0$; the cosine is not, since $\cos 0 = 1$. At the right boundary node, $j = N + 1$, we need $\sin\big((N + 1)\theta\big) = 0$, that is $(N + 1)\,\theta = m\pi$ or $\theta = m\pi/(N + 1)$, as in Equation 4.6. Remember that θ is not a node but the phase step from one node to the next: the right boundary only admits the waves for which a whole number m of half waves fits between the two boundary nodes.

Result

$$e^{i\varphi} = \cos\varphi + i\sin\varphi, \qquad \cos\varphi = \frac{e^{i\varphi} + e^{-i\varphi}}{2}, \qquad \sin\varphi = \frac{e^{i\varphi} - e^{-i\varphi}}{2i}.$$

With the complex exponential, a product of waves adds the angles, and a shift or a derivative multiplies the wave by a factor. Calculations with waves then reduce to calculations with these factors. For the model matrix the factor is the eigenvalue $2 - 2\cos\theta$, as in Equation 4.6.

4.3.2 Waves on a grid

The eigenvectors of Section 4.1.4 were sines on a domain with Dirichlet boundaries. From here on we use a **periodic** domain, as on a latitude circle, because waves are simplest there. Two things change compared with Section 4.1.4, and it helps to keep them apart:

	model matrix (Section 4.1.4)	periodic grid (this section)
grid points	$x_j = jh$, $j = 1, \ldots, N$, plus two boundary nodes	$x_j = j\Delta x$, $j = 0, \ldots, N - 1$, no boundary
waves	sines, m **half** waves, $k = m\pi/L$	cosines and sines, ℓ **whole** waves ($= 2\ell$ half waves), $k = 2\pi\ell/L$
index	$m = 1, \ldots, N$	$\ell = 0, 1, \ldots, N/2$

The question. Think of N values $u_0, \ldots, u_{N-1}$ measured at equally spaced points around a latitude circle, for example the surface temperature at N stations. Plotted, they look irregular (top panel of Figure 4.4). Fourier analysis asks: *which waves is this signal made of, and how strong is each one?* A wave here is a pattern that repeats ℓ times around the circle: $\ell = 1$ is one warm and one cold half, $\ell = 2$ two of each, and so on. $\ell = 0$ is the constant part, the mean. The answer, one amplitude for each ℓ, is the **spectrum** of the signal.

Waves on the grid points. The domain has length $L = N\Delta x$ and the grid points are $x_j = j\Delta x = jL/N$. A wave with ℓ crests in the domain has the wavenumber $k_\ell = 2\pi\ell/L$, and at the grid points

$$k_\ell x_j = \frac{2\pi\ell}{L} \cdot \frac{jL}{N} = \frac{2\pi\ell j}{N}.$$

A wave needs two numbers, its amplitude and its position. We therefore use a cosine and a sine for each ℓ, with coefficients a_ℓ and b_ℓ: a cosine has its crest at $x = 0$, a sine a quarter wavelength further on, and the mix of the two places the crest anywhere. The signal is

$$u_j = a_0 + \sum_{\ell=1}^{N/2} \left[a_\ell \cos\frac{2\pi\ell j}{N} + b_\ell \sin\frac{2\pi\ell j}{N} \right]. \tag{4.15}$$

Why only up to $\ell = N/2$? N grid values cannot determine more than N coefficients, and $\ell = N/2$ is already the shortest wave, which changes sign from point to point. Shorter waves cannot be told apart from longer ones; this is aliasing, Section 4.3.4.

4.3.3 The discrete Fourier transform

The goal: compute the coefficients a_ℓ and b_ℓ from the data u_j.

The idea: multiply and sum. To find out how much of the wave with n crests the signal contains, we multiply the data point by point with that wave, the **test wave**, and add up:

$$\sum_{j=0}^{N-1} u_j \cos\frac{2\pi n j}{N}.$$

Figure 4.4 shows what happens. The signal there contains a cosine with $n = 2$ crests (amplitude 1) and a sine with 5 crests (amplitude 0.6).

- Multiplied with the matching test wave, $n = 2$, the products are mostly positive: where the wave is positive, the test wave is positive too, and where it is negative, so is the test wave. The products add up, here to 12.

- Multiplied with a test wave that does not match, $n = 3$, the products are positive at some points and negative at others, and they cancel. The sum is zero.

So the sum picks out exactly the part of the signal that has the shape of the test wave, and ignores all other waves. This is the whole idea of the Fourier transform.

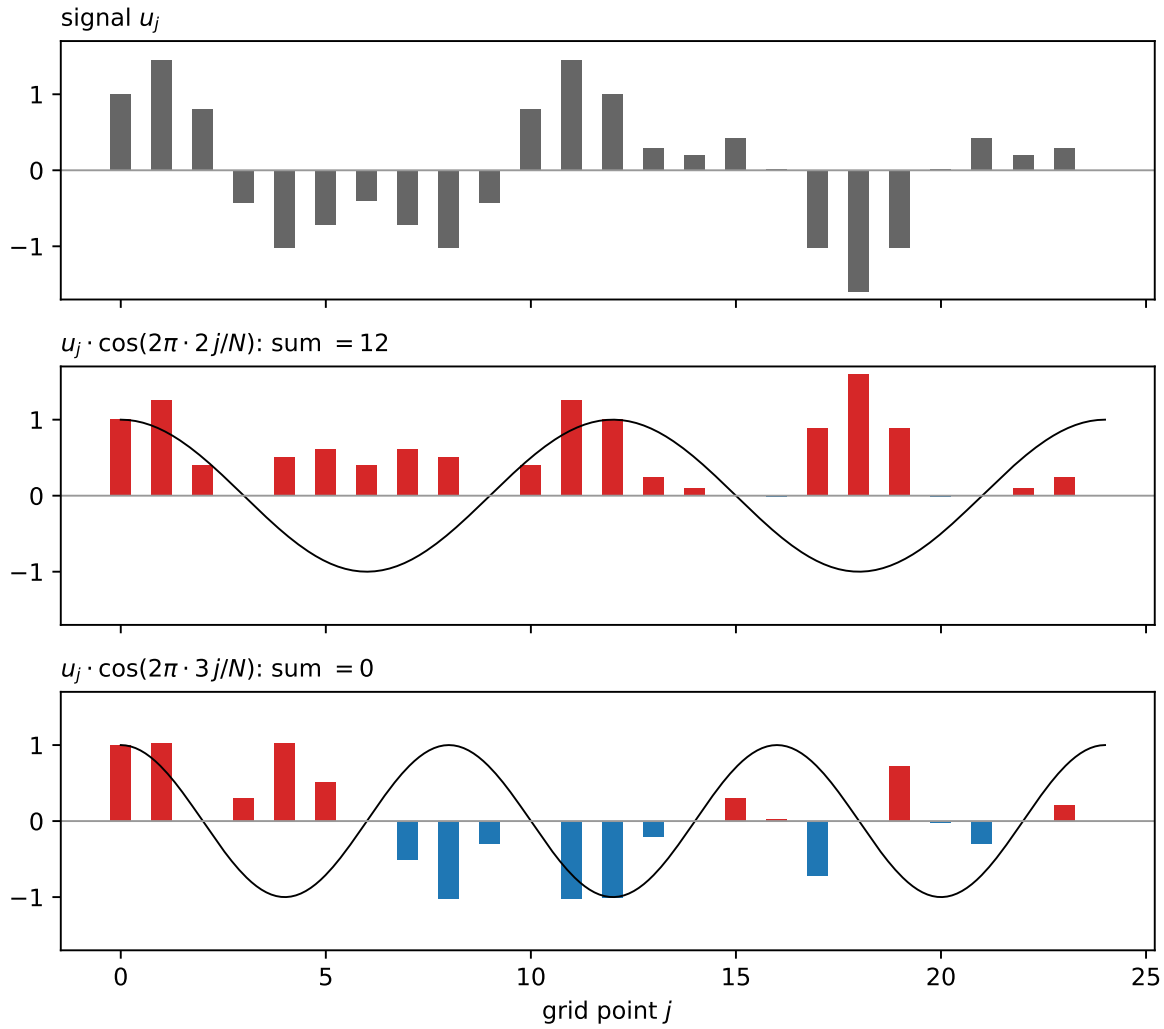


Figure 4.4: The idea of the Fourier transform. Top: a signal at $N = 24$ points, $\cos(2\pi \cdot 2j/N) + 0.6\sin(2\pi \cdot 5j/N)$. Middle: multiplied with the matching test wave $\cos(2\pi \cdot 2j/N)$ (line), the products (bars) are mostly positive and add up. Bottom: multiplied with $\cos(2\pi \cdot 3j/N)$, which is not in the signal, they cancel.

From the sum to the coefficient. Put the signal Equation 4.15 into the sum. All waves except the matching one drop out (why, we show next), and what is left is the matching cosine

times itself:

$$\sum_{j=0}^{N-1} u_j \cos\frac{2\pi nj}{N} = a_n \sum_{j=0}^{N-1} \cos^2\frac{2\pi nj}{N} = a_n \frac{N}{2},$$

because $\cos^2\alpha = \frac{1}{2}(1+\cos 2\alpha)$: on average the square is $\frac{1}{2}$, so N terms give $N/2$ (for $0 < n < N/2$).

Let us check this with Figure 4.4. There $N = 24$ and the test wave has $n = 2$, so the formula predicts the sum $a_2 \cdot 24/2 = 12\,a_2$. The figure shows that the sum is 12. Hence $12\,a_2 = 12$, or $a_2 = 1$: exactly the amplitude of the cosine we built into the signal. The method recovers it.

Solving for the coefficient, and the same with a sine as test wave,

$$a_n = \frac{2}{N}\sum_{j=0}^{N-1} u_j \cos\frac{2\pi nj}{N}, \qquad b_n = \frac{2}{N}\sum_{j=0}^{N-1} u_j \sin\frac{2\pi nj}{N}, \qquad 0 < n < \frac{N}{2}. \tag{4.16}$$

For $n = 0$ the test wave is the constant 1, and $a_0 = \frac{1}{N}\sum_j u_j$ is the mean. This is the **discrete Fourier transform** (DFT) in real form. It is the same idea as $\beta_m = \vec{v}_m^T\vec{b}$ in Equation 4.7: the waves are orthogonal vectors, and the coefficient of each is the product of the data with that wave.

Why the other waves cancel. The product of two waves with n and ℓ crests is, by the product rules for cosine and sine, a combination of waves with $n+\ell$ and $n-\ell$ crests. So everything reduces to one question: why is the sum of a wave over all grid points zero? Here the complex exponential helps. The values $e^{2\pi i\,pj/N}$, $j = 0, \ldots, N-1$, are points on the unit circle, p/N of a turn apart (Figure 4.5). For $p = 1$ they are the corners of a regular polygon, spread evenly around the circle. Added as arrows from the origin, they balance: the sum is zero. Its real part is the sum of the cosines and its imaginary part the sum of the sines, so both are zero. For $p = 3$ the points are visited in a different order, but they are still spread evenly, and the sum is again zero. Only if p is a multiple of N does every point land on 1, and the sum is N:

$$\sum_{j=0}^{N-1} e^{2\pi i\,pj/N} = \begin{cases} N & \text{if } p = 0, \pm N, \pm 2N, \ldots, \\ 0 & \text{otherwise.} \end{cases} \tag{4.17}$$

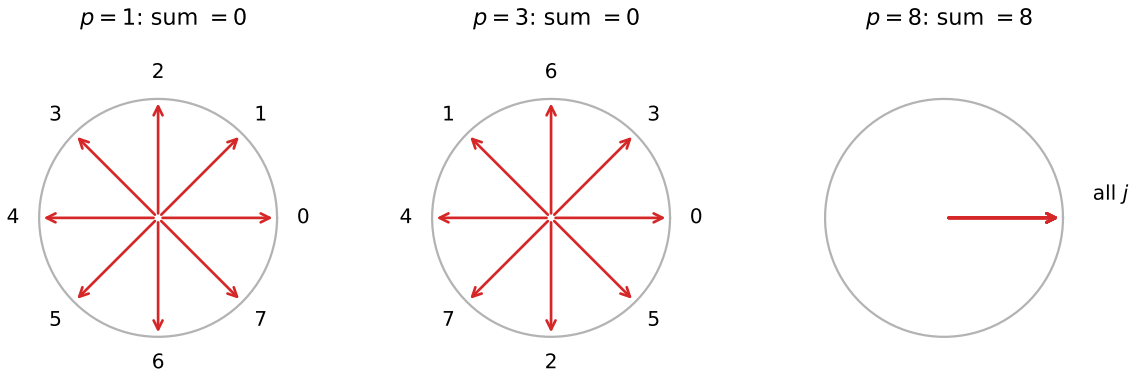


Figure 4.5: The points $e^{2\pi i\,pj/N}$ for $N = 8$, $j = 0, \ldots, 7$, numbered by j. For $p = 1$ and $p = 3$ they are spread evenly around the circle, and as arrows from the origin they add up to zero. For $p = N = 8$ every point is 1, and the sum is 8.

The arrows balance because turning the whole picture by one step, multiplying every point by $q = e^{2\pi ip/N}$, only renumbers the points and leaves the sum S unchanged: $qS = S$. With $q \neq 1$ this is possible only for $S = 0$. (Written out, this is the geometric series $\sum_j q^j = (1-q^N)/(1-q)$ with $q^N = 1$.)

The complex form. The two real sums of Equation 4.16 can be packed into one complex number, with $e^{-i\alpha} = \cos\alpha - i\sin\alpha$:

$$\hat{u}_n = \frac{1}{N}\sum_{j=0}^{N-1} u_j\, e^{-2\pi i\, nj/N} = \frac{1}{N}\sum_j u_j \cos\frac{2\pi nj}{N} - \frac{i}{N}\sum_j u_j \sin\frac{2\pi nj}{N} = \frac{a_n - i\,b_n}{2}. \tag{4.18}$$

This is the DFT as it is usually written, and as NumPy computes it. Nothing new happens in it: the real part holds the cosine coefficient, the imaginary part the sine coefficient. The amplitude of the wave, $\sqrt{a_n^2 + b_n^2}$, is therefore

$$\sqrt{a_n^2 + b_n^2} = 2\,|\hat{u}_n|,$$

and the angle of $\hat{u}_n$ in the complex plane gives the position of the crest. Why the factor 2? A real wave is the sum of two rotations, $e^{+i\alpha}$ and $e^{-i\alpha}$ (Equation 4.14), and $\hat{u}_n$ measures only one of them; the other half sits in $\hat{u}_{N-n}$, the wave turning the other way. Conversely, the signal is rebuilt from its amplitudes by

$$u_j = \sum_{\ell=0}^{N-1} \hat{u}_\ell\, e^{2\pi i\,\ell j/N}, \tag{4.19}$$

the inverse DFT, which is Equation 4.15 in complex form.

A worked example. Four values, $N = 4$, $u = (1, 2, 1, 0)$. The test waves with $n = 1$ are $\cos(\pi j/2) = (1, 0, -1, 0)$ and $\sin(\pi j/2) = (0, 1, 0, -1)$. Multiply and sum:

$$\begin{aligned} a_0 &= \tfrac{1}{4}(1 + 2 + 1 + 0) = 1, \\ a_1 &= \tfrac{2}{4}(1 \cdot 1 + 2 \cdot 0 + 1 \cdot (-1) + 0 \cdot 0) = 0, \\ b_1 &= \tfrac{2}{4}(1 \cdot 0 + 2 \cdot 1 + 1 \cdot 0 + 0 \cdot (-1)) = 1. \end{aligned}$$

So $u_j = 1 + \sin(\pi j/2)$: the mean 1 plus one sine wave of amplitude 1. Check: $j = 0, 1, 2, 3$ gives $1, 2, 1, 0$. In complex form, $\hat{u}_1 = (a_1 - ib_1)/2 = -0.5i$, and $2|\hat{u}_1| = 1$ is the amplitude. NumPy gives the same:

```
u = np.array([1.0, 2.0, 1.0, 0.0])
N = len(u)
j = np.arange(N)

# by hand: multiply with the test waves and sum
a1 = 2 / N * np.sum(u * np.cos(2 * np.pi * 1 * j / N))
b1 = 2 / N * np.sum(u * np.sin(2 * np.pi * 1 * j / N))
print("mean", u.mean(), " a1", round(a1, 12), " b1", round(b1, 12))

# with the FFT: rfft returns N * u_hat for n = 0, ..., N/2
u_hat = np.fft.rfft(u) / N
print("u_hat", u_hat, " amplitude of n = 1:", 2 * abs(u_hat[1]))
```

```
mean 1.0  a1 0.0  b1 1.0
u_hat [1.+0.j  0.-0.5j 0.+0.j ]  amplitude of n = 1: 1.0
```

In practice. Computed directly, Equation 4.18 costs N operations for each of the N amplitudes, N^2 in total. The **fast Fourier transform** (FFT) obtains the same numbers with $O(N \log N)$ operations: for $N = 10^6$ about 2×10^7 instead of 10^{12}. In NumPy, `np.fft.rfft(u)` returns $N\hat{u}_n$ for $n = 0, \ldots, N/2$, without the factor $1/N$. The amplitude spectrum of a real signal is therefore `2 * abs(np.fft.rfft(u)) / N` (and the mean is `abs(np.fft.rfft(u)[0]) / N`, without the 2). Figure 4.6 applies this to a noisy signal.

```
N, L = 128, 1.0
x = np.arange(N) * L / N
rng = np.random.default_rng(1)
signal = np.sin(2 * np.pi * 3 * x) + 0.5 * np.cos(2 * np.pi * 10 * x) + 0.2 *
↪  rng.standard_normal(N)

amplitude = 2 * abs(np.fft.rfft(signal)) / N
```

```
ell = np.fft.rfftfreq(N, d=1 / N)       # number of waves in the domain

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(9, 3))
ax1.plot(x, signal)
ax1.set_xlabel("$x$")
ax1.set_ylabel("$u$")
ax2.stem(ell, amplitude, basefmt=" ", markerfmt=".")
ax2.set_xlim(0, 30)
ax2.set_xlabel(r"number of waves in the domain, $\ell$")
ax2.set_ylabel("amplitude")
plt.tight_layout()
plt.show()
```

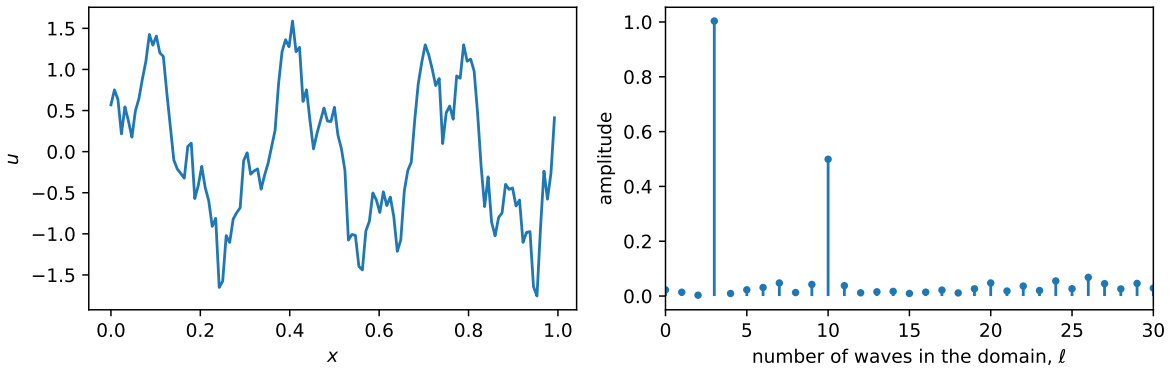


Figure 4.6: A signal composed of two waves and noise (left) and its amplitude spectrum (right). The two waves stand out as peaks with their amplitudes 1 and 0.5; the noise is spread thinly over all ℓ.

Result

To find how much of a wave is in a signal, multiply the signal with that wave and sum (Equation 4.16). All other waves cancel, because a wave summed over all grid points gives zero (Equation 4.17). In complex form, $\hat{u}_n = \frac{1}{N}\sum_j u_j e^{-2\pi i n j/N}$ (Equation 4.18) holds the cosine and the sine coefficient at once, and the amplitude of the wave is $2|\hat{u}_n|$. In NumPy: `2 * abs(np.fft.rfft(u)) / N`.

4.3.4 The shortest wave and aliasing

The goal: find the shortest wave a grid can represent, and see what happens to shorter waves.

The shortest wave that a grid can represent has one crest and one trough on two neighbouring points. Its wavelength is $2\Delta x$ and its wavenumber is the **Nyquist wavenumber** $k_{\max} = \pi/\Delta x$.

What happens to a wave with a larger wavenumber? Consider the wavenumbers k and $k + 2\pi/\Delta x$. At the grid points $x_j = j\Delta x$

$$e^{i(k+2\pi/\Delta x)\,x_j} = e^{ikx_j}\,e^{2\pi i\,j} = e^{ikx_j},$$

because $e^{2\pi ij} = 1$ for every integer j. The two waves have the same values at all grid points. On the grid they cannot be distinguished. A wave that is shorter than $2\Delta x$ therefore does not disappear when it is sampled: it is mistaken for a longer wave with a wavenumber in the range $|k| \le k_{\max}$. This is **aliasing**.

In Figure 4.7 a wave with wavelength $\frac{4}{3}\Delta x$, that is $k = \frac{3}{2}\pi/\Delta x$, appears as the wave with $k - 2\pi/\Delta x = -\frac{1}{2}\pi/\Delta x$, which has the wavelength $4\Delta x$.

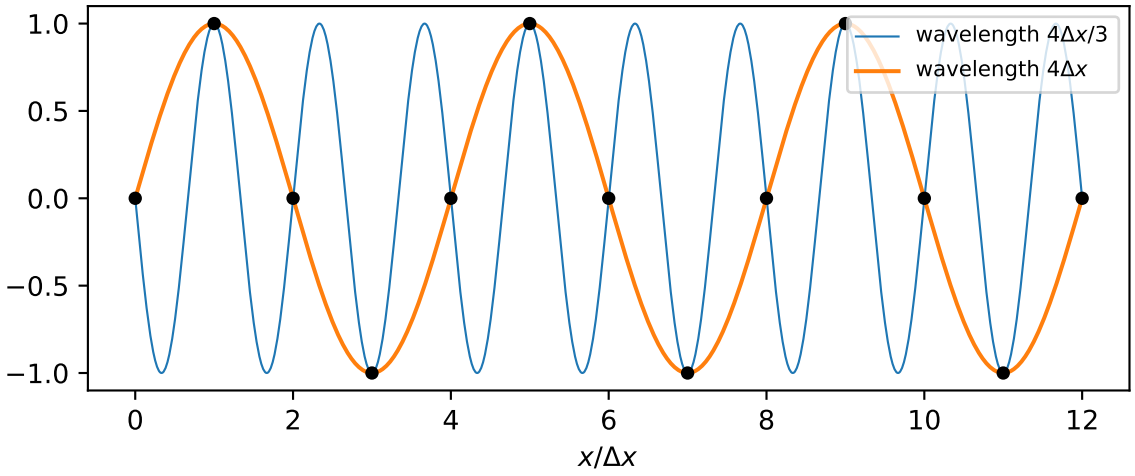


Figure 4.7: Aliasing. At the grid points (dots) a wave with wavelength $4\Delta x/3$ cannot be distinguished from a wave with wavelength $4\Delta x$.

Aliasing matters in nonlinear models. The product of two waves is

$$e^{ik_1x}\,e^{ik_2x} = e^{i(k_1+k_2)x},$$

a wave with the wavenumber $k_1 + k_2$. A nonlinear term such as $u\,\partial u/\partial x$ therefore creates shorter waves from longer ones. If $k_1 + k_2$ exceeds $k_{\max}$, the grid misrepresents the product as

a longer wave, and energy appears at scales where it does not belong. This can make a model unstable (Chapter 17).

Result

The shortest wave on a grid has the wavelength $2\Delta x$, the wavenumber $k_{\max} = \pi/\Delta x$. A wave with k and one with $k + 2\pi/\Delta x$ have the same grid values. Shorter waves therefore do not vanish but appear as longer ones (aliasing), for example when a nonlinear term creates them.

4.3.5 Why waves are the natural test for a numerical method

The goal: find out how accurately a difference formula treats a wave of a given length. For a linear equation with constant coefficients each wave evolves independently of the others. It therefore suffices to understand what a numerical method does to a single wave e^{ikx}, and that is easy to compute.

The exact derivative of a wave is

$$\frac{\mathrm{d}}{\mathrm{d}x} e^{ikx} = ik\, e^{ikx} :$$

the wave is multiplied by ik. For the central difference Equation 3.5 we factor out e^{ikx} and use $e^{i\alpha} - e^{-i\alpha} = 2i \sin\alpha$:

$$\frac{e^{ik(x+\Delta x)} - e^{ik(x-\Delta x)}}{2\Delta x} = \frac{e^{ik\Delta x} - e^{-ik\Delta x}}{2\Delta x}\, e^{ikx} = i\, \frac{\sin(k\Delta x)}{\Delta x}\, e^{ikx}.$$

The wave is again multiplied by i times a real number. The numerical derivative is that of a wave with the **modified wavenumber**

$$k_{\text{eff}} = \frac{\sin(k\Delta x)}{\Delta x}. \tag{4.20}$$

For $k\Delta x \ll 1$ the expansion $\sin(k\Delta x) \approx k\Delta x - (k\Delta x)^3/6$ gives $k_{\text{eff}} \approx k\,[1 - (k\Delta x)^2/6]$: the relative error is $(k\Delta x)^2/6$. This is the truncation error of Equation 3.5, now expressed for each wave separately.

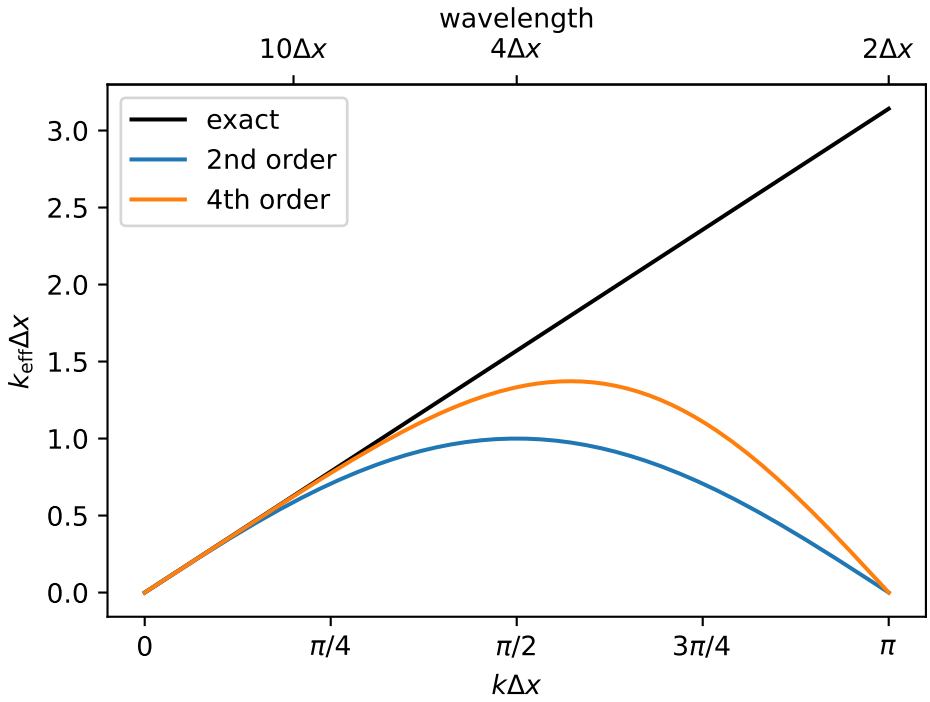


Figure 4.8: Modified wavenumber of the central differences of second and fourth order. The exact derivative corresponds to the diagonal.

Figure 4.8 shows what the statement "second order" hides. For long waves the approximation is excellent. For the $4\Delta x$ wave, $k\Delta x = \pi/2$, the derivative is too small by the factor $\sin(\pi/2)/(\pi/2) = 0.64$. For the $2\Delta x$ wave it is zero: the central difference does not see the shortest wave at all. A wave needs about ten grid points per wavelength to be treated accurately by a scheme of second order; then $k\Delta x = \pi/5$ and the error is 6 %.

This has a consequence that every user of model output should know. The smallest features that a model represents correctly are several times larger than its grid spacing. The **effective resolution** of a weather model with a grid spacing of 2 km is closer to 10 km.

Result

The central difference differentiates the wave e^{ikx} as if it had the modified wavenumber $k_{\mathrm{eff}} = \sin(k\Delta x)/\Delta x$ (Equation 4.20). The relative error is $(k\Delta x)^2/6$ for long waves, 36 % for the $4\Delta x$ wave, and 100 % for the $2\Delta x$ wave. About ten grid points per wavelength are needed for an error of a few percent.

4.3.6 The second difference and the model matrix

The goal: apply the same analysis to the second difference, and show that it gives back the eigenvalues of the model matrix. The calculation for the second difference Equation 3.6, with $e^{i\alpha} + e^{-i\alpha} = 2\cos\alpha$, gives

$$\frac{e^{ik(x+\Delta x)} - 2e^{ikx} + e^{ik(x-\Delta x)}}{\Delta x^2} = -\frac{2 - 2\cos(k\Delta x)}{\Delta x^2}\, e^{ikx} = -\frac{4}{\Delta x^2}\sin^2\frac{k\Delta x}{2}\, e^{ikx}. \tag{4.21}$$

The exact second derivative multiplies the wave by $-k^2$. The second difference multiplies it by $-\frac{4}{\Delta x^2}\sin^2\frac{k\Delta x}{2}$, which for long waves, $\sin\alpha \approx \alpha$, is again about $-k^2$.

Now compare with the model matrix. Take $\Delta x = h$ and the wavenumber $k_m = m\pi/L$ of eigenvector m, so that $k_m\Delta x = \theta_m$. Then the factor in Equation 4.21 is

$$-\frac{4}{h^2}\sin^2\frac{\theta_m}{2} = -\frac{\lambda_m}{h^2},$$

with λ_m from Equation 4.6. The second difference is $-A/h^2$, with the minus sign and the h^2 that we moved to the right hand side in Section 4.1.1. So the factor by which the second difference multiplies a wave is, up to $-1/h^2$, the eigenvalue of A.

Result: two views of one fact

Waves are the eigenvectors of difference operators with constant coefficients, and the factor by which a wave is multiplied is the eigenvalue. For the second difference,

$$\text{wave analysis: } -\frac{4}{\Delta x^2}\sin^2\frac{k\Delta x}{2} \quad\Longleftrightarrow\quad \text{matrix: } -\frac{\lambda_m}{h^2}, \quad \lambda_m = 4\sin^2\frac{\theta_m}{2},$$

with $k\Delta x = \theta_m$. The wave analysis gives the eigenvalues without setting up the matrix.

Applied to a complete scheme in space and time, this analysis shows whether a wave is damped, amplified or moved at the wrong speed. This is the von Neumann stability analysis (Chapter 12, Chapter 13).

4.4 Summary

- **The model matrix.** Discretizing $-u'' = f$ with Dirichlet values gives $A\vec{x} = \vec{b}$ with $A = \text{tridiag}(-1, 2, -1)$ and $\vec{b} = h^2\vec{f}$ (Equation 4.2). Each row is the stencil of the difference formula.
- **Sparse, symmetric, positive definite.** A has $3N - 2$ non-zeros. Its quadratic form is a sum of squares (Equation 4.3), so A is positive definite and $A\vec{x} = \vec{b}$ has exactly one solution.

- **Eigenvalues.** For $m = 1, \ldots, N$ (the number of half waves), $\lambda_m = 4\sin^2\left(m\pi/(2(N+1))\right)$ with sine waves as eigenvectors (Equation 4.6). They lie between $\lambda_1 \approx \pi^2/(N+1)^2$ and $\lambda_N \approx 4$.
- **Norm and condition number.** For a symmetric matrix $\|A\|_2 = \max|\lambda_m|$ (Equation 4.9). The condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ (Equation 4.11) bounds the amplification of relative errors. For the model matrix $\kappa \approx 4(N+1)^2/\pi^2$ grows as $1/h^2$ (Equation 4.12).
- **Waves on a grid.** The DFT (Equation 4.18) splits N grid values into N waves. The shortest wave is $2\Delta x$; shorter waves are aliased.
- **Difference formulas and waves.** A difference formula multiplies a wave by a factor. For the first derivative this gives the modified wavenumber (Equation 4.20): about ten points per wavelength are needed. For the second difference the factor is $-\lambda_m/h^2$: wave analysis and eigenvalues are two views of one fact.

4.5 Further reading

Durran (2010), chapter 3, for the wave view of finite differences; Shewchuk (1994) for eigenvalues, quadratic forms and the condition number.

Part II

Time integration of ODEs

5 One-step methods

Learning goals

After this unit you can

- read the qualitative behaviour of an ODE from its direction field,
- derive the forward and backward Euler methods in three ways and implement them,
- derive the local truncation error of a method and explain how it determines the global error,
- derive the order conditions of two-stage Runge–Kutta methods and use the classical method of fourth order,
- compare methods by the accuracy they reach for a given amount of work,
- explain how adaptive methods choose the time step.

Every prognostic model steps forward in time. After the spatial derivatives of a PDE have been discretized, what remains is a large system of ordinary differential equations in time. The methods of this unit are therefore the time stepping schemes of weather, ocean and groundwater models, studied here on problems small enough to be understood completely.

5.1 The initial value problem

We consider the initial value problem

$$\frac{\mathrm{d}u}{\mathrm{d}t} = f(t, u), \qquad u(t_0) = u_0. \tag{5.1}$$

The function f is the **tendency**: it gives the rate of change of the state for any time and any state. In much of the literature on ODEs the unknown is called y and the step h; we keep u and Δt.

The problem is well-posed (Chapter 2) if f is continuous and does not change too rapidly with u. Precisely, f must satisfy a **Lipschitz condition**: there is a constant L such that

$$|f(t, a) - f(t, b)| \le L\,|a - b| \tag{5.2}$$

for all states a, b in the region of interest. If f is differentiable, L is the largest value of $|\partial f/\partial u|$. The constant L has the unit of one over time: $1/L$ is the shortest time scale on which the tendency responds to a change of the state. It reappears in the error analysis below.

5.1.1 Direction fields

Before solving Equation 5.1 it is useful to understand how its solutions behave. At every point of the (t, u) plane the equation prescribes the slope of the solution passing through it, namely $f(t, u)$. Drawing a short line segment with this slope at many points gives the **direction field**. Solutions are the curves that follow the segments.

As an example consider

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{3 - u}{2}. \tag{5.3}$$

```
import numpy as np
import matplotlib.pyplot as plt

f = lambda t, u: (3 - u) / 2

t, u = np.meshgrid(np.linspace(0, 8, 21), np.linspace(-1, 6, 19))
slope = f(t, u)
norm = np.sqrt(1 + slope**2)

fig, ax = plt.subplots(figsize=(6.5, 3.8))
ax.quiver(t, u, 1 / norm, slope / norm, angles="xy", headwidth=1, headlength=0,
          pivot="mid", color="0.5", scale=35)
t_fine = np.linspace(0, 8, 200)
for u0 in [-1, 1.5, 6]:
    ax.plot(t_fine, 3 + (u0 - 3) * np.exp(-t_fine / 2))
ax.set_xlabel("$t$")
ax.set_ylabel("$u$")
plt.show()
```

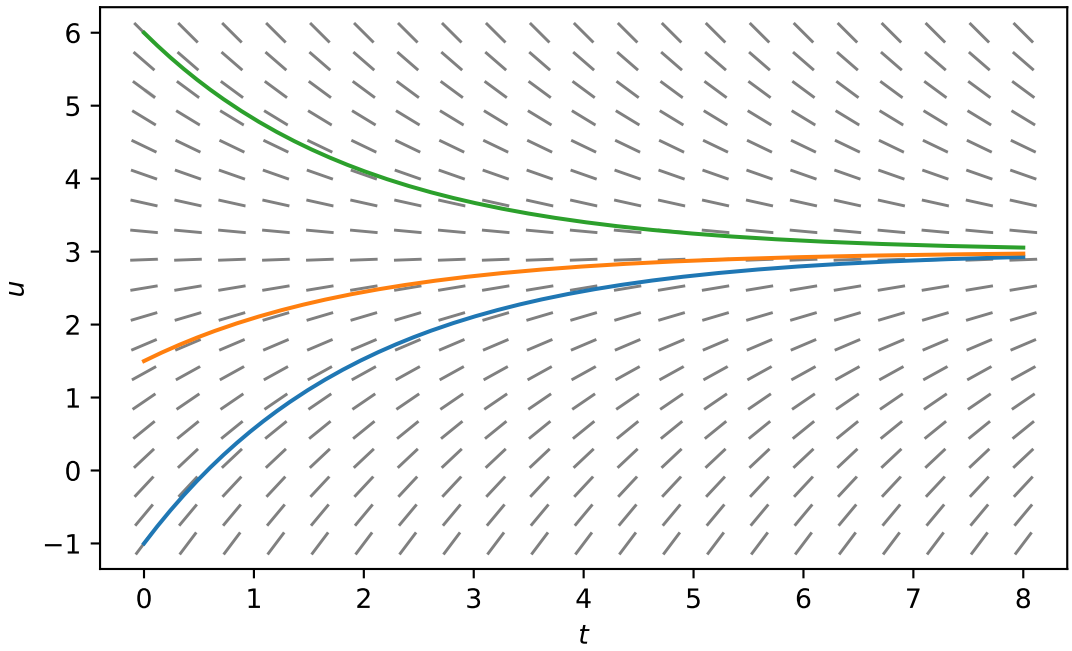


Figure 5.1: Direction field of $\mathrm{d}u/\mathrm{d}t = (3-u)/2$ with three exact solutions. All solutions approach the equilibrium $u = 3$.

Figure 5.1 shows, without any calculation, that $u = 3$ is an equilibrium, since the slope is zero there, and that it is stable: above it the slope is negative, below it positive.

5.1.2 An exact solution

Equation 5.3 is **separable**: all terms with u can be collected on one side and all terms with t on the other,

$$\frac{\mathrm{d}u}{3-u} = \frac{\mathrm{d}t}{2}.$$

Integrating both sides,

$$-\ln|3-u| = \frac{t}{2} + C \quad \Longrightarrow \quad 3-u = \tilde{C}\,e^{-t/2},$$

where $\tilde{C} = \pm e^{-C}$. The initial condition $u(0) = u_0$ gives $\tilde{C} = 3 - u_0$, and therefore

$$u(t) = 3 + (u_0 - 3)\,e^{-t/2}. \tag{5.4}$$

The distance from the equilibrium decays exponentially with the time scale 2. With $u - 3$ as the unknown this is the decay equation Equation 1.4 with $k = \frac{1}{2}$, the equation of radioactive decay with decay constant k.

Exact solutions exist only for special tendencies. For the rest we need numerical methods. Exact solutions remain indispensable, because they are what numerical methods are verified against.

5.2 The forward Euler method

We look for approximations u_n of the solution at the times $t_n = t_0 + n\,\Delta t$. The simplest method can be derived in three ways. Each of them leads to its own family of better methods.

5.2.1 First view: following the tangent

At t_0 we know the solution, u_0, and its slope, $f(t_0, u_0)$. The tangent to the solution at this point is

$$u = u_0 + f(t_0, u_0)\,(t - t_0).$$

If t_1 is close to t_0, the tangent stays close to the solution, and we take its value at t_1 as the approximation: $u_1 = u_0 + \Delta t\, f(t_0, u_0)$. At t_1 we do not know the solution, but we have u_1, and we repeat the step with the slope $f(t_1, u_1)$. In general

$$u_{n+1} = u_n + \Delta t\, f(t_n, u_n). \tag{5.5}$$

This is the **forward Euler method**. In the direction field it moves along one line segment per step (Figure 5.2).

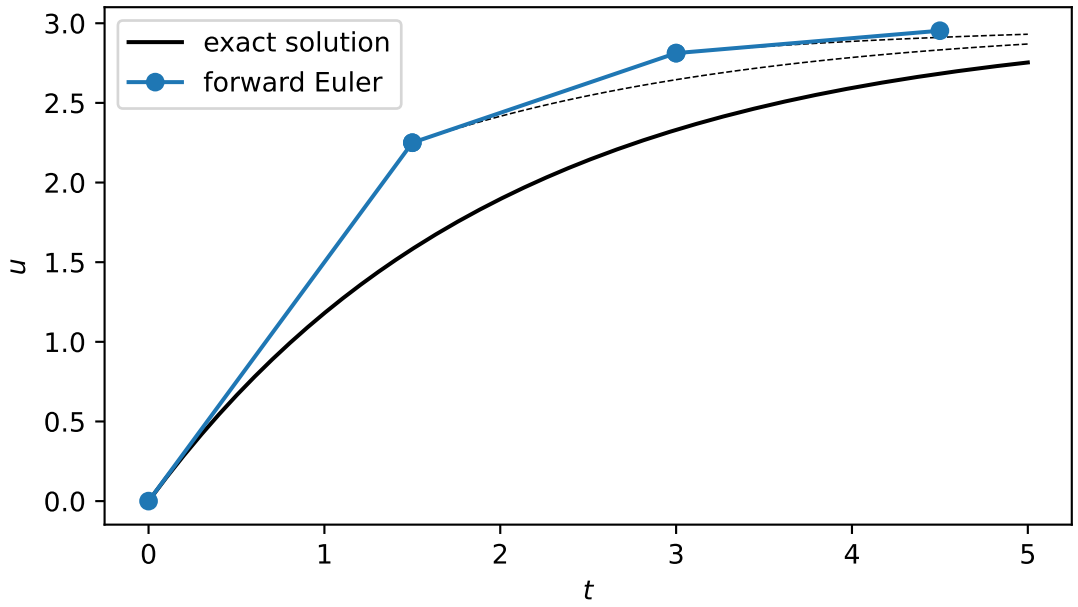


Figure 5.2: Three steps of the forward Euler method for $\mathrm{d}u/\mathrm{d}t = (3-u)/2$ with $\Delta t = 1.5$. Each step follows the tangent of the solution through the current point (dashed), not the solution that started at u_0.

5.2.2 Second view: replacing the derivative

We replace the derivative in Equation 5.1 at time t_n by the forward difference Equation 3.3,

$$\frac{u_{n+1} - u_n}{\Delta t} = f(t_n, u_n),$$

and solve for u_{n+1}. This gives Equation 5.5 again, and explains the name. It also tells us the accuracy to expect: the forward difference is of first order.

5.2.3 Third view: integrating the tendency

Integrating Equation 5.1 from t_n to t_{n+1} gives, without any approximation,

$$u(t_{n+1}) = u(t_n) + \int_{t_n}^{t_{n+1}} f\big(t, u(t)\big)\,\mathrm{d}t. \tag{5.6}$$

The integral cannot be evaluated, because the integrand contains the unknown solution. The forward Euler method approximates it by a rectangle whose height is the integrand at the left

end, $\Delta t\, f(t_n, u_n)$. Better time stepping schemes are better quadrature rules for this integral (Chapter 3).

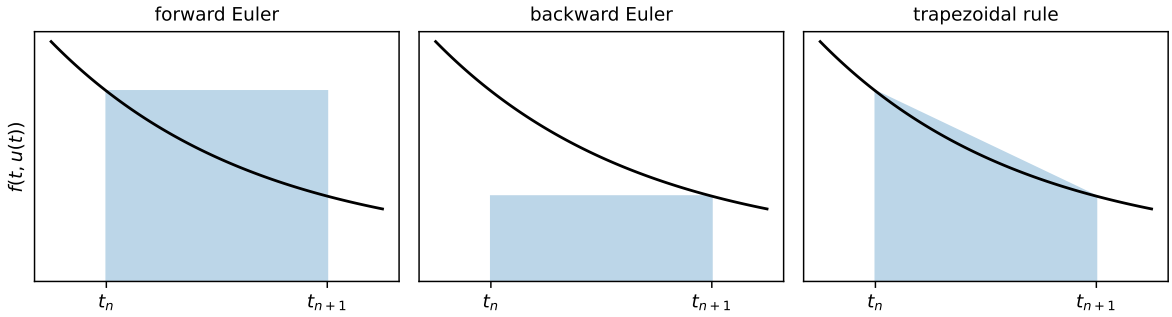


Figure 5.3: The tendency integrated over one time step. Forward Euler uses the value at the start of the step (left), backward Euler the value at the end (middle), the trapezoidal rule the mean of both (right).

5.2.4 Implementation

```
def euler_forward(f, u0, t_end, dt):
    n_steps = int(round(t_end / dt))
    t = dt * np.arange(n_steps + 1)
    u = np.empty(n_steps + 1)
    u[0] = u0
    for n in range(n_steps):
        u[n + 1] = u[n] + dt * f(t[n], u[n])
    return t, u

f = lambda t, u: (3 - u) / 2
for dt in [1.0, 0.5, 0.25]:
    t, u = euler_forward(f, 0.0, 4.0, dt)
    exact = 3 - 3 * np.exp(-2)
    print(f"dt = {dt:4.2f}: u(4) = {u[-1]:.5f}, error = {abs(u[-1] - exact):.5f}")
```

```
dt = 1.00: u(4) = 2.81250, error = 0.21851
dt = 0.50: u(4) = 2.69966, error = 0.10567
dt = 0.25: u(4) = 2.64580, error = 0.05180
```

Halving the time step roughly halves the error, as expected for a method of first order.

5.3 The backward Euler method

In the second view we can equally well use the backward difference Equation 3.4 at the new time t_{n+1},

$$\frac{u_{n+1} - u_n}{\Delta t} = f(t_{n+1}, u_{n+1}),$$

which in the third view is the rectangle with the height of the integrand at the right end (Figure 5.3). This is the **backward Euler method**

$$u_{n+1} = u_n + \Delta t\, f(t_{n+1}, u_{n+1}). \tag{5.7}$$

The unknown u_{n+1} now appears on both sides, inside the tendency. The formula does not give u_{n+1} explicitly; it is an equation that defines u_{n+1} **implicitly** and has to be solved in every time step. Methods such as Equation 5.5, which compute the new value directly from known values, are called **explicit**.

Linear tendency. For the decay equation, $f = -ku$, the equation is linear in u_{n+1} and can be solved by hand:

$$u_{n+1} = u_n - k\,\Delta t\, u_{n+1} \quad \Longrightarrow \quad u_{n+1} = \frac{u_n}{1 + k\,\Delta t}. \tag{5.8}$$

Compare this with the forward Euler result $u_{n+1} = (1 - k\Delta t)u_n$ from Equation 1.6. The factor $1/(1 + k\Delta t)$ lies between 0 and 1 for every time step: the backward Euler solution decays monotonically however large Δt is, where the forward Euler solution oscillated and grew for $k\Delta t > 2$. This is the reason to accept the extra work of an implicit method, and the subject of Chapter 6.

Nonlinear tendency. In general we write Equation 5.7 as a root finding problem for the unknown $U = u_{n+1}$,

$$G(U) = U - u_n - \Delta t\, f(t_{n+1}, U) = 0,$$

and solve it with Newton's method Equation 3.14. The derivative is $G'(U) = 1 - \Delta t\, \partial f/\partial u$, so that one Newton iteration reads

$$U^{(m+1)} = U^{(m)} - \frac{U^{(m)} - u_n - \Delta t\, f\big(t_{n+1}, U^{(m)}\big)}{1 - \Delta t\, \dfrac{\partial f}{\partial u}\big(t_{n+1}, U^{(m)}\big)},$$

started from $U^{(0)} = u_n$. Because u_n is already close to u_{n+1}, two or three iterations usually suffice. For a system of N equations the division becomes the solution of a linear system with the matrix $I - \Delta t\, J$, where J is the Jacobian of the tendency (Equation 3.16). An implicit time step of a PDE model therefore costs at least one solution of a large linear system.

```
def euler_backward(f, dfdu, u0, t_end, dt, newton_iterations=4):
    n_steps = int(round(t_end / dt))
    t = dt * np.arange(n_steps + 1)
    u = np.empty(n_steps + 1)
    u[0] = u0
    for n in range(n_steps):
        U = u[n]
        for _ in range(newton_iterations):
            G = U - u[n] - dt * f(t[n + 1], U)
            U = U - G / (1 - dt * dfdu(t[n + 1], U))
        u[n + 1] = U
    return t, u

# nonlinear test problem du/dt = -u^2, u(0) = 1, exact solution 1 / (1 + t)
f_nl = lambda t, u: -u**2
dfdu_nl = lambda t, u: -2 * u
for dt in [0.5, 0.25, 0.125]:
    t, u = euler_backward(f_nl, dfdu_nl, 1.0, 2.0, dt)
    print(f"dt = {dt:5.3f}: error at t = 2 is {abs(u[-1] - 1 / 3):.5f}")
```

```
dt = 0.500: error at t = 2 is 0.05425
dt = 0.250: error at t = 2 is 0.02867
dt = 0.125: error at t = 2 is 0.01477
```

The backward Euler method is also of first order.

5.4 Errors

5.4.1 Local and global error

Two errors must be distinguished.

The **global error** is what we care about: the difference between the exact and the numerical solution at time t_n,

$$E_n = u(t_n) - u_n.$$

The **local truncation error** τ_n is the error made in a single step that starts from the exact solution. It measures how well the exact solution satisfies the numerical formula. In Figure 5.2 it is the distance, after one step, between an Euler segment and the dashed solution through its starting point.

In addition there is the round-off error of each step. For the time steps used in practice it is much smaller than the truncation error, and we neglect it here.

5.4.2 Local truncation error of the forward Euler method

We expand the exact solution in a Taylor series around t_n,

$$u(t_{n+1}) = u(t_n) + \Delta t\, u'(t_n) + \frac{\Delta t^2}{2} u''(\xi_n),$$

where the last term is the remainder, with ξ_n somewhere between t_n and t_{n+1}. Since the exact solution satisfies the differential equation, $u'(t_n) = f\big(t_n, u(t_n)\big)$, and

$$u(t_{n+1}) = \underbrace{u(t_n) + \Delta t\, f\big(t_n, u(t_n)\big)}_{\text{one Euler step from the exact solution}} + \underbrace{\frac{\Delta t^2}{2} u''(\xi_n)}_{\tau_n}. \tag{5.9}$$

The local truncation error of the forward Euler method is proportional to Δt^2 and to the curvature of the solution. If $|u''| \le M$ on the interval of interest, then $|\tau_n| \le M\Delta t^2/2$.

The second derivative can be expressed by the tendency. Differentiating $u' = f(t,u)$ with the chain rule,

$$u'' = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial u}\, u' = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial u}\, f. \tag{5.10}$$

5.4.3 From the local to the global error

To reach a fixed time T we need $n = T/\Delta t$ steps. If the local errors simply added up, the global error would be about $n \cdot M\Delta t^2/2 = TM\Delta t/2$: one power of Δt is lost, and the method is of first order. This argument is incomplete, because an error made in one step is carried along and changed by all later steps. The complete argument takes three steps.

Step 1: a recursion for the error. We subtract the Euler formula Equation 5.5 from Equation 5.9:

$$\underbrace{u(t_{n+1}) - u_{n+1}}_{E_{n+1}} = \underbrace{u(t_n) - u_n}_{E_n} + \Delta t \left[f(t_n, u(t_n)) - f(t_n, u_n) \right] + \tau_n.$$

The bracket is the difference of the tendency at the exact and at the numerical state. By the Lipschitz condition Equation 5.2 it is at most $L\,|E_n|$ in magnitude. Hence

$$|E_{n+1}| \le (1 + L\Delta t)\,|E_n| + \frac{M\Delta t^2}{2}. \tag{5.11}$$

Each step multiplies the existing error by at most $1 + L\Delta t$ and adds a new local error.

Step 2: solving the recursion. We start without error, $E_0 = 0$, and apply Equation 5.11 repeatedly, writing $q = 1 + L\Delta t$ and $\tau = M\Delta t^2/2$:

$$|E_1| \le \tau, \qquad |E_2| \le q\tau + \tau, \qquad |E_3| \le q^2\tau + q\tau + \tau, \qquad \dots$$

After n steps the bound is a geometric series,

$$|E_n| \le \tau \sum_{j=0}^{n-1} q^j = \tau\,\frac{q^n - 1}{q - 1} = \frac{M\Delta t^2}{2} \cdot \frac{(1 + L\Delta t)^n - 1}{L\Delta t}.$$

Step 3: simplifying. Since $1 + x \le e^x$ for all x, we have $(1 + L\Delta t)^n \le e^{nL\Delta t} = e^{Lt_n}$ with $t_n = n\Delta t$ measured from $t_0 = 0$. One factor Δt cancels:

$$|E_n| \le \frac{M}{2L}\left(e^{Lt_n} - 1\right)\Delta t. \tag{5.12}$$

This is the result. At any fixed time the global error of the forward Euler method is bounded by a constant times Δt: the method **converges**, with **order one**. The rule that came out of the incomplete argument holds in general:

> A local truncation error of order Δt^{p+1} gives a global error of order Δt^p. The order of a method is p.

The constant in Equation 5.12 deserves a comment. It grows exponentially with Lt_n. For a problem whose solutions diverge from each other, this is realistic: errors grow as fast as neighbouring solutions separate, which is the predictability problem of Chapter 8. For a problem whose solutions converge, such as Equation 5.3, the bound is far too pessimistic, because the Lipschitz condition ignores the sign of $\partial f/\partial u$. Errors made early are then damped instead of amplified.

5.5 Methods of higher order

5.5.1 Heun's method

In the third view a better method needs a better quadrature rule for Equation 5.6. The trapezoidal rule Equation 3.12 uses the mean of the integrand at both ends of the step:

$$u_{n+1} = u_n + \frac{\Delta t}{2}\left[f(t_n, u_n) + f(t_{n+1}, u_{n+1})\right]. \tag{5.13}$$

This is the implicit **trapezoidal method**. To avoid the implicit equation, we replace the unknown u_{n+1} inside the tendency by a forward Euler estimate, the **predictor**, and then apply the trapezoidal formula as **corrector**:

$$\begin{aligned} k_1 &= f(t_n, u_n), \\ k_2 &= f(t_n + \Delta t,\ u_n + \Delta t\, k_1), \\ u_{n+1} &= u_n + \frac{\Delta t}{2}(k_1 + k_2). \end{aligned} \tag{5.14}$$

This is **Heun's method**, also called the improved Euler method. It is explicit and needs two evaluations of the tendency per step, two **stages**.

5.5.2 The order of two-stage methods

Is Heun's method really of second order, although the predictor is only of first order? We check this for a whole family of methods at once. Consider

$$k_1 = f(u_n), \qquad k_2 = f(u_n + c\,\Delta t\, k_1), \qquad u_{n+1} = u_n + \Delta t\,(b_1 k_1 + b_2 k_2), \tag{5.15}$$

with parameters b_1, b_2, c. To keep the algebra short we take a tendency that depends only on the state, $f = f(u)$, and write f and f' for $f(u_n)$ and its derivative with respect to u. The result holds in general.

The exact solution. Starting from u_n, the Taylor series is $u(t_n + \Delta t) = u_n + \Delta t\, u' + \frac{1}{2}\Delta t^2 u'' + O(\Delta t^3)$. With $u' = f$ and, from Equation 5.10, $u'' = f'f$:

$$u(t_n + \Delta t) = u_n + \Delta t\, f + \frac{\Delta t^2}{2} f' f + O(\Delta t^3).$$

The numerical solution. We expand the second stage around u_n:

$$k_2 = f(u_n + c\,\Delta t\, f) = f + c\,\Delta t\, f' f + O(\Delta t^2).$$

Inserting into Equation 5.15,

$$u_{n+1} = u_n + \Delta t\,(b_1 + b_2)\, f + \Delta t^2\, b_2 c\, f' f + O(\Delta t^3).$$

Comparison. The two expansions agree up to the terms with Δt^2, leaving a local truncation error of $O(\Delta t^3)$ and hence a method of second order, if

$$b_1 + b_2 = 1 \qquad \text{and} \qquad b_2\, c = \frac{1}{2}. \tag{5.16}$$

These are the **order conditions**. They are two equations for three parameters, so there is a family of second-order methods with two stages.

Method	c	b_1	b_2	Quadrature rule
Heun	1	$\frac{1}{2}$	$\frac{1}{2}$	trapezoidal
Midpoint	$\frac{1}{2}$	0	1	midpoint

The **midpoint method** takes half an Euler step and uses the tendency there for the full step: $u_{n+1} = u_n + \Delta t\, f(t_n + \frac{1}{2}\Delta t,\ u_n + \frac{1}{2}\Delta t\, k_1)$. Forward Euler is the member of the family Equation 5.15 with $b_2 = 0$; it violates the second condition and is of first order.

5.5.3 Runge–Kutta methods

Methods of the form Equation 5.15, with any number of stages, are **Runge–Kutta methods**. The most widely used one has four stages:

$$\begin{aligned} k_1 &= f(t_n, u_n), \\ k_2 &= f(t_n + \tfrac{1}{2}\Delta t,\ u_n + \tfrac{1}{2}\Delta t\, k_1), \\ k_3 &= f(t_n + \tfrac{1}{2}\Delta t,\ u_n + \tfrac{1}{2}\Delta t\, k_2), \\ k_4 &= f(t_n + \Delta t,\ u_n + \Delta t\, k_3), \\ u_{n+1} &= u_n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \tag{5.17}$$

This **classical Runge–Kutta method** (RK4) is of fourth order. Its order conditions are derived as above, by matching the Taylor series up to Δt^4; there are eight of them. If f depends only on t, the stages are the integrand at the left end, twice at the midpoint and at the right end, with the weights $\frac{1}{6}, \frac{4}{6}, \frac{1}{6}$: Equation 5.17 then is Simpson's rule.

Up to order four, a method of order p needs p stages. Beyond that more stages than the order are needed, which is one reason why RK4 is so popular.

5.5.4 Verification

We verify the orders on the nonlinear test problem $\mathrm{d}u/\mathrm{d}t = -u^2$, $u(0) = 1$, with the exact solution $u = 1/(1+t)$. Each method becomes a function that makes one step: it receives the tendency `f`, the time `t`, the state `u` and the time step `dt` and returns the state at `t + dt`. A second function, `integrate`, repeats such steps up to the end time. The method is handed to it as an argument, so the same loop serves all three methods.

```
def euler_forward_step(f, t, u, dt):
    return u + dt * f(t, u)

def heun_step(f, t, u, dt):
    k1 = f(t, u)                    # slope at the start of the step
    k2 = f(t + dt, u + dt * k1)     # slope at the end, after an Euler step
    return u + dt * (k1 + k2) / 2   # step with the mean of the two slopes

def rk4_step(f, t, u, dt):
    k1 = f(t, u)
    k2 = f(t + dt / 2, u + dt / 2 * k1)
    k3 = f(t + dt / 2, u + dt / 2 * k2)
    k4 = f(t + dt, u + dt * k3)
    return u + dt / 6 * (k1 + 2 * k2 + 2 * k3 + k4)

def integrate(step, f, u0, t_end, dt):
    # integrate from t = 0 to t_end with the one-step method "step"
    n_steps = round(t_end / dt)     # number of steps, computed once
    t_values = [0.0]                # lists of the times and states
    u_values = [u0]
    u = u0
    for n in range(n_steps):
        u = step(f, n * dt, u, dt)  # one step from t = n dt to (n + 1) dt
        t_values.append((n + 1) * dt)
        u_values.append(u)
    return np.array(t_values), np.array(u_values)   # lists -> arrays
```

As in Exercise 1, the time is computed from the step counter `n` instead of adding `dt` repeatedly. The functions work unchanged if the state `u` is a NumPy array instead of a number, because `+` and `*` act on each element of an array.

For the convergence test we integrate to $t = 2$, where the exact solution is $u = 1/3$, with $N = 4, 8, \ldots, 128$ steps.

```
def f_nl(t, u):
    return -u**2

n_list = [4, 8, 16, 32, 64, 128]   # numbers of steps
dt_list = []
for n in n_list:
    dt_list.append(2.0 / n)

def errors_at_t2(step):
    # errors at t = 2 of the method "step" for all time steps in dt_list
    errors = []
    for dt in dt_list:
        t, u = integrate(step, f_nl, 1.0, 2.0, dt)
        errors.append(abs(u[-1] - 1 / 3))   # u[-1] is the last value, at t = 2
    return errors

error_euler = errors_at_t2(euler_forward_step)
error_heun = errors_at_t2(heun_step)
error_rk4 = errors_at_t2(rk4_step)

print("    dt   order Euler   order Heun   order RK4")
for i in range(1, len(dt_list)):    # dt is halved: order = log2(error ratio)
    p_euler = np.log2(error_euler[i - 1] / error_euler[i])
    p_heun = np.log2(error_heun[i - 1] / error_heun[i])
    p_rk4 = np.log2(error_rk4[i - 1] / error_rk4[i])
    print(f"{dt_list[i]:6.4f}  {p_euler:10.2f}  {p_heun:11.2f}  {p_rk4:10.2f}")

fig, ax = plt.subplots(figsize=(6, 3.8))
ax.loglog(dt_list, error_euler, "o-", label="forward Euler")
ax.loglog(dt_list, error_heun, "o-", label="Heun")
ax.loglog(dt_list, error_rk4, "o-", label="RK4")
ax.set_xlabel(r"$\Delta t$")
ax.set_ylabel("error at $t = 2$")
ax.legend()
plt.show()
```

```
    dt   order Euler   order Heun   order RK4
0.2500         1.19         2.06        2.00
0.1250         1.06         2.09        3.82
0.0625         1.03         2.06        3.97
0.0312         1.01         2.03        3.99
0.0156         1.01         2.01        4.00
```

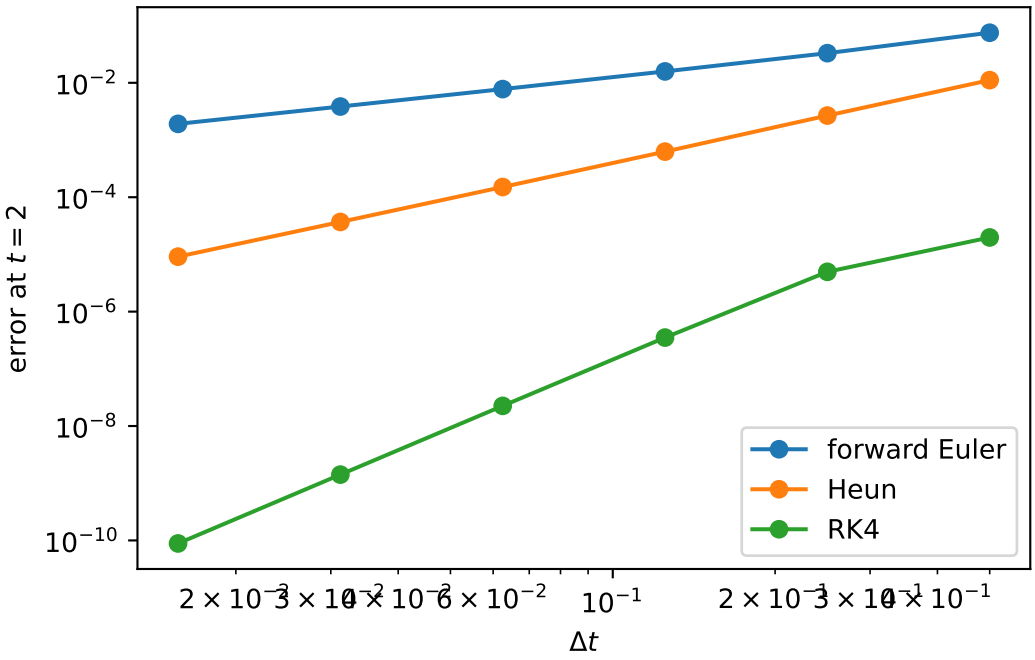


Figure 5.4: Global error at $t = 2$ as a function of the time step for $\mathrm{d}u/\mathrm{d}t = -u^2$. The slopes are the orders of the methods.

5.5.5 Accuracy for a given amount of work

A step of RK4 costs four evaluations of the tendency, a step of forward Euler one. In a weather model one evaluation of the tendency is the entire computation of all spatial derivatives and all physical processes, so the fair comparison is the error reached for a given number of evaluations. With N steps, forward Euler needs N evaluations, Heun $2N$ and RK4 $4N$.

```
evaluations_euler = []
evaluations_heun = []
evaluations_rk4 = []
for n in n_list:
    evaluations_euler.append(1 * n)
    evaluations_heun.append(2 * n)
    evaluations_rk4.append(4 * n)

fig, ax = plt.subplots(figsize=(6, 3.8))
ax.loglog(evaluations_euler, error_euler, "o-", label="forward Euler")
ax.loglog(evaluations_heun, error_heun, "o-", label="Heun")
ax.loglog(evaluations_rk4, error_rk4, "o-", label="RK4")
ax.set_xlabel("evaluations of the tendency")
```

```
ax.set_ylabel("error at $t = 2$")
ax.legend()
plt.show()
```

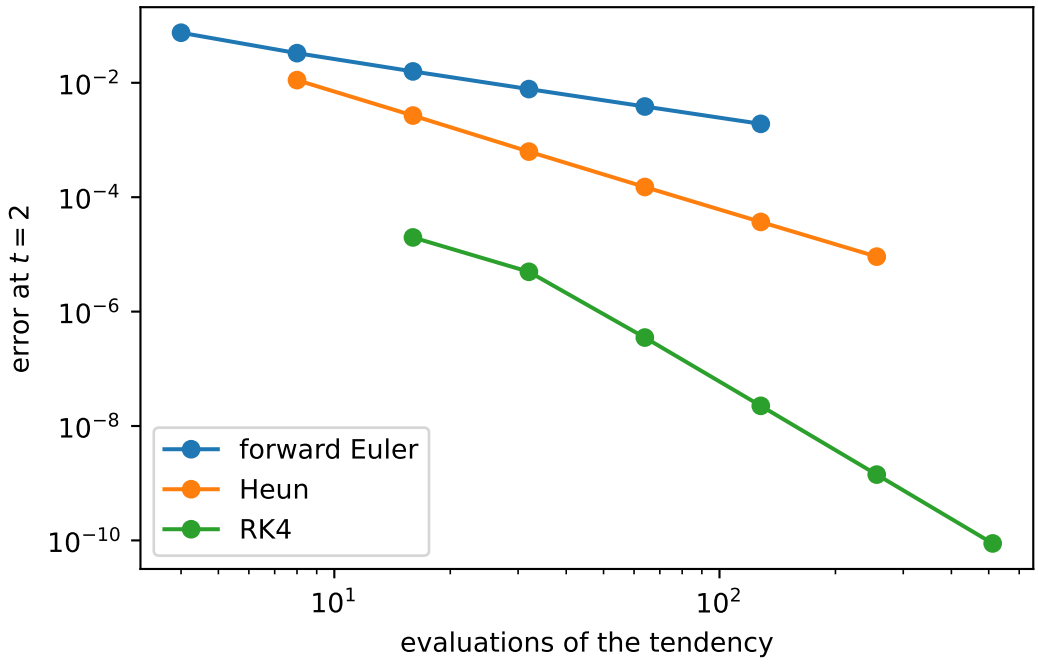


Figure 5.5: Error at $t = 2$ as a function of the number of evaluations of the tendency.

In Figure 5.5 the method of higher order wins as soon as more than a rough approximation is required: with about 100 evaluations RK4 is many orders of magnitude more accurate than forward Euler. Two qualifications apply in practice. The advantage exists only if the solution is smooth. And in models of the atmosphere the time step is often limited by stability rather than accuracy (Chapter 6); then the largest stable time step per evaluation decides.

5.6 Systems of equations

Nothing in the derivations above required u to be a single number. For a system of N equations the state is a vector $\vec{u}$ and the tendency a vector function,

$$\frac{\mathrm{d}\vec{u}}{\mathrm{d}t} = \vec{f}(t, \vec{u}),$$

and all formulas of this unit apply unchanged with vectors in the place of numbers. Equations of higher order are first rewritten as first-order systems (Chapter 2). The functions `rk4_step`, `euler_forward_step` and `integrate` defined above accept a NumPy array as the state; `integrate` then returns a two-dimensional array with one row per time and one column per variable.

As an example we integrate the predator–prey equations of Chapter 2,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = ax - \alpha xy, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = -cy + \gamma xy.$$

```
a, alpha, c, gamma = 1.0, 0.5, 0.75, 0.25

def tendency(t, u):
    x, y = u
    return np.array([a * x - alpha * x * y, -c * y + gamma * x * y])

u0 = np.array([2.0, 1.0])
t, u_rk4 = integrate(rk4_step, tendency, u0, 40.0, 0.05)
t, u_euler = integrate(euler_forward_step, tendency, u0, 40.0, 0.05)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(9, 3.2), gridspec_kw={"width_ratios":
↪  [2, 1]})
ax1.plot(t, u_rk4[:, 0], label="prey")
ax1.plot(t, u_rk4[:, 1], label="predators")
ax1.plot(t, u_euler[:, 0], "C0--", lw=0.8)
ax1.plot(t, u_euler[:, 1], "C1--", lw=0.8)
ax1.set_xlabel("$t$")
ax1.legend(fontsize=8)
ax2.plot(u_rk4[:, 0], u_rk4[:, 1], "k", label="RK4")
ax2.plot(u_euler[:, 0], u_euler[:, 1], "k--", lw=0.8, label="forward Euler")
ax2.set_xlabel("prey")
ax2.set_ylabel("predators")
ax2.legend(fontsize=8)
plt.tight_layout()
plt.show()
```

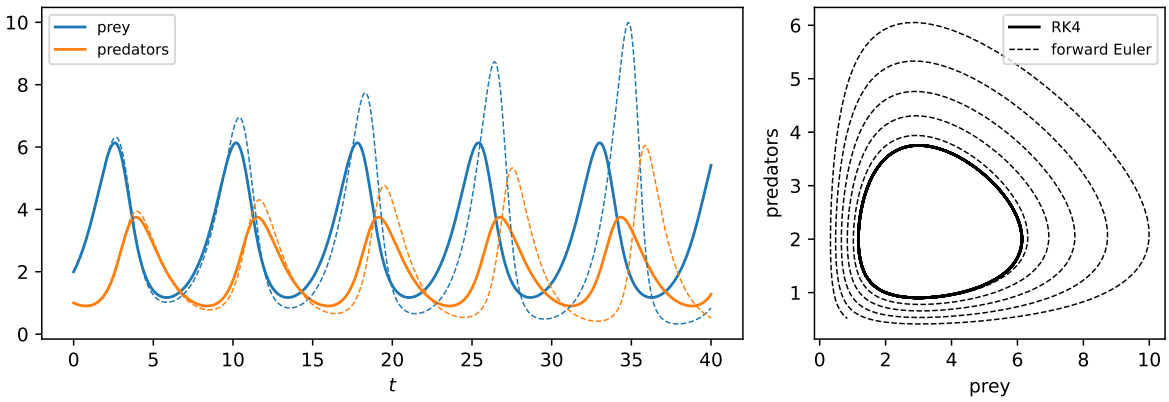


Figure 5.6: Predator–prey equations integrated with RK4 (lines) and forward Euler (dashed) with the same time step $\Delta t = 0.05$. The exact solution is periodic.

The exact solution is a closed curve in the plane of the two populations. RK4 reproduces it. The forward Euler solution spirals outward: the amplitude of the oscillation grows in every cycle. The error does not show up as random noise but as a systematic, qualitatively wrong behaviour. Why forward Euler amplifies oscillations is answered in Chapter 7.

5.7 Adaptive time steps

So far the time step was constant. The local truncation error, however, depends on the solution: for forward Euler it is proportional to u''. A constant step is too small where the solution is smooth and too large where it changes quickly. Adaptive methods choose the step such that the local error stays below a tolerance.

This requires an estimate of the local error without knowing the exact solution. The idea is to compute each step twice, with two methods of different order, p and $p+1$. The difference of the two results,

$$\mathrm{err} = \left| u_{n+1}^{(p+1)} - u_{n+1}^{(p)} \right|,$$

estimates the local error of the less accurate one, which is proportional to Δt^{p+1}. If err exceeds the tolerance tol, the step is rejected and repeated with a smaller Δt. In either case the next step size follows from the requirement that the error, scaled with Δt^{p+1}, should equal the tolerance:

$$\mathrm{tol} = \mathrm{err} \left(\frac{\Delta t_{\mathrm{new}}}{\Delta t} \right)^{p+1} \quad \Longrightarrow \quad \Delta t_{\mathrm{new}} = 0.9 \, \Delta t \left(\frac{\mathrm{tol}}{\mathrm{err}} \right)^{1/(p+1)}, \tag{5.18}$$

with a safety factor of 0.9. **Embedded Runge–Kutta pairs** make this cheap: the two methods share the same stages and differ only in the weights b_i, so the error estimate needs no additional evaluations of the tendency. The default method of `scipy.integrate.solve_ivp`, called RK45, is such a pair of orders four and five.

```
from scipy.integrate import solve_ivp

solution = solve_ivp(tendency, (0, 40), u0, method="RK45", rtol=1e-6, atol=1e-9)

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7, 4.2), sharex=True)
ax1.plot(solution.t, solution.y[0], ".-", ms=4, label="prey")
ax1.plot(solution.t, solution.y[1], ".-", ms=4, label="predators")
ax1.legend(fontsize=8)
ax2.semilogy(solution.t[1:], np.diff(solution.t), ".-", ms=4)
ax2.set_xlabel("$t$")
ax2.set_ylabel(r"$\Delta t$")
plt.tight_layout()
plt.show()
print(f"{solution.t.size - 1} steps, {solution.nfev} evaluations of the tendency")
```

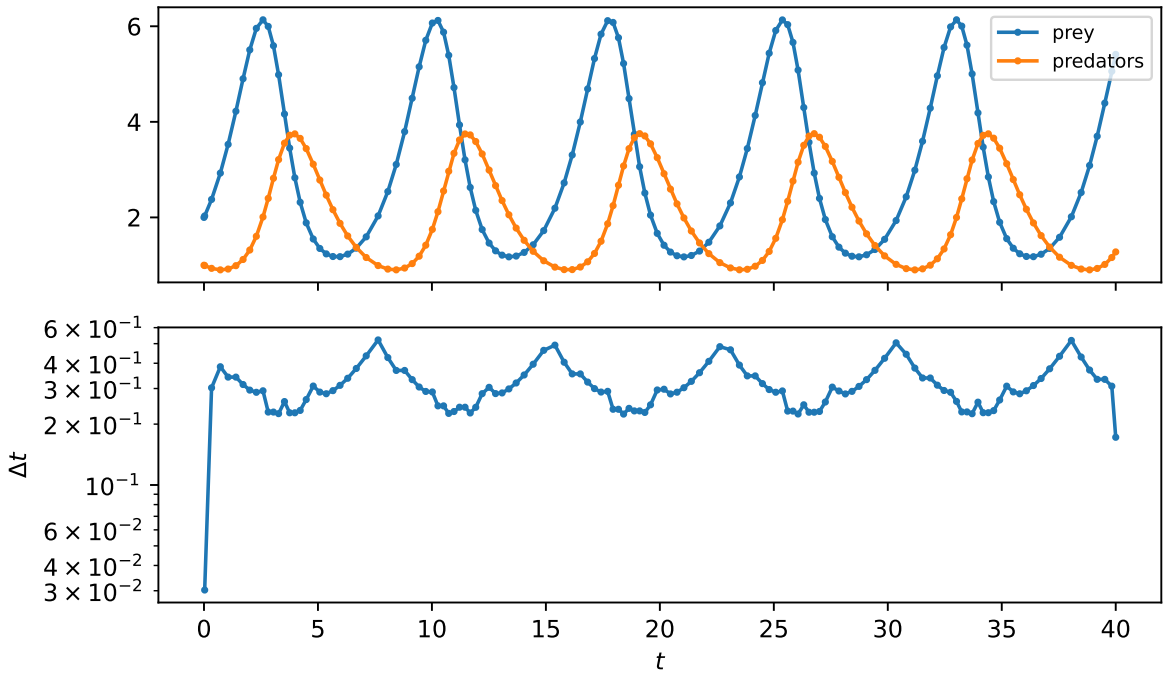


Figure 5.7: Predator–prey equations solved with the adaptive method RK45. Dots mark the time steps chosen by the method (top). The step is small where the solution changes rapidly (bottom).

```
131 steps, 950 evaluations of the tendency
```

Adaptive methods are the standard for ODE problems of moderate size, and for these `solve_ivp` should be preferred to a method written by hand. Large models of the atmosphere and ocean, in contrast, run with a fixed time step. Their step is dictated by stability, the cost of a rejected step is high, and a fixed step keeps the model reproducible and simple to couple with other components.

5.8 Summary

- The forward Euler method follows the tangent, replaces the derivative by a forward difference, or integrates the tendency with a rectangle. It is explicit and of first order.
- The backward Euler method evaluates the tendency at the new time. It is implicit: every step requires the solution of an equation, for nonlinear problems with Newton's method.
- The local truncation error is the error of one step from the exact solution. A local error of order Δt^{p+1} gives a global error of order Δt^p.
- Runge–Kutta methods reach higher order with several evaluations of the tendency per step. Their coefficients follow from matching the Taylor series of the exact solution. Heun's method and the midpoint method are of second order, the classical Runge–Kutta method of fourth order.
- Methods are compared by the accuracy reached for a given number of evaluations of the tendency.
- Adaptive methods estimate the local error from two methods of different order and adjust the time step.

5.9 Further reading

Boyce and DiPrima (2009), chapters 2 and 8, for direction fields, the Euler method and its errors; LeVeque (2007), chapter 5, for one-step methods and their analysis; Durran (2010), chapter 2, for time stepping in geophysical models.

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